Project 2 Data 624

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Contents

1 Project 2

1.1 Importing the data

Recap located at the end of the section. Skip there to jump past the code/workflow.

```
library(tidyverse)
library(VIM)

test_data <- read.csv("https://raw.githubusercontent.com/samanthabarbaro/data624/refs/heads/main/testda

training_data <- read.csv("https://raw.githubusercontent.com/samanthabarbaro/data624/refs/heads/main/tr
```

```r
glimpse(test_data)
```

```
## Rows: 267
## Columns: 33
## $ Brand.Code        <chr> "D", "A", "B", "B", "B", "B", "A", "B", "A", "D", "B~
## $ Carb.Volume       <dbl> 5.480000, 5.393333, 5.293333, 5.266667, 5.406667, 5.~
## $ Fill.Ounces       <dbl> 24.03333, 23.95333, 23.92000, 23.94000, 24.20000, 24~
## $ PC.Volume         <dbl> 0.2700000, 0.2266667, 0.3033333, 0.1860000, 0.160000~
## $ Carb.Pressure     <dbl> 65.4, 63.2, 66.4, 64.8, 69.4, 73.4, 65.2, 67.4, 66.8~
## $ Carb.Temp         <dbl> 134.6, 135.0, 140.4, 139.0, 142.2, 147.2, 134.6, 139~
## $ PSC               <dbl> 0.236, 0.042, 0.068, 0.004, 0.040, 0.078, 0.088, 0.0~
## $ PSC.Fill          <dbl> 0.40, 0.22, 0.10, 0.20, 0.30, 0.22, 0.14, 0.10, 0.48~
## $ PSC.CO2           <dbl> 0.04, 0.08, 0.02, 0.02, 0.06, NA, 0.00, 0.04, 0.04, ~
## $ Mnf.Flow          <dbl> -100, -100, -100, -100, -100, -100, -100, -100, -100~
## $ Carb.Pressure1    <dbl> 116.6, 118.8, 120.2, 124.8, 115.0, 118.6, 117.6, 121~
## $ Fill.Pressure     <dbl> 46.0, 46.2, 45.8, 40.0, 51.4, 46.4, 46.2, 40.0, 43.8~
## $ Hyd.Pressure1     <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure2     <dbl> NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ Hyd.Pressure3     <dbl> NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ Hyd.Pressure4     <int> 96, 112, 98, 132, 94, 94, 108, 108, 110, 106, 98, 96~
## $ Filler.Level      <dbl> 129.4, 120.0, 119.4, 120.2, 116.0, 120.4, 119.6, 131~
## $ Filler.Speed      <int> 3986, 4012, 4010, NA, 4018, 4010, 4010, NA, 4010, 10~
## $ Temperature       <dbl> 66.0, 65.6, 65.6, 74.4, 66.4, 66.6, 66.8, NA, 65.8, ~
## $ Usage.cont        <dbl> 21.66, 17.60, 24.18, 18.12, 21.32, 18.00, 17.68, 12.~
## $ Carb.Flow         <int> 2950, 2916, 3056, 28, 3214, 3064, 3042, 1972, 2502, ~
## $ Density           <dbl> 0.88, 1.50, 0.90, 0.74, 0.88, 0.84, 1.48, 1.60, 1.52~
## $ MFR               <dbl> 727.6, 735.8, 734.8, NA, 752.0, 732.0, 729.8, NA, 74~
## $ Balling           <dbl> 1.398, 2.942, 1.448, 1.056, 1.398, 1.298, 2.894, 3.3~
## $ Pressure.Vacuum   <dbl> -3.8, -4.4, -4.2, -4.0, -4.0, -3.8, -4.2, -4.4, -4.4~
## $ PH                <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ Oxygen.Filler     <dbl> 0.022, 0.030, 0.046, NA, 0.082, 0.064, 0.042, 0.096,~
## $ Bowl.Setpoint     <int> 130, 120, 120, 120, 120, 120, 120, 120, 120, 120, 12~
## $ Pressure.Setpoint <dbl> 45.2, 46.0, 46.0, 46.0, 50.0, 46.0, 46.0, 46.0, 46.0~
## $ Air.Pressurer     <dbl> 142.6, 147.2, 146.6, 146.4, 145.8, 146.0, 145.0, 146~
## $ Alch.Rel          <dbl> 6.56, 7.14, 6.52, 6.48, 6.50, 6.50, 7.18, 7.16, 7.14~
## $ Carb.Rel          <dbl> 5.34, 5.58, 5.34, 5.50, 5.38, 5.42, 5.46, 5.42, 5.44~
## $ Balling.Lvl       <dbl> 1.48, 3.04, 1.46, 1.48, 1.46, 1.44, 3.02, 3.00, 3.10~
```

```r
glimpse(training_data)
```

```
## Rows: 2,571
## Columns: 33
## $ Brand.Code        <chr> "B", "A", "B", "A", "A", "A", "A", "B", "B", "B", "B~
## $ Carb.Volume       <dbl> 5.340000, 5.426667, 5.286667, 5.440000, 5.486667, 5.~
## $ Fill.Ounces       <dbl> 23.96667, 24.00667, 24.06000, 24.00667, 24.31333, 23~
## $ PC.Volume         <dbl> 0.2633333, 0.2386667, 0.2633333, 0.2933333, 0.111333~
## $ Carb.Pressure     <dbl> 68.2, 68.4, 70.8, 63.0, 67.2, 66.6, 64.2, 67.6, 64.2~
## $ Carb.Temp         <dbl> 141.2, 139.6, 144.8, 132.6, 136.8, 138.4, 136.8, 141~
## $ PSC               <dbl> 0.104, 0.124, 0.090, NA, 0.026, 0.090, 0.128, 0.154,~
## $ PSC.Fill          <dbl> 0.26, 0.22, 0.34, 0.42, 0.16, 0.24, 0.40, 0.34, 0.12~
## $ PSC.CO2           <dbl> 0.04, 0.04, 0.16, 0.04, 0.12, 0.04, 0.04, 0.04, 0.14~
## $ Mnf.Flow          <dbl> -100, -100, -100, -100, -100, -100, -100, -100, -100~
```

```
## $ Carb.Pressure1    <dbl> 118.8, 121.6, 120.2, 115.2, 118.4, 119.6, 122.2, 124~
## $ Fill.Pressure     <dbl> 46.0, 46.0, 46.0, 46.4, 45.8, 45.6, 51.8, 46.8, 46.0~
## $ Hyd.Pressure1     <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure2     <dbl> NA, NA, NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure3     <dbl> NA, NA, NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure4     <int> 118, 106, 82, 92, 92, 116, 124, 132, 90, 108, 94, 86~
## $ Filler.Level      <dbl> 121.2, 118.6, 120.0, 117.8, 118.6, 120.2, 123.4, 118~
## $ Filler.Speed      <int> 4002, 3986, 4020, 4012, 4010, 4014, NA, 1004, 4014, ~
## $ Temperature       <dbl> 66.0, 67.6, 67.0, 65.6, 65.6, 66.2, 65.8, 65.2, 65.4~
## $ Usage.cont        <dbl> 16.18, 19.90, 17.76, 17.42, 17.68, 23.82, 20.74, 18.~
## $ Carb.Flow         <int> 2932, 3144, 2914, 3062, 3054, 2948, 30, 684, 2902, 3~
## $ Density           <dbl> 0.88, 0.92, 1.58, 1.54, 1.54, 1.52, 0.84, 0.84, 0.90~
## $ MFR               <dbl> 725.0, 726.8, 735.0, 730.6, 722.8, 738.8, NA, NA, 74~
## $ Balling           <dbl> 1.398, 1.498, 3.142, 3.042, 3.042, 2.992, 1.298, 1.2~
## $ Pressure.Vacuum   <dbl> -4.0, -4.0, -3.8, -4.4, -4.4, -4.4, -4.4, -4.4, -4.4~
## $ PH                <dbl> 8.36, 8.26, 8.94, 8.24, 8.26, 8.32, 8.40, 8.38, 8.38~
## $ Oxygen.Filler     <dbl> 0.022, 0.026, 0.024, 0.030, 0.030, 0.024, 0.066, 0.0~
## $ Bowl.Setpoint     <int> 120, 120, 120, 120, 120, 120, 120, 120, 120, 120, 12~
## $ Pressure.Setpoint <dbl> 46.4, 46.8, 46.6, 46.0, 46.0, 46.0, 46.0, 46.0, 46.0~
## $ Air.Pressurer     <dbl> 142.6, 143.0, 142.0, 146.2, 146.2, 146.6, 146.2, 146~
## $ Alch.Rel          <dbl> 6.58, 6.56, 7.66, 7.14, 7.14, 7.16, 6.54, 6.52, 6.52~
## $ Carb.Rel          <dbl> 5.32, 5.30, 5.84, 5.42, 5.44, 5.44, 5.38, 5.34, 5.34~
## $ Balling.Lvl       <dbl> 1.48, 1.56, 3.28, 3.04, 3.04, 3.02, 1.44, 1.44, 1.44~
```

```r
str(training_data)
```

```
## 'data.frame':    2571 obs. of  33 variables:
##  $ Brand.Code       : chr  "B" "A" "B" "A" ...
##  $ Carb.Volume      : num  5.34 5.43 5.29 5.44 5.49 ...
##  $ Fill.Ounces      : num  24 24 24.1 24 24.3 ...
##  $ PC.Volume        : num  0.263 0.239 0.263 0.293 0.111 ...
##  $ Carb.Pressure    : num  68.2 68.4 70.8 63 67.2 66.6 64.2 67.6 64.2 72 ...
##  $ Carb.Temp        : num  141 140 145 133 137 ...
##  $ PSC              : num  0.104 0.124 0.09 NA 0.026 0.09 0.128 0.154 0.132 0.014 ...
##  $ PSC.Fill         : num  0.26 0.22 0.34 0.42 0.16 0.24 0.4 0.34 0.12 0.24 ...
##  $ PSC.CO2          : num  0.04 0.04 0.16 0.04 0.12 0.04 0.04 0.04 0.14 0.06 ...
##  $ Mnf.Flow         : num  -100 -100 -100 -100 -100 -100 -100 -100 -100 -100 ...
##  $ Carb.Pressure1   : num  119 122 120 115 118 ...
##  $ Fill.Pressure    : num  46 46 46 46.4 45.8 45.6 51.8 46.8 46 45.2 ...
##  $ Hyd.Pressure1    : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ Hyd.Pressure2    : num  NA NA NA 0 0 0 0 0 0 0 ...
##  $ Hyd.Pressure3    : num  NA NA NA 0 0 0 0 0 0 0 ...
##  $ Hyd.Pressure4    : int  118 106 82 92 92 116 124 132 90 108 ...
##  $ Filler.Level     : num  121 119 120 118 119 ...
##  $ Filler.Speed     : int  4002 3986 4020 4012 4010 4014 NA 1004 4014 4028 ...
##  $ Temperature      : num  66 67.6 67 65.6 65.6 66.2 65.8 65.2 65.4 66.6 ...
##  $ Usage.cont       : num  16.2 19.9 17.8 17.4 17.7 ...
##  $ Carb.Flow        : int  2932 3144 2914 3062 3054 2948 30 684 2902 3038 ...
##  $ Density          : num  0.88 0.92 1.58 1.54 1.54 1.52 0.84 0.84 0.9 0.9 ...
##  $ MFR              : num  725 727 735 731 723 ...
##  $ Balling          : num  1.4 1.5 3.14 3.04 3.04 ...
##  $ Pressure.Vacuum  : num  -4 -4 -3.8 -4.4 -4.4 -4.4 -4.4 -4.4 -4.4 -4.4 ...
##  $ PH               : num  8.36 8.26 8.94 8.24 8.26 8.32 8.4 8.38 8.38 8.5 ...
##  $ Oxygen.Filler    : num  0.022 0.026 0.024 0.03 0.03 0.024 0.066 0.046 0.064 0.022 ...
```

```
##  $ Bowl.Setpoint    : int  120 120 120 120 120 120 120 120 120 120 ...
##  $ Pressure.Setpoint: num  46.4 46.8 46.6 46 46 46 46 46 46 46 ...
##  $ Air.Pressurer    : num  143 143 142 146 146 ...
##  $ Alch.Rel         : num  6.58 6.56 7.66 7.14 7.14 7.16 6.54 6.52 6.52 6.54 ...
##  $ Carb.Rel         : num  5.32 5.3 5.84 5.42 5.44 5.44 5.38 5.34 5.34 5.34 ...
##  $ Balling.Lvl      : num  1.48 1.56 3.28 3.04 3.04 3.02 1.44 1.44 1.44 1.38 ...
```

All the rows with numbers are formatted as numbers. All the variables are numeric except for brand code, likely the only categorical variable.

1.1.1 Recap

- Loaded the training and evaluation sets from the project GitHub repo
- Inspected the structure with `glimpse()` and `str()`
- All variables came in as numeric except `Brand.Code`, which is the only categorical

1.2 Exploring the data

We'll look at:

- The number of cases
- Missing values
- Data types (numerical vs. categorical)

1.2.1 Number of cases

```r
nrow(training_data)
```

```
## [1] 2571
```

```r
nrow(test_data)
```

```
## [1] 267
```

The training data has 2571 cases and the test data has 267.

1.2.2 Variables

Counting the number of columns/variables:

```r
ncol(test_data)
```

```
## [1] 33
```

```r
ncol(training_data)
```

```
## [1] 33
```

Both data sets have 33 columns.

Are all the variables represented in both data sets?

```r
colnames(test_data)
```

```
##  [1] "Brand.Code"        "Carb.Volume"       "Fill.Ounces"
##  [4] "PC.Volume"         "Carb.Pressure"     "Carb.Temp"
##  [7] "PSC"               "PSC.Fill"          "PSC.CO2"
## [10] "Mnf.Flow"          "Carb.Pressure1"    "Fill.Pressure"
## [13] "Hyd.Pressure1"     "Hyd.Pressure2"     "Hyd.Pressure3"
## [16] "Hyd.Pressure4"     "Filler.Level"      "Filler.Speed"
## [19] "Temperature"       "Usage.cont"        "Carb.Flow"
## [22] "Density"           "MFR"               "Balling"
## [25] "Pressure.Vacuum"   "PH"                "Oxygen.Filler"
## [28] "Bowl.Setpoint"     "Pressure.Setpoint" "Air.Pressurer"
## [31] "Alch.Rel"          "Carb.Rel"          "Balling.Lvl"
```

```r
colnames(training_data)
```

```
##  [1] "Brand.Code"        "Carb.Volume"       "Fill.Ounces"
##  [4] "PC.Volume"         "Carb.Pressure"     "Carb.Temp"
##  [7] "PSC"               "PSC.Fill"          "PSC.CO2"
## [10] "Mnf.Flow"          "Carb.Pressure1"    "Fill.Pressure"
## [13] "Hyd.Pressure1"     "Hyd.Pressure2"     "Hyd.Pressure3"
## [16] "Hyd.Pressure4"     "Filler.Level"      "Filler.Speed"
## [19] "Temperature"       "Usage.cont"        "Carb.Flow"
## [22] "Density"           "MFR"               "Balling"
## [25] "Pressure.Vacuum"   "PH"                "Oxygen.Filler"
## [28] "Bowl.Setpoint"     "Pressure.Setpoint" "Air.Pressurer"
## [31] "Alch.Rel"          "Carb.Rel"          "Balling.Lvl"
```

Both data sets have the same columns, so we can proceed.

1.2.3 Missing data

How many missing values are there? Are there any trends? How many complete cases are there?

1.2.3.1 Test data

Count nulls/missing values in the test data:

```r
sort(colSums(is.na(test_data)), decreasing = TRUE)
```

```
##                PH               MFR      Filler.Speed        Brand.Code
##               267                31                10                 8
##       Fill.Ounces               PSC           PSC.CO2         PC.Volume
##                 6                 5                 5                 4
```

```
##    Carb.Pressure1      Hyd.Pressure4          PSC.Fill     Oxygen.Filler
##                 4                  4                 3                 3
##          Alch.Rel      Fill.Pressure      Filler.Level       Temperature
##                 3                  2                 2                 2
##        Usage.cont Pressure.Setpoint          Carb.Rel       Carb.Volume
##                 2                  2                 2                 1
##         Carb.Temp      Hyd.Pressure2     Hyd.Pressure3           Density
##                 1                  1                 1                 1
##           Balling    Pressure.Vacuum     Bowl.Setpoint     Air.Pressurer
##                 1                  1                 1                 1
##     Carb.Pressure           Mnf.Flow     Hyd.Pressure1         Carb.Flow
##                 0                  0                 0                 0
##       Balling.Lvl
##                 0
```

First off, all the **pH values are missing**. This is not a test set —it's what we will predict. The training data will have to be split on its own into a training and test set.

Most columns are missing a few (under 10) values or none at all. MFR is missing the most values – 31 out of 267, or about 12%. Filler speed is next, missing only 10 values.

Next, we'll look at the data row-wise and see if there are full or close-to-full missing cases.

```r
rowSums(is.na(test_data))
```

```
##   [1]  3  1  1  4  1  2  1  4  1  2  3  1  1  1  1  4  1  1  1  1  1  2  3  1  1
##  [26]  1  1  1  2  1  1  2  2  2  2  1  1  1  1  2  1  2  1  1  1  1  1  1  1  1
##  [51]  3  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
##  [76]  1  2  1  1  1  1  1  1  1  5  2  1  1  1  1  2 15  1  2  2  1  3  1  2  1
## [101]  2  1  1  1  1  1  1  1  1  2  1  1  1  1  3  1  1  2  1  1  2  1  1  1  1
## [126]  1  2  2  1  1  2  2  2  3  1  1  3  2  2  2  1  1  1  2  1  1  1  1  2  1
## [151]  1  1  1  1  1  1  1  1  1  1  2  2  1  1  1  2  2  1  1  1  2  2  1  1  1
## [176]  1  1  1  2  1  1  1  1  1  2  1  1  1  1  1  1  1  1  2  1  1  1  1  1  1
## [201]  1  1  2  1  1  1  1  2  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [226]  1  3  1  1  1  1  2  1  1  1  2  1  1  1  1  1  1  2  1  1  1  1  1  3  1
## [251]  2  1  3  1  1  9  2  1  1  1  1  1  1  1  1  1  1
```

```r
#totals by row (1 does not count)
table(rowSums(is.na(test_data)))
```

```
##
##   1   2   3   4   5   9  15
## 200  50  11   3   1   1   1
```

```r
#total NAs (subtracting 267 for the missing pH values)
sum(is.na(test_data)) - 267
```

```
## [1] 107
```

Overall, a data set with 267 rows and 33 columns is missing 107 values, or about 1.3%. 67 out of 267 cases are missing at least one value (about 25%).

One case has 14 missing values, so it may make sense to drop that; another is missing 8. Otherwise, most rows with missing values are missing only 1-3.

1.2.3.2 Training data

Which variables are missing data?

```r
sort(colSums(is.na(training_data)), decreasing = TRUE)
```

```
##               MFR        Brand.Code      Filler.Speed         PC.Volume
##               212               120                57                39
##           PSC.CO2       Fill.Ounces               PSC    Carb.Pressure1
##                39                38                33                32
##     Hyd.Pressure4     Carb.Pressure         Carb.Temp          PSC.Fill
##                30                27                26                23
##     Fill.Pressure      Filler.Level     Hyd.Pressure2     Hyd.Pressure3
##                22                20                15                15
##       Temperature     Oxygen.Filler Pressure.Setpoint     Hyd.Pressure1
##                14                12                12                11
##       Carb.Volume          Carb.Rel          Alch.Rel        Usage.cont
##                10                10                 9                 5
##                PH          Mnf.Flow         Carb.Flow     Bowl.Setpoint
##                 4                 2                 2                 2
##           Density           Balling       Balling.Lvl   Pressure.Vacuum
##                 1                 1                 1                 0
##      Air.Pressurer
##                 0
```

MFR is missing 212 out of 2571 cases, about 8.2%. Brand code, the only categorical variable, is missing 120 times, or about 4.7% of the time. Filler speed is missing 57 times, about 2.2%, and everything else is under 2%.

Let's see if anything strange is happening across rows:

```r
#nothing out of the ordinary across rows. Most rows are complete.
#Rows with NAs typically 1 or 2 NAs, wich
rowSums(is.na(training_data))
```

```
##    [1]  2  2  2  1  0  0  2  1  0  0  0  0  1  0  0  0  0  0  0  0  0  2  0  0
##   [25]  0  1  0  0  0  0  0  0  0  0  0  1  1  0  0  0  1  0  0  0  0  0  0  1
##   [49]  0  2  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0
##   [73]  0  0  0  0  0  0  0  2  0  0  0  0  0  0  0  0  0  0  0  0  0  2  0  0
##   [97]  0  1  0  0  0  0  0  0  0  0  0  2  5  0  0  0  0  0  1  0  0  0  0  0
##  [121]  0  0  0  0  2  0  0  1  0  0  0  0  0  1  0  0  0  2  0  0  0  0  0  0
##  [145]  0  0  0  0  0  0  1  1  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
##  [169]  0  0  0  0  0  0  0  0  1  0  2  0  0  0  0  0  2  0  0  0  0  0  0  0
##  [193]  0  0  0  1  0  0  0  0  0  0  0  0  0  1  1  1  1  1  1  1  1  0  0  0
##  [217]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  2  1  0  0  0  0  1
##  [241]  0  1  6  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0  1  2
##  [265]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
##  [289]  0  0  0  0  0  0  0  0  1  1  1  1  1  0  0  0  0  3  0  0  0  0  0  2
##  [313]  0  0  0  0  0  0  0  1  0  0  0  1  1  0  1  0  0  0  1  0  0  1  2  0
##  [337]  0  1  0  2  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0
##  [361]  0  0  0  0  2  2  0  2  0  0  0  1  0  2  1  0  1  0  0  0  0  0  0  0
##  [385]  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  2  0  0  0  0  0  0  0
##  [409]  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  2  2  1  1  0
##  [433]  0  0  2  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0
##  [457]  0  0  1  0  0  0  1  0  0  0  0  0  0  0  2  1  0  0  4  1  1  1  1  1
```

```
##  [481]  1  1  1  0  0  0  0  0  0  0  0  1  0  1  0  3  0  1  1  0  0  0  0  0
##  [505]  0  0  0  1  0  0  0  0  0  0  0  0  1  0  1  0  0  0  0  0  0  0  0  0
##  [529]  2  0  0  1  0  0  0  0  0  0  0  0  0  0  1  0  0  1  0  0  0  0  1  0
##  [553]  0  0  0  0  0  1  0  1  0  0  2  0  0  0  0  0  0  0  0  0  0  0  0  0
##  [577]  1  0  0  0  0  0  0  1  0  0  0  3  0  0  1  0  0  0  0  0  0  0  0  0
##  [601]  0  0  0  0  0  0  0  0  2  4  0  0  0  0  0  0  0  0  0  0  0  0  1  0
##  [625]  0  0  0  0  3  0  0  0  0  0  0  0  0  0  0  0  0  3  0  0  0  0  0  0
##  [649]  1  0  0  0  0  0  1  0  2  2  1  0  0  0  0  0  0  0  0  0  0  0  1  0
##  [673]  0  1  0  0  0  0  0  0  0  0  0  0  0  2  0  0  0  5  2  0  0  1  2  3
##  [697]  3  0  1  1  1  1  1  1  1  0  0  0  0  0  0  0  0  0  0  1  0  0  3  0
##  [721]  0  2  0  0  0  0  0  0  0  0  0  1  2  0  0  0  0  9  4  0  0  0  0  0
##  [745]  0  0  0  0  0  1  0  0  1  0  3  0  0  1  0  0  0  0  0  0  1  0  0  0
##  [769]  0  0  2  2  3  3  3  3  4  3  3  3  3  3  3  3  4  0  0  0  0  0  0  0
##  [793]  0  0  2  0  0  0  1  2  4  1  0  0  0  0  0  0  2  1  0  0  0  0  0  3
##  [817]  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  2  1  0  0  0  1  0  0  0
##  [841]  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
##  [865]  2  0  0  0  0  0  0  0  0  0  0  0  0  0  2  2  0  0  0  0  1  0  0  0
##  [889]  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  6  0  0  0
##  [913]  2  0  0  0  0  0  0  0  0  0  0  0  0  0  1  1  1  0  0  0  0  0  1  0
##  [937]  0  0  0  0  0  0  0  0  0  0  0  0  0  1  1  0  0  0  1  1  1  1  1  1
##  [961]  2  1  1  2  2  0  0  0  1  0  2  0  0  0  0  0  0  0  0  0  0  0  0  0
##  [985]  0  0  0  0  0  0  0  0  0  0  2  1  1  1  0  3  0  0  0  0  0  0  0  0
## [1009]  0  0  0  0  0  2  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  2  0  0
## [1033]  0  0  2  0  0  2  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  3  0  0
## [1057]  0  0  0  0  0  0  0  0  1  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0
## [1081]  0  0  0  0  0  0  0  1  0  0  0  0  0  8  2  0  1  0  0  0  0  1  0  0
## [1105]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  1
## [1129]  0  0  0  0  0  0  0  0  3  1  0  0  0  2  0  0  0  0  0  0  0  0  0  0
## [1153]  0  0  0  0  1  0  0  2  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0
## [1177]  0  0  0  0  0  0  1  0  0  0  0  0  1  0  0  0  0  1  0  0  0  0  0  0
## [1201]  1  0  1  0  0  0  0  0  1  1  1  1  3  2  3  2  4  1  1  1  1  1  0  0
## [1225]  0  0  0  0  0  0  0  0  0  0  2  0  0  0  0  0  0  4  2  0  0  1  0  0
## [1249]  0  0  0  0  3  0  0  0  0  0  0  2  0  0  0  0  1  0  0  1  1  1  0  0
## [1273]  1  0  0  0  0  0  1  3  0  0  0  0  1  0  0  0  0  0  0  0  2  0  2  1
## [1297]  0  1  0  4  0  1  0  1  0  0  1  1  1  0  0  0  0  0  1  1  1  2  1  1
## [1321]  1  1  1  2  1  1  0  0  0  0  1  1  1  1  1  1  2  1  1  1  2  1  1  1
## [1345]  1  1  2  1  1  1  1  1  1  1  2  1  2  0  1  1  1  1  1  1  3  1  1  2
## [1369]  2  1  0  0  0  0  0  2  0  0  0  0  0  0  0  1  1  1  1  0  0  0  0  0
## [1393]  0  0  0  2  0  0  0  1  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## [1417]  0  0  0  0  3  2  0  0  0  0  0  1  0  0  3  0  0  0  0  0  0  0  0  0
## [1441]  0  0  0  0  0  0  0  0  3  0  0  0  0  0  0  0  0  0  0  0  3  0  0  0
## [1465]  0  0  1  0  0  0  1  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0
## [1489]  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0  0  1  0  0  0
## [1513]  0  0  0  0  0  0  0  2  0  0  0  0  0  0  0  4  1  0  0  0  0  0  6  0
## [1537]  0  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  1  0  1  1  1  1
## [1561]  2  3  0  0  0  0  0  0  0  0  0  0  0  1  3  0  0  0  0  0  0  0  0  0
## [1585]  0  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## [1609]  0  1  0  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  2  0  0  0
## [1633]  0  1  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## [1657]  0  0  0  0  0  0  0  0  1  1  0  0  0  0  1  0  1  0  0  0  0  0  1  0
## [1681]  0  0  1  0  0  1  1  0  1  1  0  0  0  0  0  0  0  0  0  0  0  0  1  0
## [1705]  0  0  1  0  0  0  0  0  0  0  0  0  0  0 12  1  0  0  1  0  3  1  2  0
## [1729]  0  0  0  1  0  0  0  2  0  0  0  0  0  0  0  0  0  0  0  2  0  0  0  0
## [1753]  0  0  0  0  0  0  0  0  0  0  0  0  0  2  0  0  0  0  0  1  1  0  1  0
```

```
## [1777]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  1  0  0  2  1  1
## [1801]  4  1  1  1  1  1  3  1  0  0  0  0  0  0  0  2  2  0  0  0  0  0  0  0
## [1825]  2  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  5  0  0  0
## [1849]  0  0  1  0  0  2  0  0  0  0  0  2  0  0  0  0  0  0  0  0  0  0  0  0
## [1873]  0  0  0  0  0  0  3  0  0  0  0  0  0  0  0  0  0  0  0  0  2  0  0  0
## [1897]  0  0  0  2  0  2  0  0  0  0  0  1  0  0  0  0  0  1  0  0  0  0  1  0
## [1921]  1  0  0  0  0  2  0  1  0  0  1  0  0  0  0  0  0  0  0  1  0  1  0  0
## [1945]  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  2  1  1  1  1  1  0  0
## [1969]  1  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  1  0  0  0
## [1993]  0  0  0  0  0  1  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  1
## [2017]  3  1  1  1  3  1  1  3  1  1  0  0  0  0  0  1  0  0  0  0  0  0  0  0
## [2041]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## [2065]  0  2  0  1  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0  0  2  0  0
## [2089]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  1  0
## [2113]  0  0  0  0  0  0  0  0  0  0  0  4  0  0  0  0  1  0  0  0  0  0  0  0
## [2137]  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  2  0  0  0
## [2161]  0  0  0  0  0  0  0  0  0  0  0  0  0  1  1  0  0  0  0  0  0  0  0  0
## [2185]  1  1  1  0  0  2  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0
## [2209]  0  0  0  0  0  5  0  0  0  1  0  0  0  0  0  0  2  0  0  0  0  0  0  0
## [2233]  0  0  1  0  0  0  0  1  0  0  2  0  0  0  0  0  0  0  0  0  0  0  0  0
## [2257]  2  1  1  1  0  0  0  0  0  0  0  0  0  0  0  4  0  0  0  0  0  0  2  0
## [2281]  0  1  0  0  0  0  1  0  0  0  0  2  0  0  0  0  0  1  0  0  0  2  0  0
## [2305]  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  2  0  0  1  2
## [2329]  0  1  1  0  1  0  0  0  0  0  1  2  0  0  0  0  0  0  0  0  0  0  0  0
## [2353]  0  0  0  0  2  0  3  0  0  0  0  0  0  0  0  0  0  1  0  0  3  0  0  0
## [2377]  0  0  0  2  0  0  0  0  2  3  0  0  1  0  0  0  0  0  0  1  0  0  0  0
## [2401]  0  0  0  0  0  0  0  0  1  0  0  0  2  2  0  0  0  0  0  0  0  0  0  0
## [2425]  0  0  1  1  1  0  0  0  0  0  0  2  0  1  0  0  0  0  0  0  0  0  1  0
## [2449]  0  0  0  0  0  0  0  0  1  0  0  0  1  0  0  0  0  0  0  0  0  0  1  1
## [2473]  0  0  0  0  2  0  2  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0  0
## [2497]  0  0  0  0  0  0  0  1  0  0  0  7  1  0  0  0  0  0  2  0  0  0  0  0
## [2521]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0
## [2545]  0  0  0  0  0  1  0  0  0  0  0  0  0  0  2  0  0  0  0  1  0  0  1  1
## [2569]  0  0  3
```

```r
table(rowSums(is.na(training_data)))
```

```
##
##    0    1    2    3    4    5    6    7    8    9   12
## 2038  345  119   45   13    4    3    1    1    1    1
```

```r
sum(is.na(test_data))
```

```
## [1] 374
```

Overall, 374 values are missing (about 14.5% total missingness). 2038/2571 cases are complete. That's about 79%; 21% of cases are incomplete. Let's also check to see if the data is MCAR.

```r
library(naniar)
numeric_only <- training_data |>
  select(-Brand.Code)
```

```r
numeric_only <- numeric_only |>
  dplyr::select(where(is.numeric)) |>
  mutate(across(everything(), as.double))

mcar_test(numeric_only)
```

```
## # A tibble: 1 x 4
##   statistic    df p.value missing.patterns
##       <dbl> <dbl>   <dbl>            <int>
## 1    10791.  2871       0               99
```

The p-value is 0, which means the data most likely isn't missing completely at random (there are patterns in the missing data).

1.3 Data cleaning

We'll cover:

- Variables with zero or near-zero variance
- Highly correlated variables
- Outliers/range of variables/scaling (if needed)

1.3.1 Checking for variables with near-zero or zero variance

This data set has 33 variables. Let's check to see if they're all (potentially) providing useful predictive information by testing for near-zero or zero variance.

```r
library(caret)
nzv_results <- nearZeroVar(training_data, saveMetrics = TRUE)

nzv_results
```

```
##                    freqRatio percentUnique zeroVar   nzv
## Brand.Code          2.014634     0.1555815   FALSE FALSE
## Carb.Volume         1.055556     3.9284325   FALSE FALSE
## Fill.Ounces         1.163043     3.5783742   FALSE FALSE
## PC.Volume           1.100000    17.6584986   FALSE FALSE
## Carb.Pressure       1.016393     4.1229094   FALSE FALSE
## Carb.Temp           1.016667     4.7841307   FALSE FALSE
## PSC                 1.211538     5.0175029   FALSE FALSE
## PSC.Fill            1.091892     1.2446519   FALSE FALSE
## PSC.CO2             1.078303     0.5056398   FALSE FALSE
## Mnf.Flow            1.048443    18.9420459   FALSE FALSE
## Carb.Pressure1      1.031746     5.4453520   FALSE FALSE
## Fill.Pressure       1.762931     4.2007001   FALSE FALSE
## Hyd.Pressure1      31.111111     9.5293660   FALSE  TRUE
## Hyd.Pressure2       7.271028     8.0513419   FALSE FALSE
## Hyd.Pressure3      11.450704     7.4679113   FALSE FALSE
## Hyd.Pressure4       1.008264     1.5558149   FALSE FALSE
```

```
## Filler.Level       1.086957    11.2018670   FALSE FALSE
## Filler.Speed       1.045161     9.4904706   FALSE FALSE
## Temperature        1.040000     2.1781408   FALSE FALSE
## Usage.cont         1.105263    18.7086737   FALSE FALSE
## Carb.Flow          1.586207    20.7312330   FALSE FALSE
## Density            1.108374     3.0338390   FALSE FALSE
## MFR                1.222222    22.8315830   FALSE FALSE
## Balling            1.193182     8.4402956   FALSE FALSE
## Pressure.Vacuum    1.389728     0.6223259   FALSE FALSE
## PH                 1.078125     2.0225593   FALSE FALSE
## Oxygen.Filler      1.266667    13.1466356   FALSE FALSE
## Bowl.Setpoint      2.990847     0.4278491   FALSE FALSE
## Pressure.Setpoint  1.319361     0.3111630   FALSE FALSE
## Air.Pressurer      1.093407     1.2446519   FALSE FALSE
## Alch.Rel           1.098315     2.0614547   FALSE FALSE
## Carb.Rel           1.011583     1.6336056   FALSE FALSE
## Balling.Lvl        1.294872     3.1894205   FALSE FALSE
```

Hyd.Pressure1 has a near-zero variance, so we could eliminate that one.

1.3.2 Looking for highly correlated variables

We'll create a correlation grid, so we can see actual correlation between variables. We'll also create a correlation plot and use findCorrelation from the caret package to find variables that have 90% or higher correlation.

```
correlation_grid <- training_data |>
  drop_na() |>
  select(-Brand.Code) |>
  cor()

correlation_grid <- as.data.frame(correlation_grid)

correlation_grid
```

```
##                    Carb.Volume  Fill.Ounces    PC.Volume Carb.Pressure
## Carb.Volume        1.000000000  0.022648036 -0.243234517  0.4278586099
## Fill.Ounces        0.022648036  1.000000000 -0.172701966  0.0044669914
## PC.Volume         -0.243234517 -0.172701966  1.000000000 -0.1300941804
## Carb.Pressure      0.427858610  0.004466991 -0.130094180  1.0000000000
## Carb.Temp         -0.135080260 -0.020073097  0.008599427  0.8243350333
## PSC               -0.020625187  0.043056041  0.190961266 -0.0520086907
## PSC.Fill          -0.012364141  0.069748653 -0.055914847 -0.0179275381
## PSC.CO2           -0.051977547  0.014705610 -0.055356933 -0.0174011136
## Mnf.Flow           0.100525364 -0.005787251 -0.203032882  0.0189865815
## Carb.Pressure1     0.081146212 -0.004186326 -0.238496181  0.0221844474
## Fill.Pressure     -0.096123314  0.068377098 -0.095744421 -0.0850458606
## Hyd.Pressure1     -0.051894268 -0.146359721  0.271171741 -0.0704043564
## Hyd.Pressure2      0.033567314 -0.143047986  0.028768135 -0.0267152675
## Hyd.Pressure3      0.057610071 -0.107816452 -0.024741466 -0.0082140513
## Hyd.Pressure4     -0.567908627  0.056086557  0.132413786 -0.3229128556
## Filler.Level      -0.018377226  0.010856355  0.220434988  0.0097134297
## Filler.Speed       0.003233452  0.044277046 -0.073766021  0.0320230919
```

```
## Temperature       -0.189760851 -0.011208233  0.142065435 -0.0490025241
## Usage.cont         0.073297402  0.085289046 -0.269204705  0.0004415267
## Carb.Flow         -0.088782919 -0.116991410  0.254535534 -0.0111938869
## Density            0.801808803 -0.080414770 -0.167310345  0.4591743002
## MFR                0.010144634  0.043445999 -0.066678296  0.0292986178
## Balling            0.818778791 -0.068439337 -0.204785622  0.4539307252
## Pressure.Vacuum   -0.079190180  0.052165821 -0.051605143  0.0051279147
## PH                 0.073251595 -0.118207504  0.097746859  0.0788617427
## Oxygen.Filler     -0.112682435 -0.032064258  0.212285069 -0.0506609705
## Bowl.Setpoint      0.001993607  0.011188522  0.210972310  0.0187581570
## Pressure.Setpoint -0.160640146  0.065078231 -0.022210124 -0.1171228371
## Air.Pressurer     -0.024854172  0.058101103 -0.036605483  0.0219237613
## Alch.Rel           0.824787575 -0.110432476 -0.177865067  0.4590380468
## Carb.Rel           0.841072250 -0.118666586 -0.167433257  0.4736224720
## Balling.Lvl        0.817993347 -0.063820071 -0.210081156  0.4589466196
##                      Carb.Temp          PSC     PSC.Fill      PSC.CO2
## Carb.Volume       -0.135080260 -0.020625187 -0.012364141 -0.051977547
## Fill.Ounces       -0.020073097  0.043056041  0.069748653  0.014705610
## PC.Volume          0.008599427  0.190961266 -0.055914847 -0.055356933
## Carb.Pressure      0.824335033 -0.052008691 -0.017927538 -0.017401114
## Carb.Temp          1.000000000 -0.044380293 -0.010968792  0.023660514
## PSC               -0.044380293  1.000000000  0.193242059  0.053922116
## PSC.Fill          -0.010968792  0.193242059  1.000000000  0.201967429
## PSC.CO2            0.023660514  0.053922116  0.201967429  1.000000000
## Mnf.Flow          -0.030110321  0.042982180 -0.033801564  0.049931986
## Carb.Pressure1    -0.014312694 -0.011071066 -0.029772433  0.038316144
## Fill.Pressure     -0.028433189  0.030908100 -0.010528126  0.078519840
## Hyd.Pressure1     -0.045431217 -0.019758828 -0.053220514 -0.049037869
## Hyd.Pressure2     -0.039543090 -0.033813839 -0.080467321 -0.021838338
## Hyd.Pressure3     -0.035120030 -0.020448188 -0.068958062  0.005757630
## Hyd.Pressure4     -0.002573081  0.036485173  0.019810466  0.054940092
## Filler.Level       0.011038984 -0.016061814  0.048227359 -0.042306839
## Filler.Speed       0.029355933 -0.002395360 -0.017295907 -0.008421406
## Temperature        0.061914927  0.014217909  0.030080929  0.037007308
## Usage.cont        -0.043584359  0.049860720 -0.025727952  0.032502663
## Carb.Flow          0.045050619 -0.037361416  0.017579095 -0.013058433
## Density            0.021593836 -0.067168725 -0.024708871 -0.045382139
## MFR                0.023442657  0.007659111 -0.008663604 -0.004263878
## Balling            0.006005105 -0.064450826 -0.019108545 -0.043036807
## Pressure.Vacuum    0.037327836  0.055246194  0.033093793 -0.018438819
## PH                 0.034594375 -0.069906559 -0.031374041 -0.088759922
## Oxygen.Filler      0.012010201  0.002765929 -0.005769535 -0.016971816
## Bowl.Setpoint      0.007550616 -0.020808347  0.049191096 -0.044820225
## Pressure.Setpoint -0.027029545  0.042979144 -0.005800041  0.082772777
## Air.Pressurer      0.038177076  0.061433424 -0.017995701 -0.006984004
## Alch.Rel           0.004966509 -0.060809035 -0.017045886 -0.050859393
## Carb.Rel           0.009020802 -0.065354969 -0.015365172 -0.066475871
## Balling.Lvl        0.010458674 -0.061576860 -0.016064731 -0.047296997
##                       Mnf.Flow Carb.Pressure1 Fill.Pressure Hyd.Pressure1
## Carb.Volume        0.100525364    0.081146212   -0.09612331  -0.051894268
## Fill.Ounces       -0.005787251   -0.004186326    0.06837710  -0.146359721
## PC.Volume         -0.203032882   -0.238496181   -0.09574442   0.271171741
## Carb.Pressure      0.018986582    0.022184447   -0.08504586  -0.070404356
## Carb.Temp         -0.030110321   -0.014312694   -0.02843319  -0.045431217
```

```
## PSC                0.042982180  -0.011071066   0.03090810  -0.019758828
## PSC.Fill          -0.033801564  -0.029772433  -0.01052813  -0.053220514
## PSC.CO2            0.049931986   0.038316144   0.07851984  -0.049037869
## Mnf.Flow           1.000000000   0.465355315   0.49111329   0.298997690
## Carb.Pressure1     0.465355315   1.000000000   0.22002705  -0.012711716
## Fill.Pressure      0.491113286   0.220027054   1.00000000   0.137594249
## Hyd.Pressure1      0.298997690  -0.012711716   0.13759425   1.000000000
## Hyd.Pressure2      0.648337926   0.284710255   0.31836940   0.711851345
## Hyd.Pressure3      0.764142833   0.375658047   0.43637028   0.605589019
## Hyd.Pressure4      0.073124200   0.038936494   0.29144885   0.051658157
## Filler.Level      -0.608380668  -0.425646963  -0.45892889  -0.029684260
## Filler.Speed       0.024363769   0.012389362  -0.21347499  -0.022850088
## Temperature       -0.063477350  -0.070067643   0.06397894  -0.073341745
## Usage.cont         0.560192617   0.354875344   0.30257079   0.118691986
## Carb.Flow         -0.497211539  -0.273632871  -0.18715299  -0.169136801
## Density            0.021343082   0.040888366  -0.20887500  -0.003601856
## MFR                0.019593355   0.013931715  -0.21290076  -0.035646540
## Balling            0.112393982   0.073508154  -0.15676227   0.025863408
## Pressure.Vacuum   -0.559943315  -0.284881843  -0.30810832  -0.324133005
## PH                -0.457218313  -0.112108310  -0.31311751  -0.048631533
## Oxygen.Filler     -0.550066399  -0.278795209  -0.24769592  -0.118317983
## Bowl.Setpoint     -0.611580420  -0.432671459  -0.42609290  -0.029844050
## Pressure.Setpoint  0.482530739   0.234201877   0.81528556   0.190079720
## Air.Pressurer     -0.041525175   0.061506659   0.03280945  -0.209227461
## Alch.Rel           0.027999719   0.009047055  -0.20467008   0.006667887
## Carb.Rel          -0.027880871   0.004804776  -0.20272201   0.034387945
## Balling.Lvl        0.042267589   0.033601804  -0.17885789  -0.007754220
##                   Hyd.Pressure2 Hyd.Pressure3 Hyd.Pressure4 Filler.Level
## Carb.Volume          0.03356731   0.057610071  -0.567908627  -0.01837723
## Fill.Ounces         -0.14304799  -0.107816452   0.056086557   0.01085635
## PC.Volume            0.02876814  -0.024741466   0.132413786   0.22043499
## Carb.Pressure       -0.02671527  -0.008214051  -0.322912856   0.00971343
## Carb.Temp           -0.03954309  -0.035120030  -0.002573081   0.01103898
## PSC                 -0.03381384  -0.020448188   0.036485173  -0.01606181
## PSC.Fill            -0.08046732  -0.068958062   0.019810466   0.04822736
## PSC.CO2             -0.02183834   0.005757630   0.054940092  -0.04230684
## Mnf.Flow             0.64833793   0.764142833   0.073124200  -0.60838067
## Carb.Pressure1       0.28471026   0.375658047   0.038936494  -0.42564696
## Fill.Pressure        0.31836940   0.436370279   0.291448848  -0.45892889
## Hyd.Pressure1        0.71185135   0.605589019   0.051658157  -0.02968426
## Hyd.Pressure2        1.00000000   0.917661875   0.022980025  -0.41103521
## Hyd.Pressure3        0.91766187   1.000000000   0.046649319  -0.52796590
## Hyd.Pressure4        0.02298003   0.046649319   1.000000000  -0.10010344
## Filler.Level        -0.41103521  -0.527965898  -0.100103442   1.00000000
## Filler.Speed         0.07028157   0.031692930  -0.259685780   0.06319357
## Temperature         -0.13725053  -0.120288087   0.283933984   0.05040323
## Usage.cont           0.37224053   0.419761856   0.070916924  -0.35740823
## Carb.Flow           -0.32187768  -0.363691033  -0.009527157   0.01743320
## Density              0.05842429   0.040354370  -0.584381995   0.00772950
## MFR                  0.04937645   0.012841039  -0.280014344   0.06935680
## Balling              0.10480326   0.125869294  -0.594996647   0.01188558
## Pressure.Vacuum     -0.62816754  -0.672258326  -0.054652623   0.35866815
## PH                  -0.21571713  -0.260535183  -0.184518087   0.34879540
## Oxygen.Filler       -0.29438355  -0.363660021   0.009692254   0.22540536
```

```
## Bowl.Setpoint       -0.41457787  -0.529215855  -0.117019292   0.97724166
## Pressure.Setpoint    0.34183180   0.459918732   0.288770453  -0.44595292
## Air.Pressurer       -0.17237591  -0.067700529   0.021731381  -0.12573419
## Alch.Rel             0.03916606   0.049948952  -0.685576013   0.04355856
## Carb.Rel             0.03150549   0.013543248  -0.581827500   0.13107856
## Balling.Lvl          0.02885940   0.041595480  -0.575044299   0.05317923
##                   Filler.Speed Temperature    Usage.cont    Carb.Flow
## Carb.Volume         0.003233452 -0.18976085  0.0732974017 -0.088782919
## Fill.Ounces         0.044277046 -0.01120823  0.0852890456 -0.116991410
## PC.Volume          -0.073766021  0.14206544 -0.2692047046  0.254535534
## Carb.Pressure       0.032023092 -0.04900252  0.0004415267 -0.011193887
## Carb.Temp           0.029355933  0.06191493 -0.0435843590  0.045050619
## PSC                -0.002395360  0.01421791  0.0498607203 -0.037361416
## PSC.Fill           -0.017295907  0.03008093 -0.0257279520  0.017579095
## PSC.CO2            -0.008421406  0.03700731  0.0325026634 -0.013058433
## Mnf.Flow            0.024363769 -0.06347735  0.5601926171 -0.497211539
## Carb.Pressure1      0.012389362 -0.07006764  0.3548753438 -0.273632871
## Fill.Pressure      -0.213474995  0.06397894  0.3025707885 -0.187152994
## Hyd.Pressure1      -0.022850088 -0.07334175  0.1186919862 -0.169136801
## Hyd.Pressure2       0.070281568 -0.13725053  0.3722405332 -0.321877677
## Hyd.Pressure3       0.031692930 -0.12028809  0.4197618556 -0.363691033
## Hyd.Pressure4      -0.259685780  0.28393398  0.0709169237 -0.009527157
## Filler.Level        0.063193573  0.05040323 -0.3574082260  0.017433196
## Filler.Speed        1.000000000 -0.06794451  0.0466749382 -0.062641899
## Temperature        -0.067944506  1.00000000 -0.0950428265  0.140119431
## Usage.cont          0.046674938 -0.09504283  1.0000000000 -0.488854072
## Carb.Flow          -0.062641899  0.14011943 -0.4888540721  1.000000000
## Density             0.025454377 -0.19369890 -0.0045531798  0.023058872
## MFR                 0.951264445 -0.08156825  0.0397443802 -0.074148067
## Balling             0.037938064 -0.23844781  0.0692694347 -0.102299824
## Pressure.Vacuum     0.045187804  0.04313148 -0.3196915871  0.290026894
## PH                 -0.047596639 -0.18152910 -0.3577042377  0.240757744
## Oxygen.Filler      -0.044126285  0.09338308 -0.3164959408  0.385692082
## Bowl.Setpoint       0.027688288  0.05010937 -0.3584563273  0.012221618
## Pressure.Setpoint -0.068525700  0.05904194  0.2626733751 -0.166362628
## Air.Pressurer      -0.006199674  0.06692883 -0.1025056258  0.137849013
## Alch.Rel           -0.005598672 -0.24776232 -0.0237408078  0.007527013
## Carb.Rel            0.005300760 -0.13595194 -0.0390798757 -0.006940815
## Balling.Lvl         0.003823258 -0.22497504  0.0232069975 -0.055365859
##                        Density          MFR      Balling Pressure.Vacuum
## Carb.Volume         0.801808803  0.010144634  0.818778791    -0.079190180
## Fill.Ounces        -0.080414770  0.043445999 -0.068439337     0.052165821
## PC.Volume          -0.167310345 -0.066678296 -0.204785622    -0.051605143
## Carb.Pressure       0.459174300  0.029298618  0.453930725     0.005127915
## Carb.Temp           0.021593836  0.023442657  0.006005105     0.037327836
## PSC                -0.067168725  0.007659111 -0.064450826     0.055246194
## PSC.Fill           -0.024708871 -0.008663604 -0.019108545     0.033093793
## PSC.CO2            -0.045382139 -0.004263878 -0.043036807    -0.018438819
## Mnf.Flow            0.021343082  0.019593355  0.112393982    -0.559943315
## Carb.Pressure1      0.040888366  0.013931715  0.073508154    -0.284881843
## Fill.Pressure      -0.208875002 -0.212900763 -0.156762274    -0.308108315
## Hyd.Pressure1      -0.003601856 -0.035646540  0.025863408    -0.324133005
## Hyd.Pressure2       0.058424287  0.049376448  0.104803262    -0.628167539
## Hyd.Pressure3       0.040354370  0.012841039  0.125869294    -0.672258326
```

```
## Hyd.Pressure4     -0.584381995 -0.280014344 -0.594996647    -0.054652623
## Filler.Level       0.007729500  0.069356797  0.011885578     0.358668150
## Filler.Speed       0.025454377  0.951264445  0.037938064     0.045187804
## Temperature       -0.193698895 -0.081568252 -0.238447810     0.043131483
## Usage.cont        -0.004553180  0.039744380  0.069269435    -0.319691587
## Carb.Flow          0.023058872 -0.074148067 -0.102299824     0.290026894
## Density            1.000000000  0.039383665  0.953127632    -0.083762802
## MFR                0.039383665  1.000000000  0.051171065     0.043987724
## Balling            0.953127632  0.051171065  1.000000000    -0.167052546
## Pressure.Vacuum   -0.083762802  0.043987724 -0.167052546     1.000000000
## PH                 0.095340535 -0.052377217  0.072297357     0.213173584
## Oxygen.Filler     -0.046621720 -0.048876320 -0.113757056     0.278478283
## Bowl.Setpoint      0.019792536  0.042312091  0.027415802     0.358696671
## Pressure.Setpoint -0.250884909 -0.064390559 -0.210098742    -0.298520301
## Air.Pressurer     -0.082976446 -0.008182360 -0.102307854     0.175090738
## Alch.Rel           0.922993173 -0.001003949  0.945725964    -0.054788103
## Carb.Rel           0.860322233  0.010884909  0.858990647    -0.008434181
## Balling.Lvl        0.956417297  0.016039036  0.988412082    -0.051626926
##                             PH Oxygen.Filler Bowl.Setpoint Pressure.Setpoint
## Carb.Volume        0.073251595  -0.112682435   0.001993607      -0.160640146
## Fill.Ounces       -0.118207504  -0.032064258   0.011188522       0.065078231
## PC.Volume          0.097746859   0.212285069   0.210972310      -0.022210124
## Carb.Pressure      0.078861743  -0.050660970   0.018758157      -0.117122837
## Carb.Temp          0.034594375   0.012010201   0.007550616      -0.027029545
## PSC               -0.069906559   0.002765929  -0.020808347       0.042979144
## PSC.Fill          -0.031374041  -0.005769535   0.049191096      -0.005800041
## PSC.CO2           -0.088759922  -0.016971816  -0.044820225       0.082772777
## Mnf.Flow          -0.457218313  -0.550066399  -0.611580420       0.482530739
## Carb.Pressure1    -0.112108310  -0.278795209  -0.432671459       0.234201877
## Fill.Pressure     -0.313117513  -0.247695921  -0.426092896       0.815285564
## Hyd.Pressure1     -0.048631533  -0.118317983  -0.029844050       0.190079720
## Hyd.Pressure2     -0.215717133  -0.294383550  -0.414577866       0.341831797
## Hyd.Pressure3     -0.260535183  -0.363660021  -0.529215855       0.459918732
## Hyd.Pressure4     -0.184518087   0.009692254  -0.117019292       0.288770453
## Filler.Level       0.348795399   0.225405359   0.977241656      -0.445952917
## Filler.Speed      -0.047596639  -0.044126285   0.027688288      -0.068525700
## Temperature       -0.181529097   0.093383081   0.050109366       0.059041937
## Usage.cont        -0.357704238  -0.316495941  -0.358456327       0.262673375
## Carb.Flow          0.240757744   0.385692082   0.012221618      -0.166362628
## Density            0.095340535  -0.046621720   0.019792536      -0.250884909
## MFR               -0.052377217  -0.048876320   0.042312091      -0.064390559
## Balling            0.072297357  -0.113757056   0.027415802      -0.210098742
## Pressure.Vacuum    0.213173584   0.278478283   0.358696671      -0.298520301
## PH                 1.000000000   0.162016479   0.358504933      -0.316950863
## Oxygen.Filler      0.162016479   1.000000000   0.224734144      -0.241221158
## Bowl.Setpoint      0.358504933   0.224734144   1.000000000      -0.440555036
## Pressure.Setpoint -0.316950863  -0.241221158  -0.440555036       1.000000000
## Air.Pressurer     -0.008276544   0.116495267  -0.130939728       0.073805550
## Alch.Rel           0.156053375  -0.050111016   0.063332637      -0.268800293
## Carb.Rel           0.189575611  -0.039206163   0.152117160      -0.261257572
## Balling.Lvl        0.103279163  -0.074544984   0.070442468      -0.242343282
##                   Air.Pressurer     Alch.Rel     Carb.Rel  Balling.Lvl
## Carb.Volume        -0.024854172  0.824787575  0.841072250  0.817993347
## Fill.Ounces         0.058101103 -0.110432476 -0.118666586 -0.063820071
```

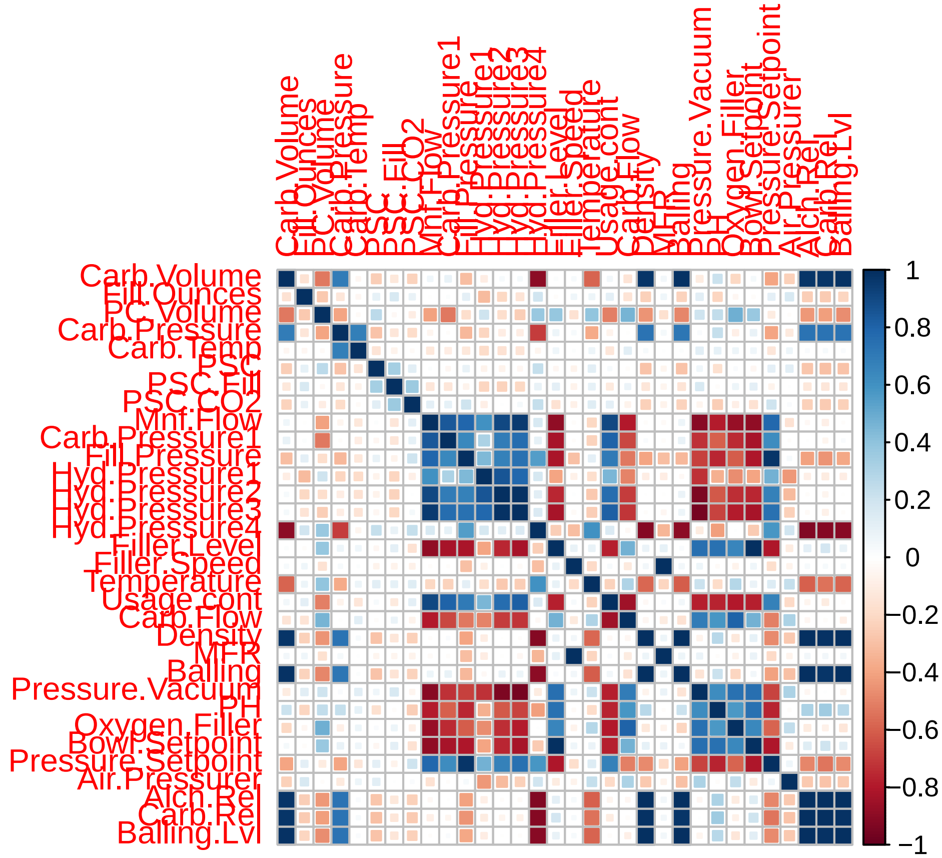
```
## PC.Volume          -0.036605483 -0.177865067 -0.167433257 -0.210081156
## Carb.Pressure       0.021923761  0.459038047  0.473622472  0.458946620
## Carb.Temp           0.038177076  0.004966509  0.009020802  0.010458674
## PSC                 0.061433424 -0.060809035 -0.065354969 -0.061576860
## PSC.Fill           -0.017995701 -0.017045886 -0.015365172 -0.016064731
## PSC.CO2            -0.006984004 -0.050859393 -0.066475871 -0.047296997
## Mnf.Flow           -0.041525175  0.027999719 -0.027880871  0.042267589
## Carb.Pressure1      0.061506659  0.009047055  0.004804776  0.033601804
## Fill.Pressure       0.032809452 -0.204670077 -0.202722006 -0.178857893
## Hyd.Pressure1      -0.209227461  0.006667887  0.034387945 -0.007754220
## Hyd.Pressure2      -0.172375910  0.039166055  0.031505491  0.028859401
## Hyd.Pressure3      -0.067700529  0.049948952  0.013543248  0.041595480
## Hyd.Pressure4       0.021731381 -0.685576013 -0.581827500 -0.575044299
## Filler.Level       -0.125734192  0.043558564  0.131078561  0.053179225
## Filler.Speed       -0.006199674 -0.005598672  0.005300760  0.003823258
## Temperature         0.066928829 -0.247762315 -0.135951942 -0.224975043
## Usage.cont         -0.102505626 -0.023740808 -0.039079876  0.023206997
## Carb.Flow           0.137849013  0.007527013 -0.006940815 -0.055365859
## Density            -0.082976446  0.922993173  0.860322233  0.956417297
## MFR                -0.008182360 -0.001003949  0.010884909  0.016039036
## Balling            -0.102307854  0.945725964  0.858990647  0.988412082
## Pressure.Vacuum     0.175090738 -0.054788103 -0.008434181 -0.051626926
## PH                 -0.008276544  0.156053375  0.189575611  0.103279163
## Oxygen.Filler       0.116495267 -0.050111016 -0.039206163 -0.074544984
## Bowl.Setpoint      -0.130939728  0.063332637  0.152117160  0.070442468
## Pressure.Setpoint   0.073805550 -0.268800293 -0.261257572 -0.242343282
## Air.Pressurer       1.000000000 -0.082451451 -0.107386102 -0.085434755
## Alch.Rel           -0.082451451  1.000000000  0.882791450  0.947343341
## Carb.Rel           -0.107386102  0.882791450  1.000000000  0.872148303
## Balling.Lvl        -0.085434755  0.947343341  0.872148303  1.000000000
```

There are many highly correlated variables. A quick scan shows:

- hyd.pressure 2 and 3 have a .91 correlation
- filler.level and bowl.set.point have a .98 correlation
- fillers.speed and MFR have a .95 correlation
- density and balling.lvl have a .95 correlation
- balling and balling.lvl have a .98 correlation; both are strongly correlated with density and alch rel.
- alch.rel and balling.lvl have a .95 correlation; it also has a .95 correlation with balling, balling.lvl, and .93 with density

Here's a correlation plot so we can see all the positively and negatively correlated variables:

```r
library(corrplot)

grid <- training_data |>
  drop_na() |>
  select(-Brand.Code) |>
  cor()

corrplot(cor(grid),method = "square")
```



There are many negatively correlated variables here that do not need to be removed, but may be helpful for analysis. We can see that the response variable, pH, has a strong negative correlation with some variables, including fill.pressure1, usage.cont, and Air.Pressurer.

The findCorrelation feature will decide which variables to drop (it leaves the variable that best represents the group):

```
library(caret)

#if we remove colinear variables at 95% correlation, we would drop 10 variables
indexestodrop95 <- findCorrelation(cor(grid), cutoff = 0.95)
colnames(grid)[indexestodrop95]
```

```
##  [1] "Pressure.Setpoint" "Mnf.Flow"          "Hyd.Pressure3"
##  [4] "Bowl.Setpoint"     "Carb.Rel"          "Balling"
##  [7] "Alch.Rel"          "Balling.Lvl"       "Density"
## [10] "Filler.Speed"
```

```
#this will leave us with 22 predictor variables and the response variable.

#if we remove colinear variables at 90% correlation, we would drop 12 variables
indexestodrop90 <- findCorrelation(cor(grid), cutoff = 0.90)
colnames(grid)[indexestodrop90]
```

```
##  [1] "Pressure.Setpoint" "Mnf.Flow"          "Hyd.Pressure3"
```

```
##  [4] "Pressure.Vacuum"   "Hyd.Pressure4"     "Bowl.Setpoint"
##  [7] "Carb.Rel"          "Balling"           "Alch.Rel"
## [10] "Balling.Lvl"       "Density"           "Filler.Speed"
```

```
#this will leave us with 20 predictor variables and the response variable.
```

1.3.3 Removing variables

Removing some of the 33 variables will help simplify this data set. We will remove the variable with NZV and highly colinear variables (greater than or about .90, as defined above).

the findcorrelation function says to drop Filler.Speed instead of MFR. However, we will opt to drop MFR instead, given that it has the highest number of missing values (filler.speed has 57 as opposed to MFR's 212). Having to impute all these values will makes the variable less reliable, and dropping it would mean having to drop more cases.

```
#removing the variable with NZV
training_data_new <- training_data |>
  select(-Hyd.Pressure1)

#removing correlated variables
training_data_new <- training_data_new |>
  select(-Pressure.Setpoint, -MFR, -Hyd.Pressure3,
         -Pressure.Vacuum, -Hyd.Pressure4, -Bowl.Setpoint, -Carb.Rel,
         -Balling, -Alch.Rel, - Balling.Lvl, - Density)

#donig the same for the test set
test_data_new <- test_data |>
  select(-Hyd.Pressure1, -Pressure.Setpoint, -MFR, -Hyd.Pressure3,
         -Pressure.Vacuum, -Hyd.Pressure4, -Bowl.Setpoint, -Carb.Rel,
         -Balling, -Alch.Rel, - Balling.Lvl, - Density)
```

1.3.4 Checking for outliers

Before we deal with missing values, let's see if there are any glaring outliers or data entry errors.

```
#may be an outlier in oxygen filler - the max calues is much higher than the

summary(test_data_new)
```

```
##   Brand.Code          Carb.Volume     Fill.Ounces      PC.Volume
##  Length:267         Min.   :5.147   Min.   :23.75   Min.   :0.09867
##  Class :character   1st Qu.:5.287   1st Qu.:23.92   1st Qu.:0.23333
##  Mode  :character   Median :5.340   Median :23.97   Median :0.27533
##                     Mean   :5.369   Mean   :23.97   Mean   :0.27769
##                     3rd Qu.:5.465   3rd Qu.:24.01   3rd Qu.:0.32200
##                     Max.   :5.667   Max.   :24.20   Max.   :0.46400
##                     NA's   :1       NA's   :6       NA's   :4
##  Carb.Pressure     Carb.Temp          PSC              PSC.Fill
##  Min.   :60.20   Min.   :130.0   Min.   :0.00400   Min.   :0.0200
##  1st Qu.:65.30   1st Qu.:138.4   1st Qu.:0.04450   1st Qu.:0.1000
##  Median :68.00   Median :140.8   Median :0.07600   Median :0.1800
```

```
##  Mean   :68.25   Mean   :141.2   Mean   :0.08545   Mean   :0.1903  
##  3rd Qu.:70.60   3rd Qu.:143.8   3rd Qu.:0.11200   3rd Qu.:0.2600  
##  Max.   :77.60   Max.   :154.0   Max.   :0.24600   Max.   :0.6200  
##                  NA's   :1       NA's   :5         NA's   :3       
##     PSC.CO2           Mnf.Flow       Carb.Pressure1  Fill.Pressure  
##  Min.   :0.00000   Min.   :-100.20   Min.   :113.0   Min.   :37.80  
##  1st Qu.:0.02000   1st Qu.:-100.00   1st Qu.:120.2   1st Qu.:46.00  
##  Median :0.04000   Median :   0.20   Median :123.4   Median :47.80  
##  Mean   :0.05107   Mean   :  21.03   Mean   :123.0   Mean   :48.14  
##  3rd Qu.:0.06000   3rd Qu.: 141.30   3rd Qu.:125.5   3rd Qu.:50.20  
##  Max.   :0.24000   Max.   : 220.40   Max.   :136.0   Max.   :60.20  
##  NA's   :5                           NA's   :4       NA's   :2      
##  Hyd.Pressure2     Filler.Level    Filler.Speed   Temperature   
##  Min.   :-50.00   Min.   : 69.2   Min.   :1006   Min.   :63.80  
##  1st Qu.:  0.00   1st Qu.:100.6   1st Qu.:3812   1st Qu.:65.40  
##  Median : 26.80   Median :118.6   Median :3978   Median :65.80  
##  Mean   : 20.11   Mean   :110.3   Mean   :3581   Mean   :66.23  
##  3rd Qu.: 34.80   3rd Qu.:120.2   3rd Qu.:3996   3rd Qu.:66.60  
##  Max.   : 61.40   Max.   :153.2   Max.   :4020   Max.   :75.40  
##  NA's   :1        NA's   :2       NA's   :10     NA's   :2      
##    Usage.cont      Carb.Flow       PH          Oxygen.Filler    
##  Min.   :12.90   Min.   :   0   Mode:logical   Min.   :0.00240  
##  1st Qu.:18.12   1st Qu.:1083   NA's:267       1st Qu.:0.01960  
##  Median :21.44   Median :3038                  Median :0.03370  
##  Mean   :20.90   Mean   :2409                  Mean   :0.04666  
##  3rd Qu.:23.74   3rd Qu.:3215                  3rd Qu.:0.05440  
##  Max.   :24.60   Max.   :3858                  Max.   :0.39800  
##  NA's   :2                                     NA's   :3        
##  Air.Pressurer  
##  Min.   :141.2  
##  1st Qu.:142.2  
##  Median :142.6  
##  Mean   :142.8  
##  3rd Qu.:142.8  
##  Max.   :147.2  
##  NA's   :1      
```

```r
#there's also at least one really large value on oxygen filler here
summary(training_data_new)
```

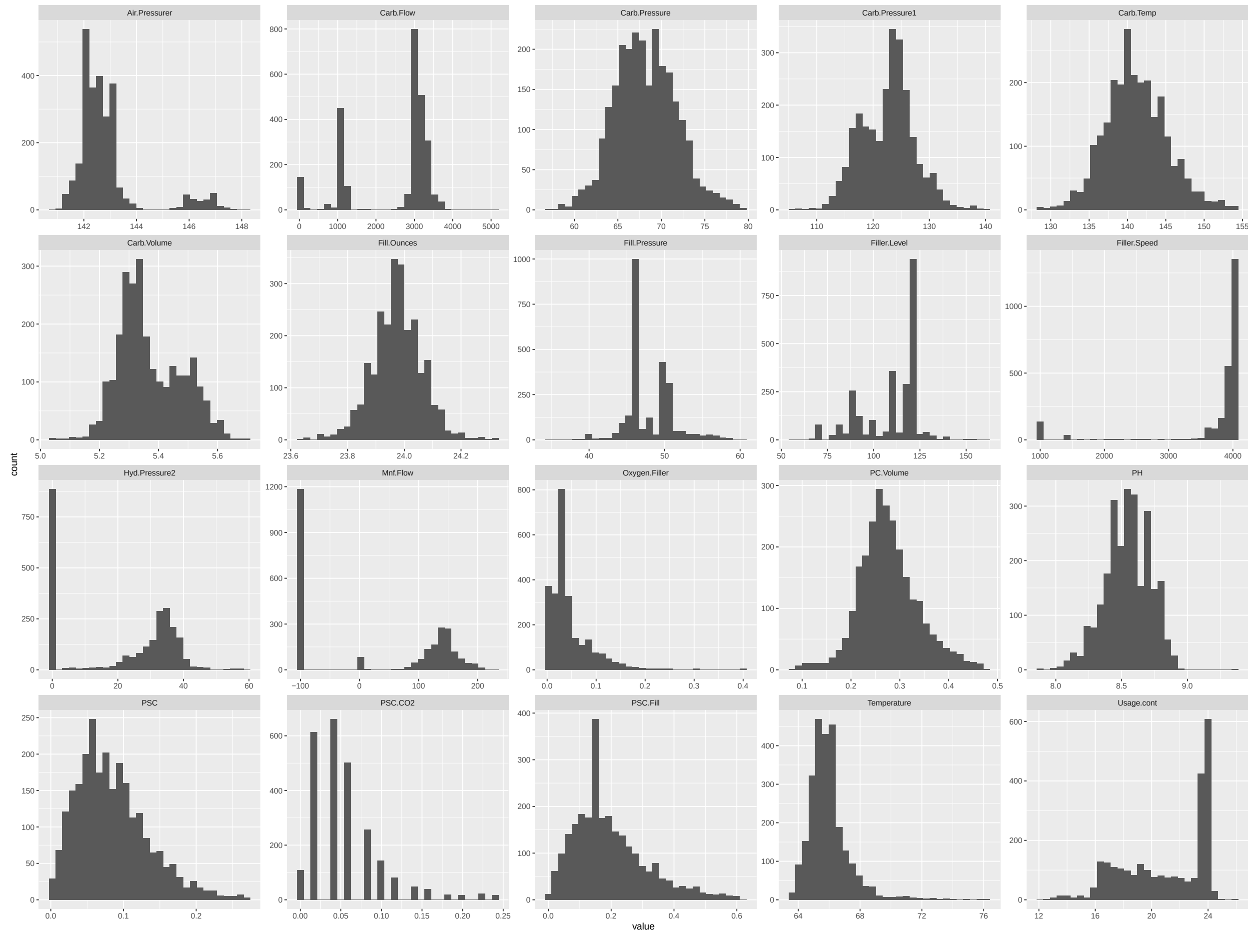
```
##   Brand.Code         Carb.Volume     Fill.Ounces      PC.Volume      
##  Length:2571        Min.   :5.040   Min.   :23.63   Min.   :0.07933  
##  Class :character   1st Qu.:5.293   1st Qu.:23.92   1st Qu.:0.23917  
##  Mode  :character   Median :5.347   Median :23.97   Median :0.27133  
##                     Mean   :5.370   Mean   :23.97   Mean   :0.27712  
##                     3rd Qu.:5.453   3rd Qu.:24.03   3rd Qu.:0.31200  
##                     Max.   :5.700   Max.   :24.32   Max.   :0.47800  
##                     NA's   :10      NA's   :38      NA's   :39       
##  Carb.Pressure     Carb.Temp          PSC             PSC.Fill     
##  Min.   :57.00   Min.   :128.6   Min.   :0.00200   Min.   :0.0000  
##  1st Qu.:65.60   1st Qu.:138.4   1st Qu.:0.04800   1st Qu.:0.1000  
##  Median :68.20   Median :140.8   Median :0.07600   Median :0.1800  
##  Mean   :68.19   Mean   :141.1   Mean   :0.08457   Mean   :0.1954  
##  3rd Qu.:70.60   3rd Qu.:143.8   3rd Qu.:0.11200   3rd Qu.:0.2600  
```

```
## Max.    :79.40   Max.   :154.0   Max.   :0.27000   Max.   :0.6200
## NA's    :27      NA's   :26      NA's   :33        NA's   :23
##    PSC.CO2          Mnf.Flow       Carb.Pressure1  Fill.Pressure
## Min.    :0.00000   Min.   :-100.20   Min.   :105.6   Min.   :34.60
## 1st Qu.:0.02000   1st Qu.:-100.00   1st Qu.:119.0   1st Qu.:46.00
## Median :0.04000   Median :  65.20   Median :123.2   Median :46.40
## Mean   :0.05641   Mean   :  24.57   Mean   :122.6   Mean   :47.92
## 3rd Qu.:0.08000   3rd Qu.: 140.80   3rd Qu.:125.4   3rd Qu.:50.00
## Max.   :0.24000   Max.   : 229.40   Max.   :140.2   Max.   :60.40
## NA's   :39        NA's   :2         NA's   :32      NA's   :22
## Hyd.Pressure2    Filler.Level    Filler.Speed   Temperature      Usage.cont
## Min.   : 0.00   Min.   : 55.8   Min.   : 998   Min.   :63.60   Min.   :12.08
## 1st Qu.: 0.00   1st Qu.: 98.3   1st Qu.:3888   1st Qu.:65.20   1st Qu.:18.36
## Median :28.60   Median :118.4   Median :3982   Median :65.60   Median :21.79
## Mean   :20.96   Mean   :109.3   Mean   :3687   Mean   :65.97   Mean   :20.99
## 3rd Qu.:34.60   3rd Qu.:120.0   3rd Qu.:3998   3rd Qu.:66.40   3rd Qu.:23.75
## Max.   :59.40   Max.   :161.2   Max.   :4030   Max.   :76.20   Max.   :25.90
## NA's   :15      NA's   :20      NA's   :57     NA's   :14      NA's   :5
##   Carb.Flow          PH        Oxygen.Filler     Air.Pressurer
## Min.   :  26   Min.   :7.880   Min.   :0.00240   Min.   :140.8
## 1st Qu.:1144   1st Qu.:8.440   1st Qu.:0.02200   1st Qu.:142.2
## Median :3028   Median :8.540   Median :0.03340   Median :142.6
## Mean   :2468   Mean   :8.546   Mean   :0.04684   Mean   :142.8
## 3rd Qu.:3186   3rd Qu.:8.680   3rd Qu.:0.06000   3rd Qu.:143.0
## Max.   :5104   Max.   :9.360   Max.   :0.40000   Max.   :148.2
## NA's   :2      NA's   :4       NA's   :12
```

After quickly browsing the remaining variables, while some of the fields have a wide range of values, none of the values look like errors (there are clusters of similar low/high values for each variable). While they may count as outliers (we'll calculate later) they are likely legitimate data ant not input errors.

Let's check histograms to see if anything stands out:

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



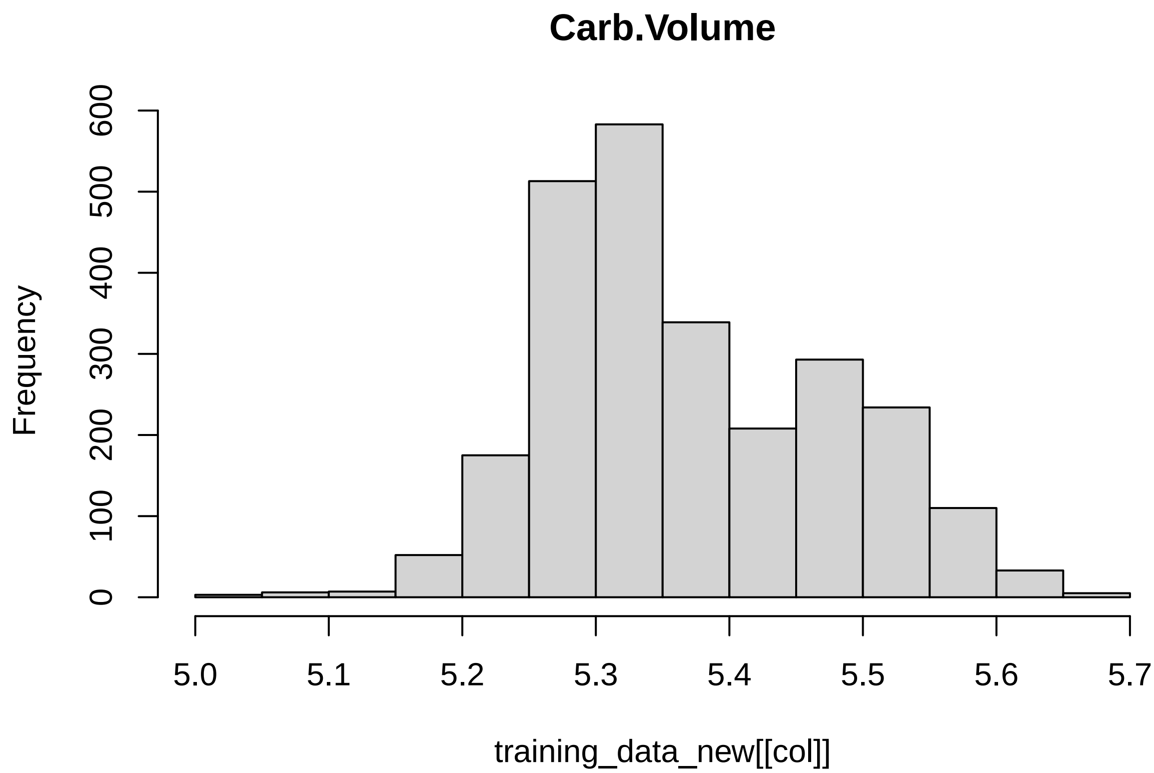

```
##   Brand.Code    n
## 1          A  293
## 2          B 1239
## 3          C  304
## 4          D  615
## 5       <NA>  120
```

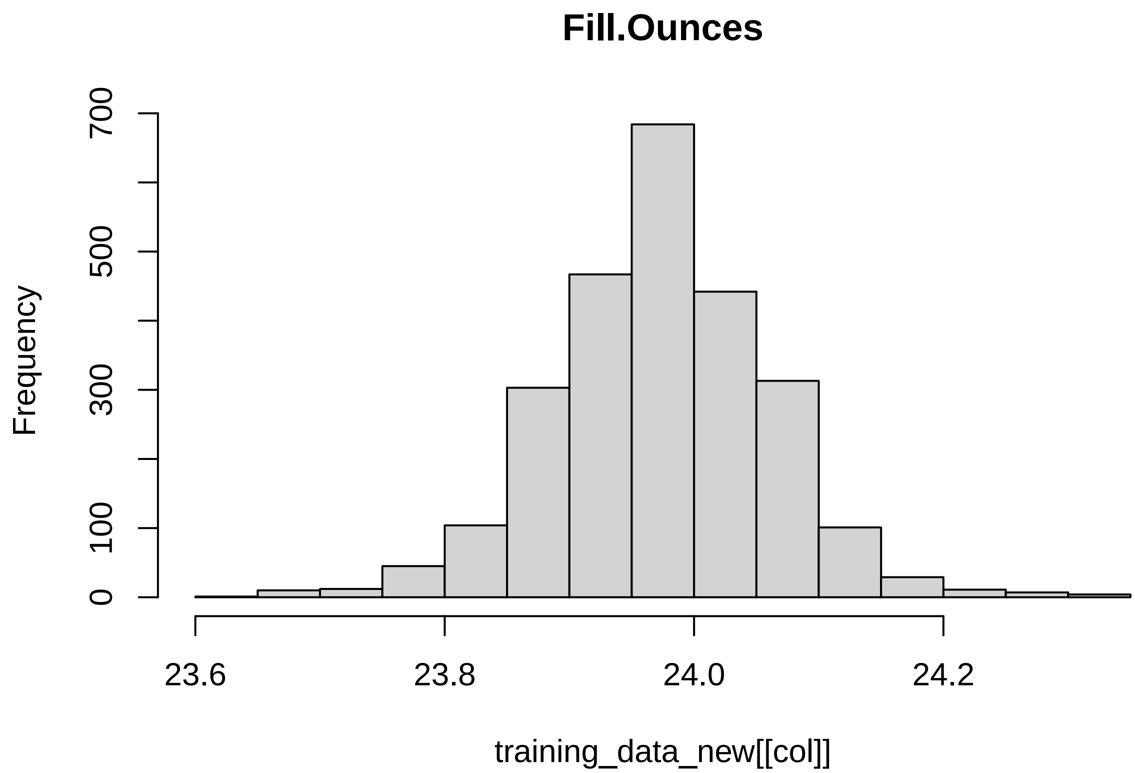
Not all the data is normally distributed. pH, the response variable is normally distributed. Other variables close to a normal distribution include carb pressure, carb pressure 1 (slightly bimodal), carb temp, fill ounces, carb volume (also slightly bimodal), psc (skewed slightly right), pc volume, temperature (skewed slightly right), and psc fill (skewed right with a large peak).

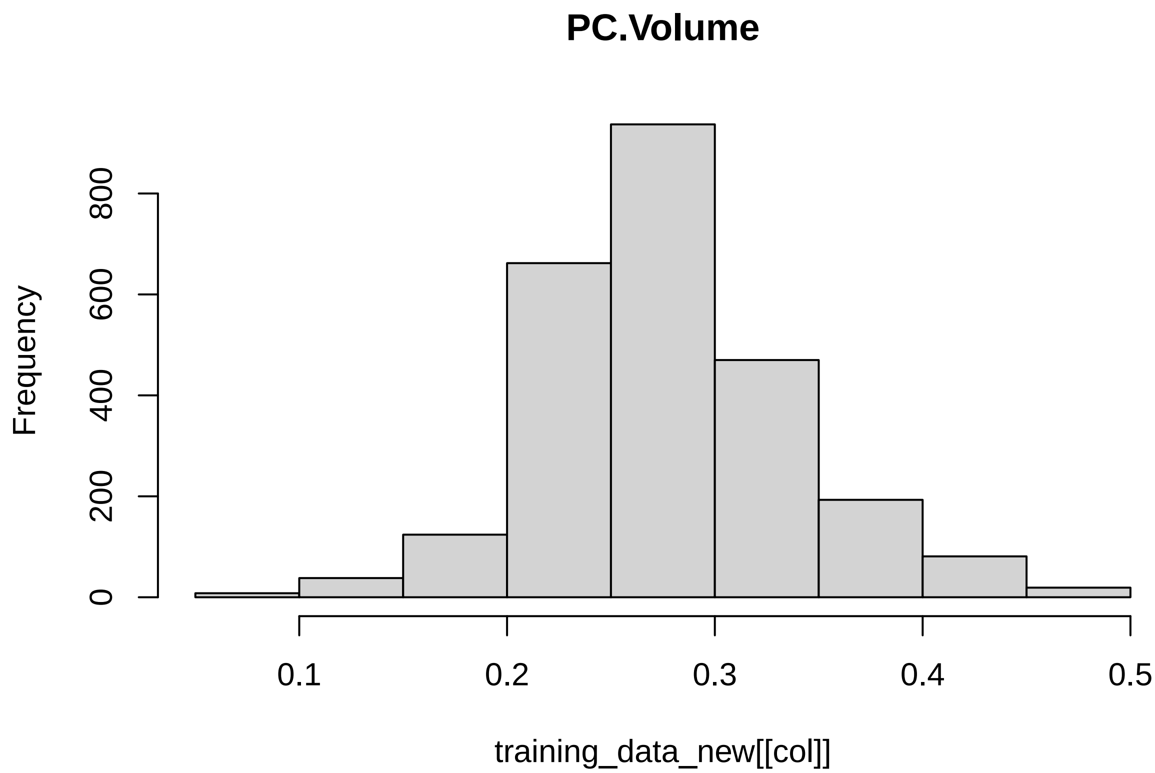
A few are more clearly bimodal/have very different peaks on either side of the graph: air pressurer, carb flow, fill pressure, filler level, filler speed, , hyd pressure 2, mnf flow, and usage cont.

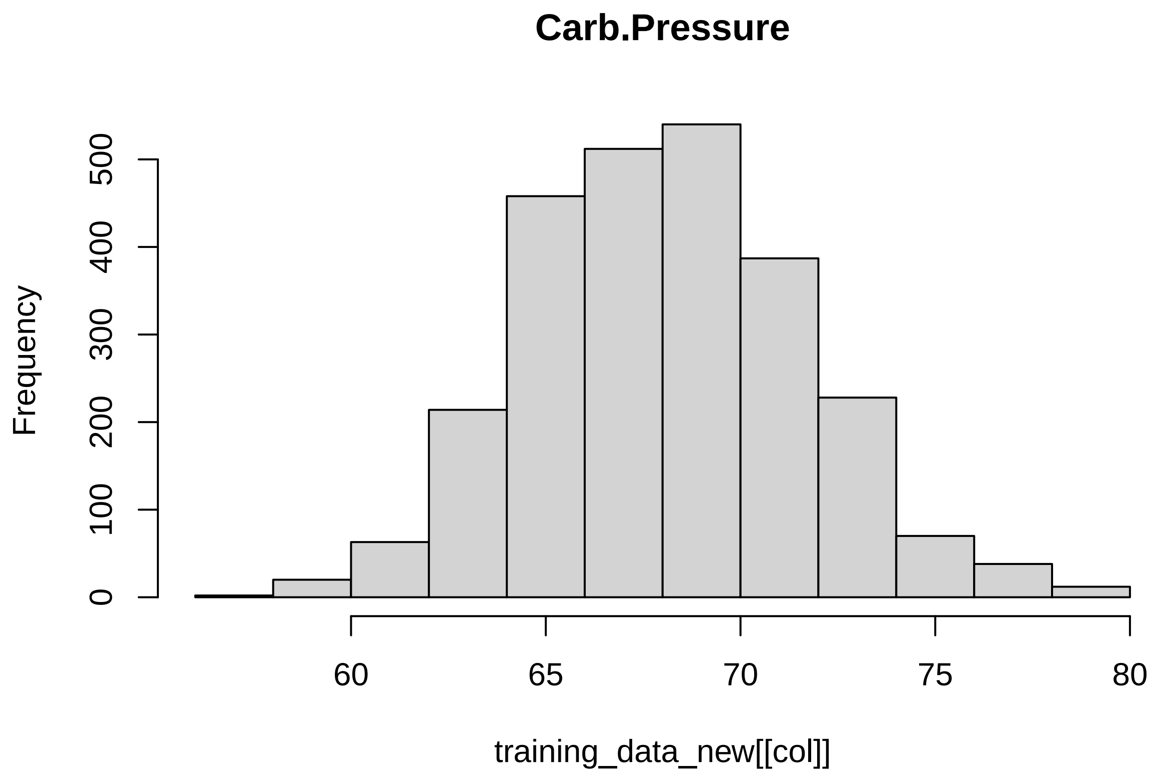
Finally, some are skewed right: in addition to the varaibles mentioned above, oxygen filler looks like it may have some outliers, and psc co2 has a strange distribution, where the variable clusters around certain values, almost like a discrete variable.

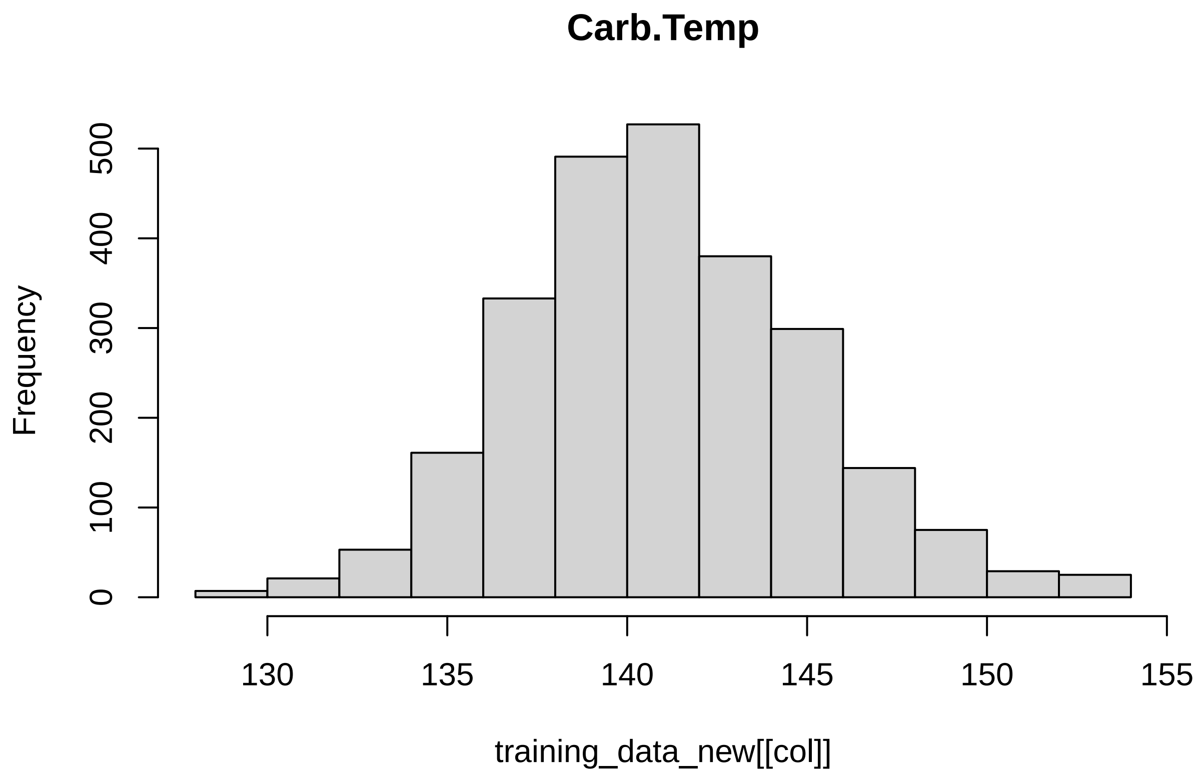
```r
#histograms

for (col in names(training_data_new %>% select(-Brand.Code))) {
  hist(training_data_new[[col]], main=col)
}
```


Carb.Volume
Frequency
0
100
200
300
400
500
600
5.0
5.1
5.2
5.3
5.4
5.5
5.6
5.7
training_data_new[[col]]

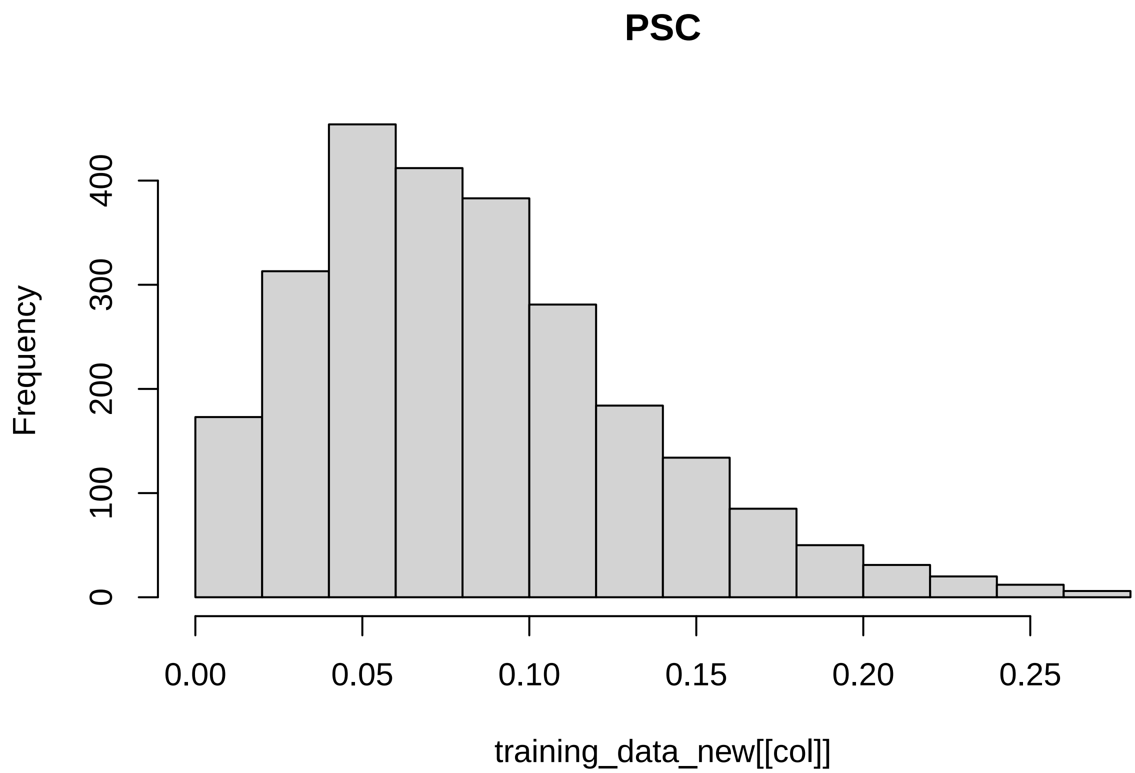

Fill.Ounces
Frequency
0
100
300
500
700
23.6
23.8
24.0
24.2
training_data_new[[col]]

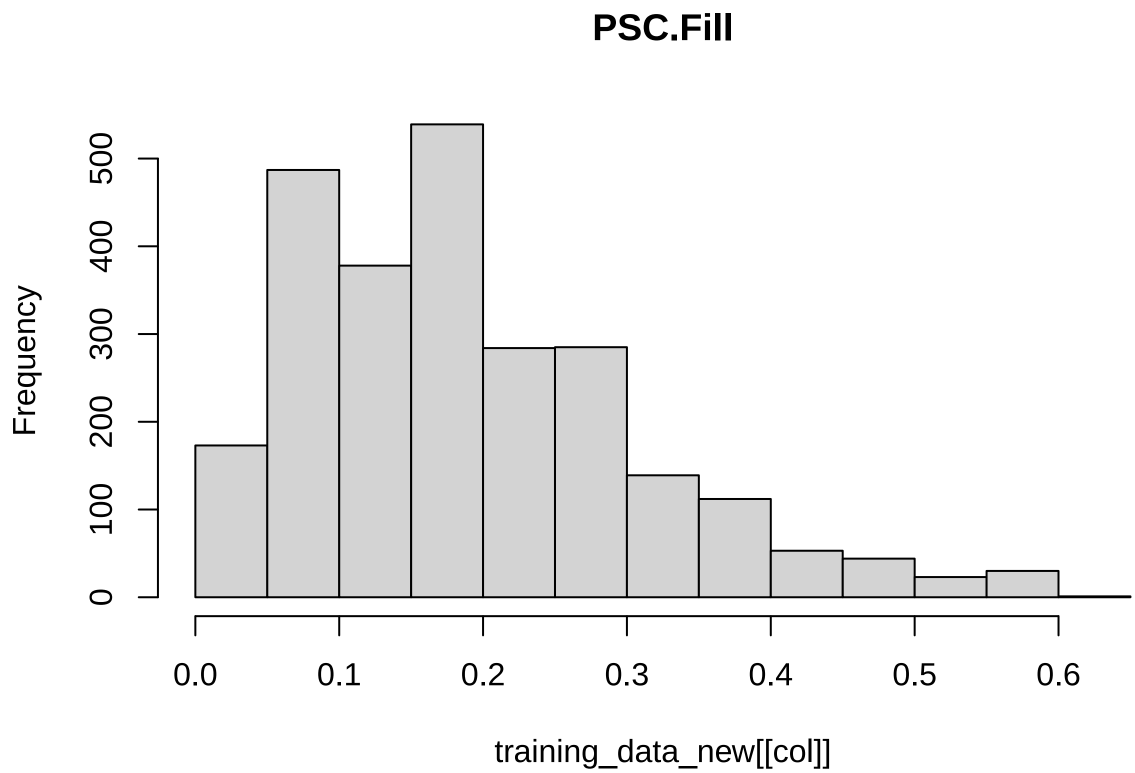

PC.Volume
Frequency
0
200
400
600
800
0.1
0.2
0.3
0.4
0.5
training_data_new[[col]]

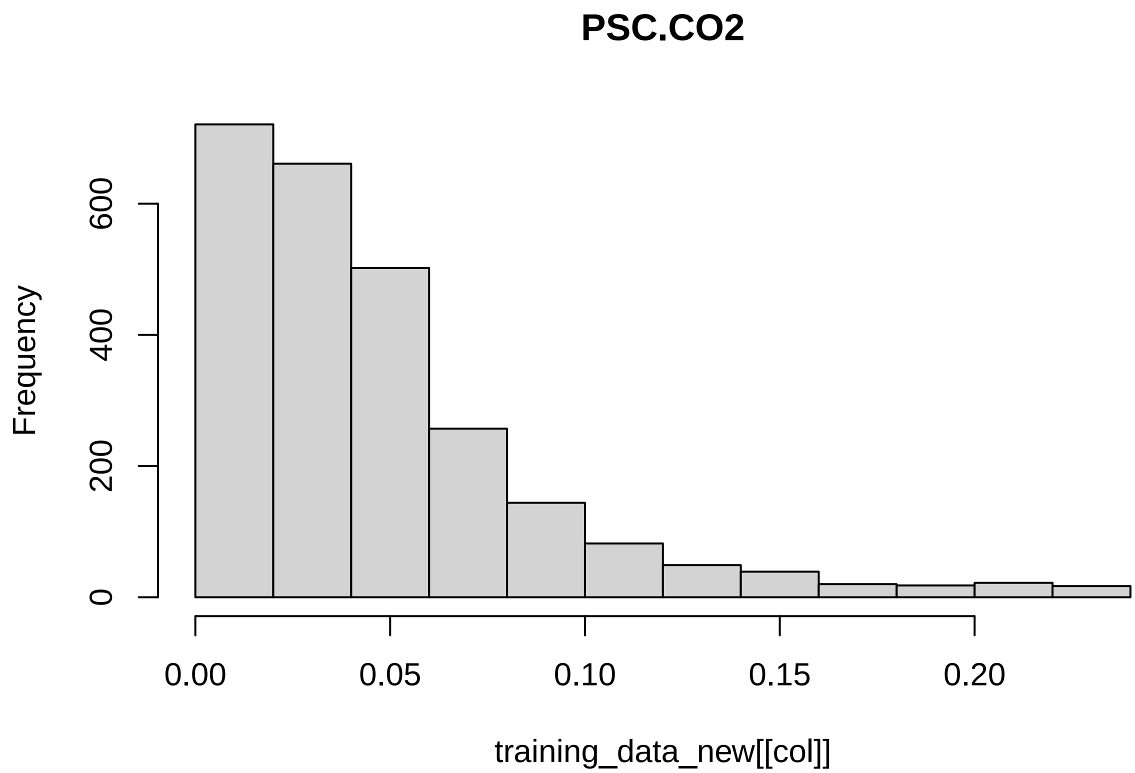

Carb.Pressure
Frequency
0
100
200
300
400
500
60
65
70
75
80
training_data_new[[col]]

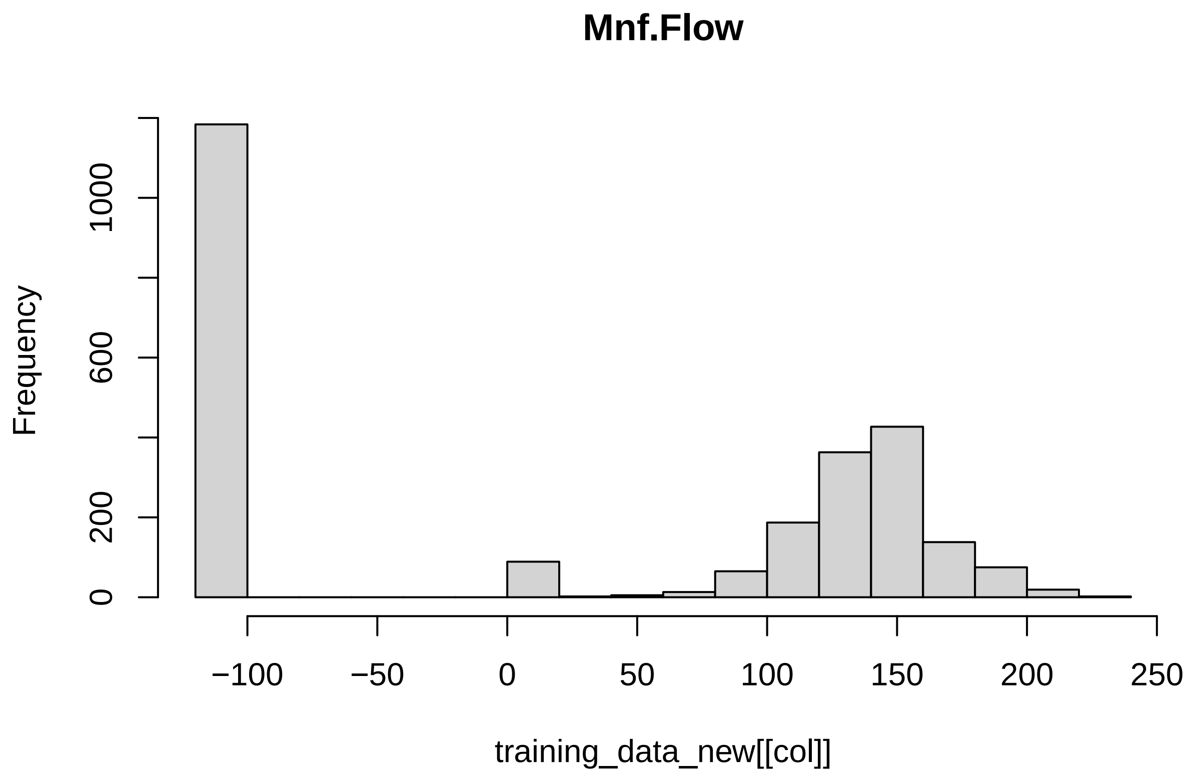


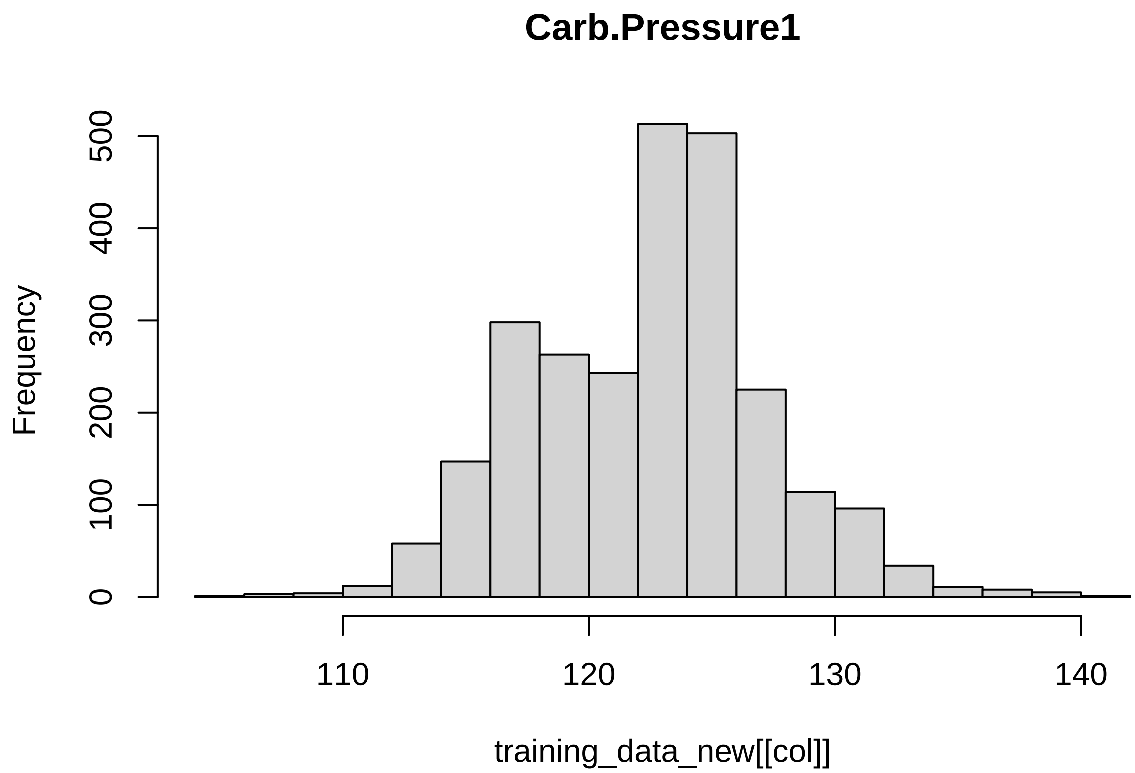
Carb.Temp
Frequency
0
100
200
300
400
500
130
135
140
145
150
155
training_data_new[[col]]

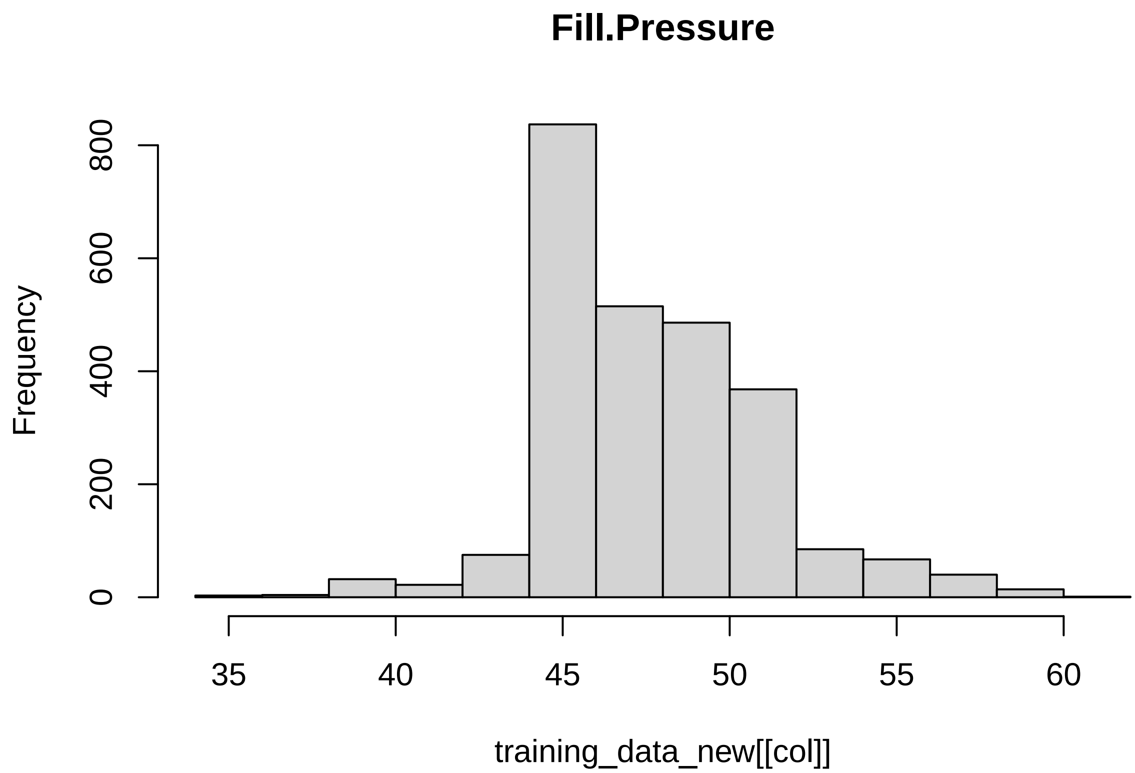

PSC
Frequency
0
100
200
300
400
0.00
0.05
0.10
0.15
0.20
0.25
training_data_new[[col]]


PSC.Fill
Frequency
0
100
200
300
400
500
0.0
0.1
0.2
0.3
0.4
0.5
0.6
training_data_new[[col]]

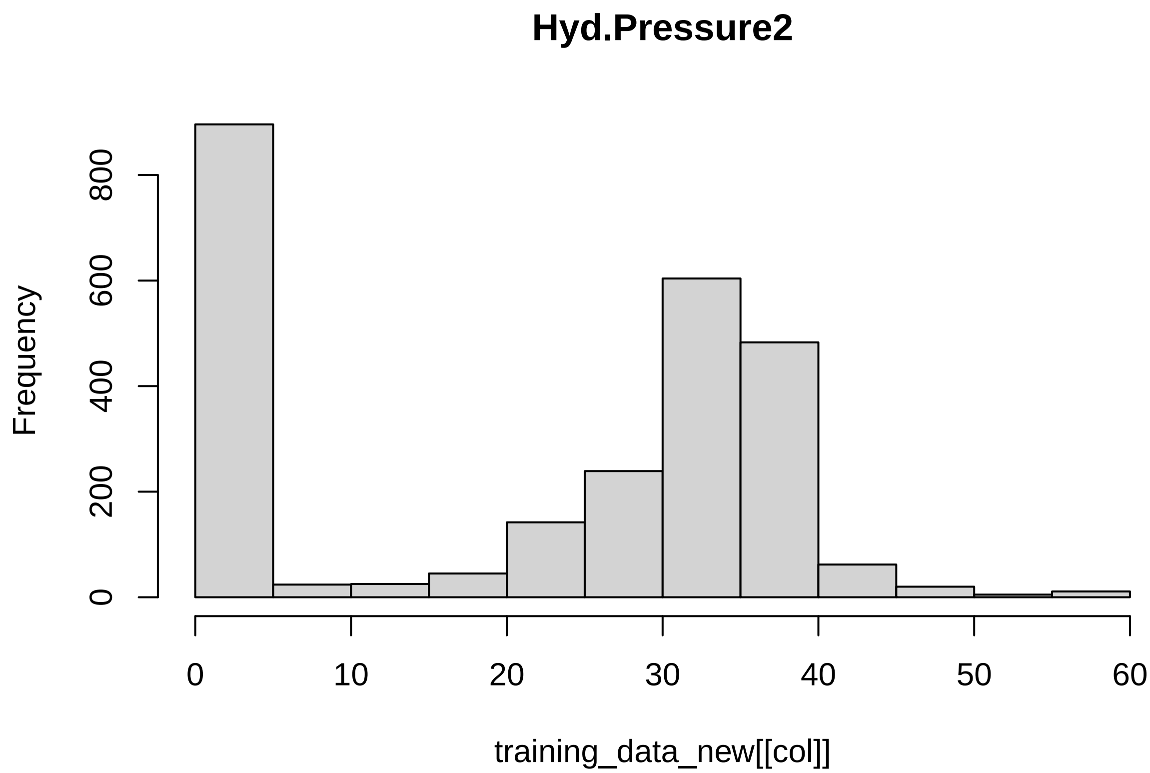

PSC.CO2
Frequency
0
200
400
600
0.00
0.05
0.10
0.15
0.20
training_data_new[[col]]

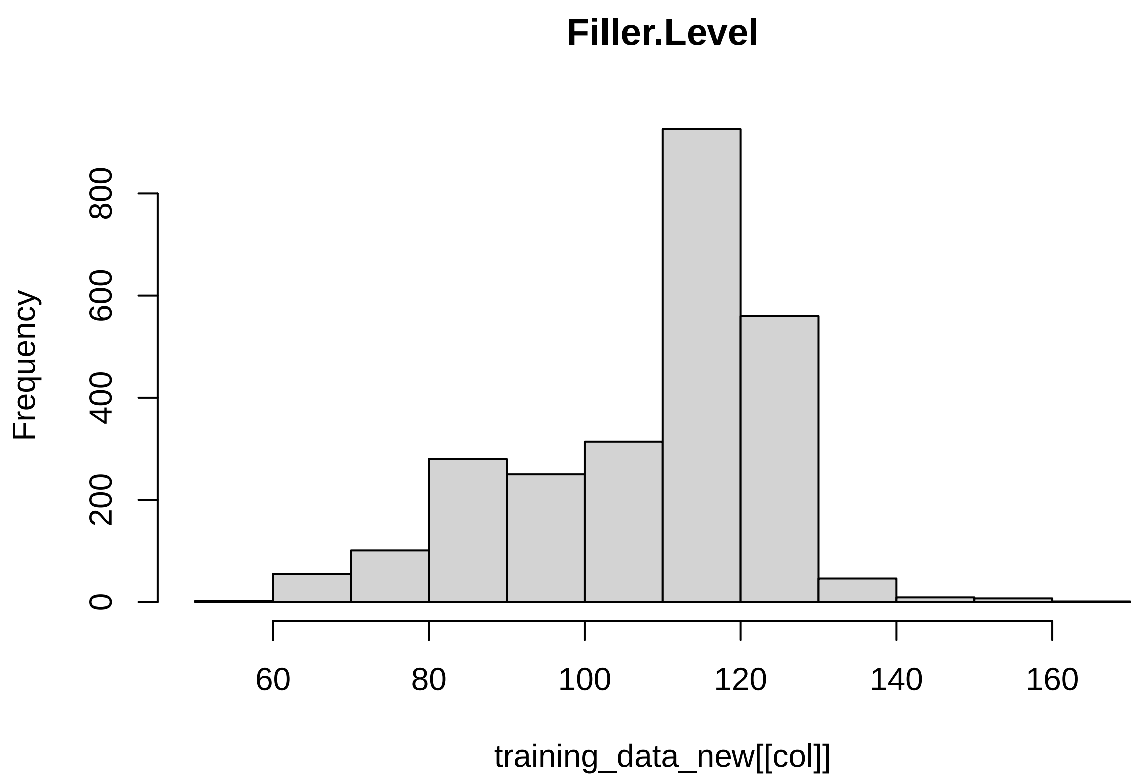

Mnf.Flow
Frequency
0
200
600
1000
−100
−50
0
50
100
150
200
250
training_data_new[[col]]

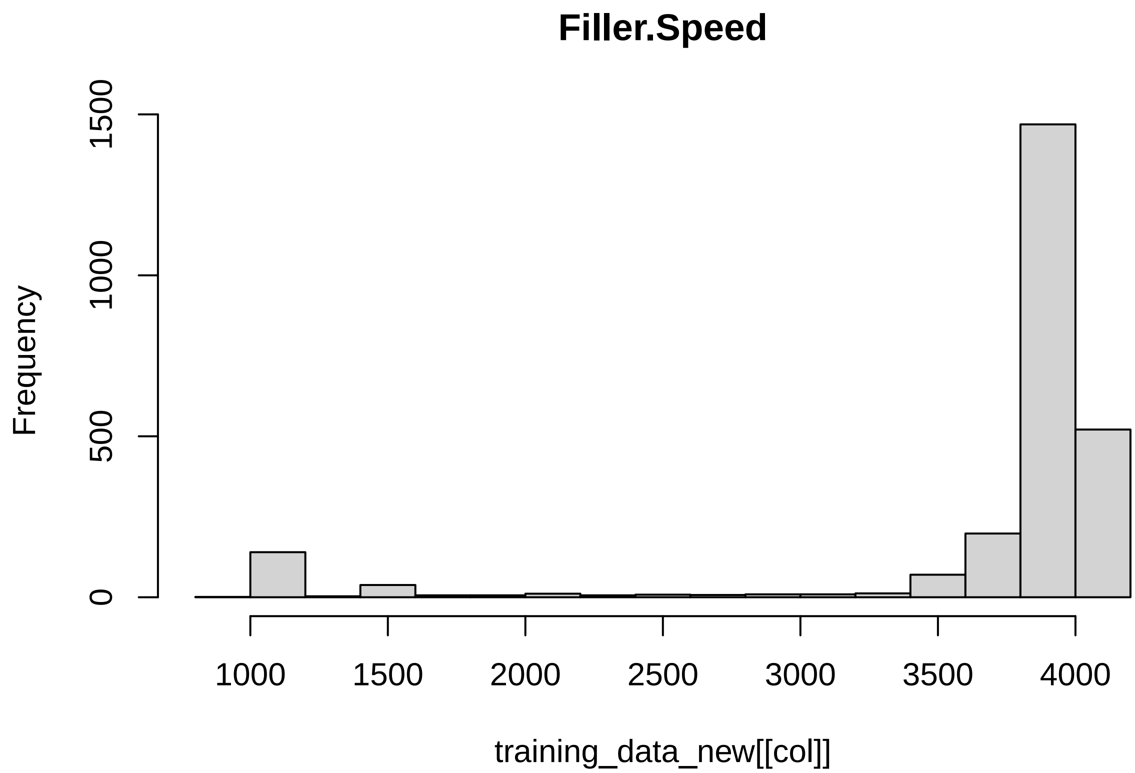

Carb.Pressure1
Frequency
0
100
200
300
400
500
110
120
130
140
training_data_new[[col]]

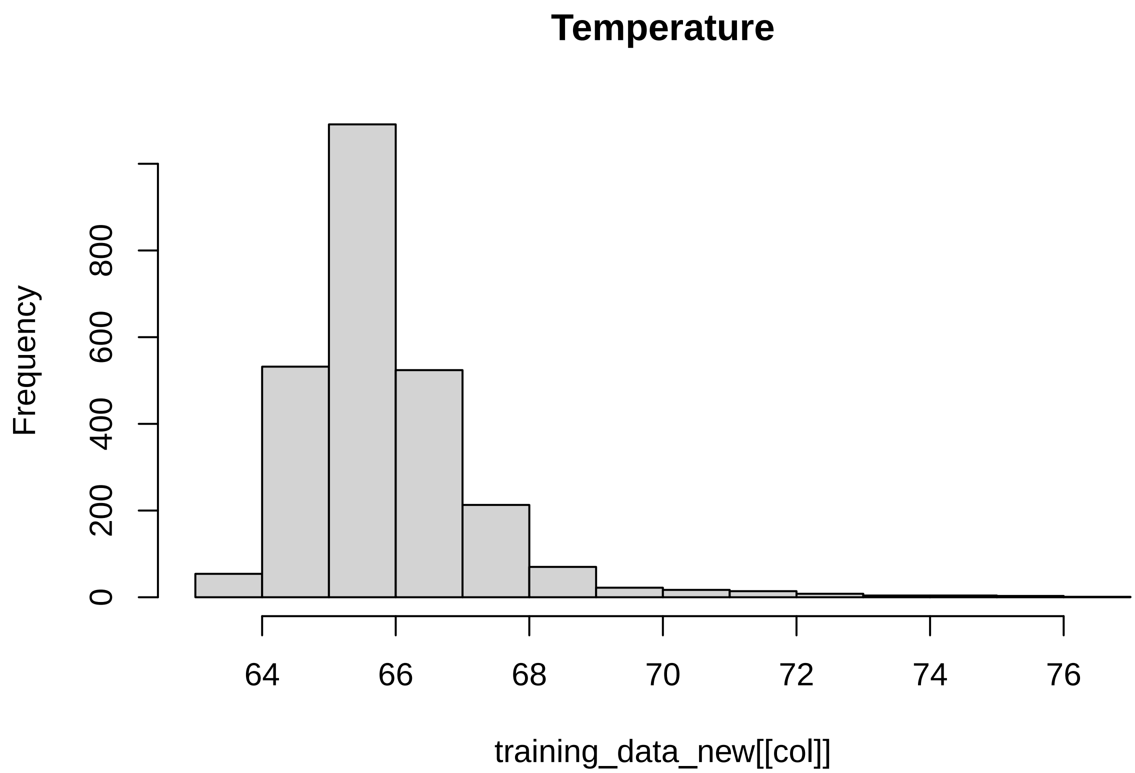


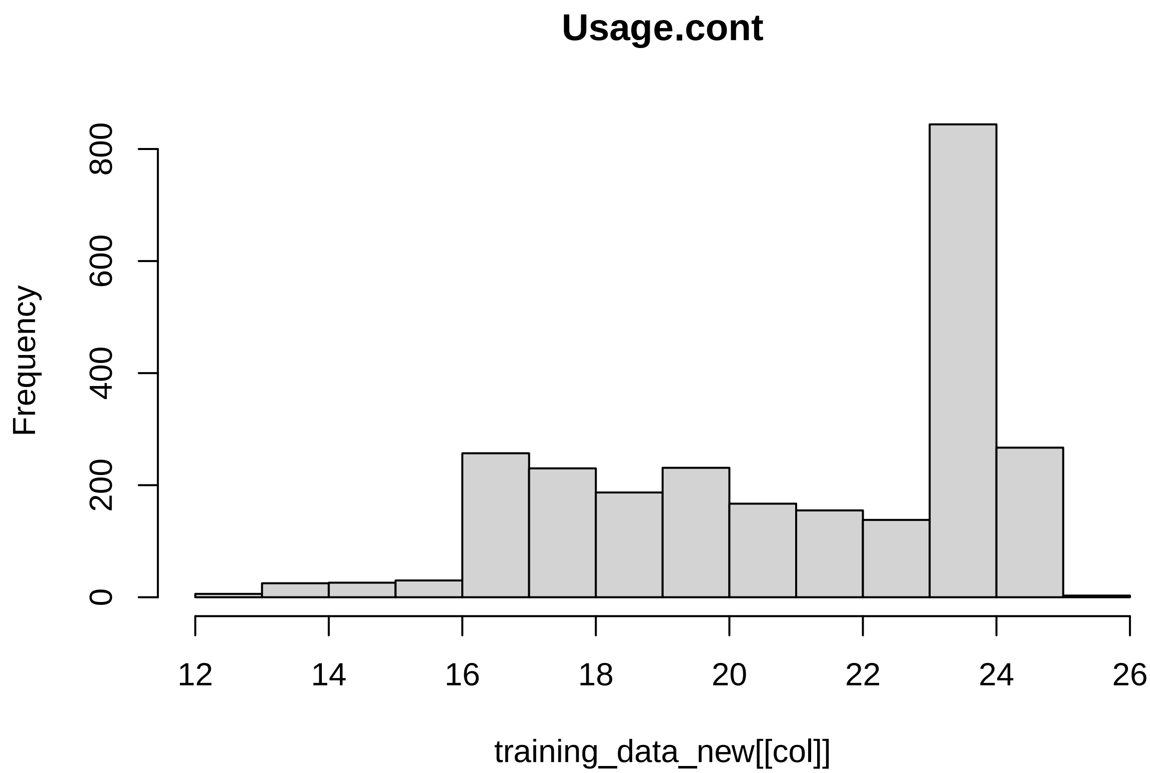
Fill.Pressure
Frequency
0
200
400
600
800
35
40
45
50
55
60
training_data_new[[col]]

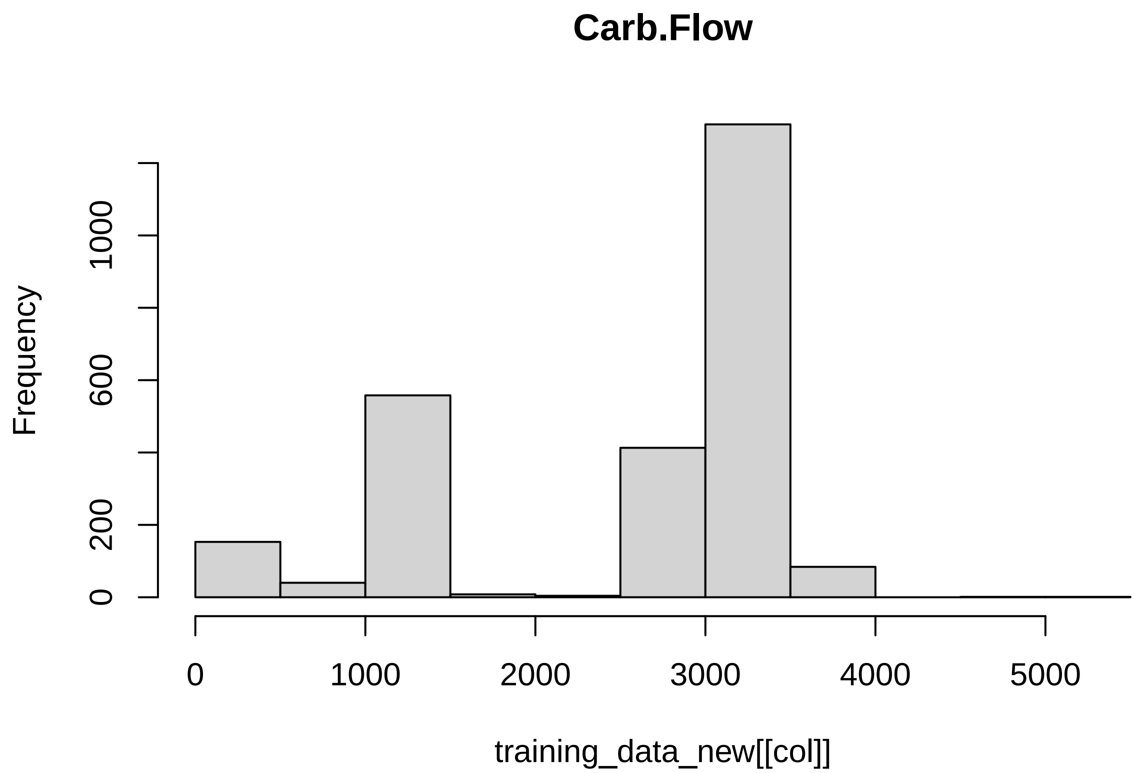

Hyd.Pressure2
Frequency
0
200
400
600
800
0
10
20
30
40
50
60
training_data_new[[col]]


Filler.Level
Frequency
0
200
400
600
800
60
80
100
120
140
160
training_data_new[[col]]

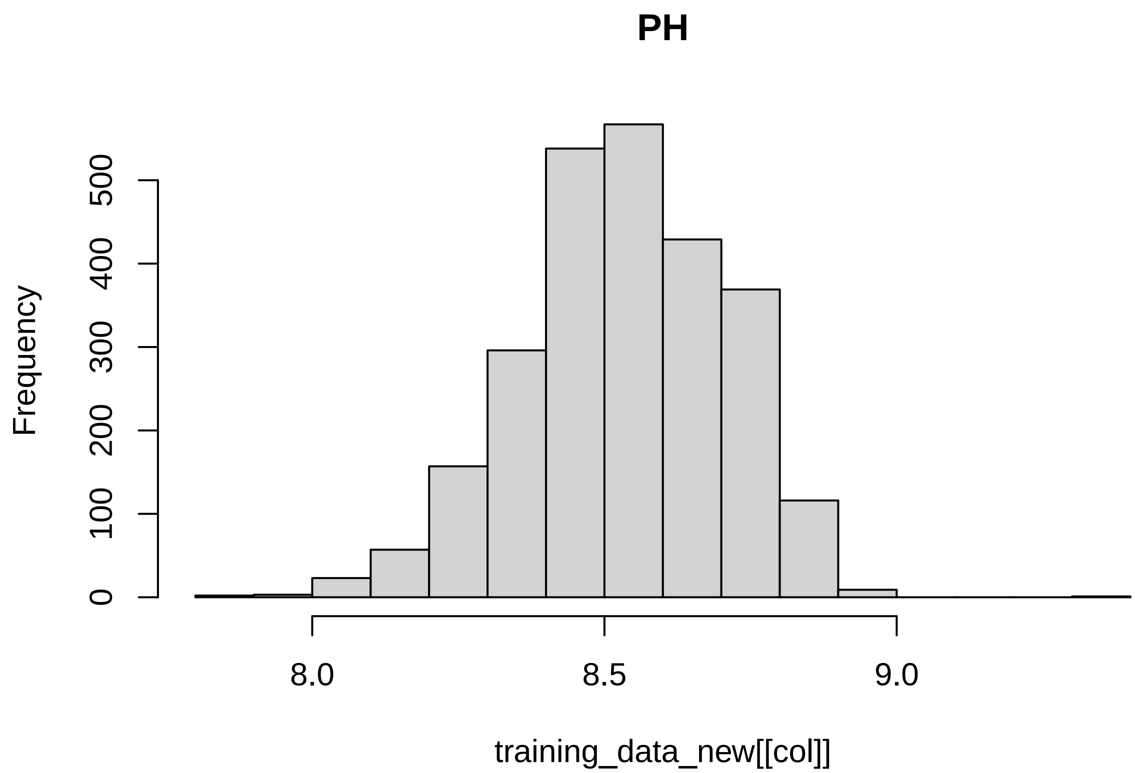

Filler.Speed
Frequency
0
500
1000
1500
1000
1500
2000
2500
3000
3500
4000
training_data_new[[col]]

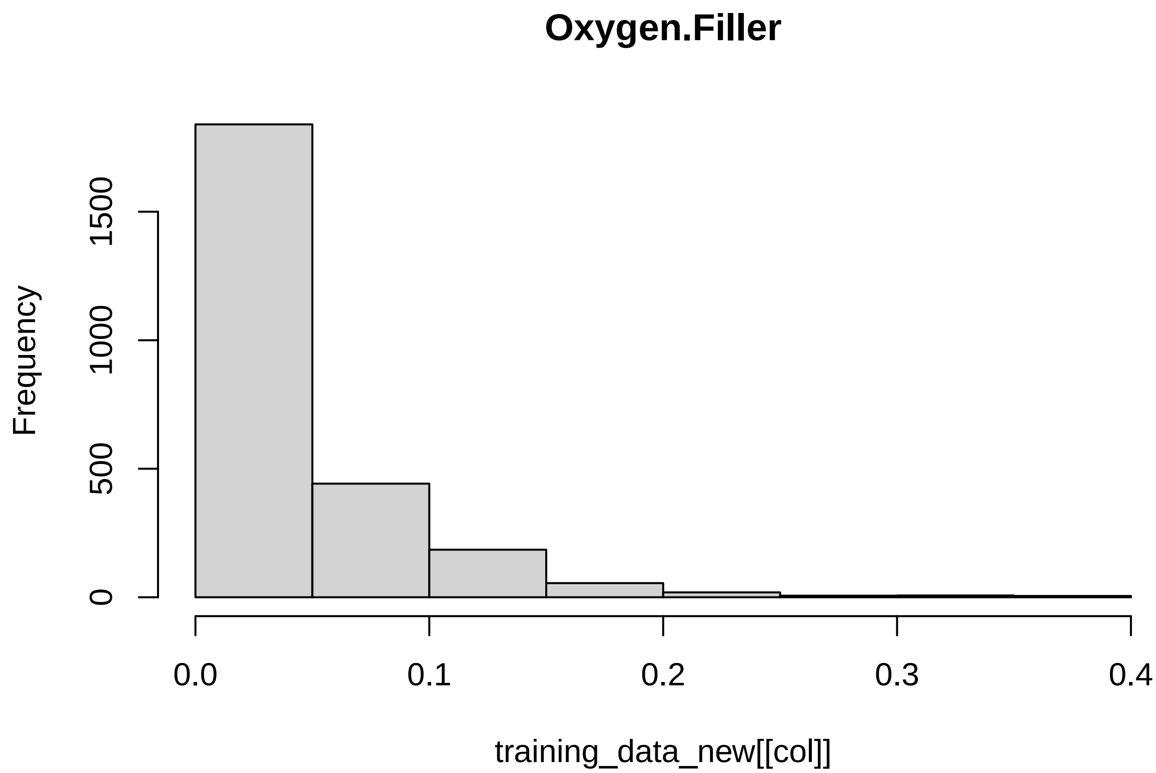

Temperature
Frequency
0
200
400
600
800
64
66
68
70
72
74
76
training_data_new[[col]]

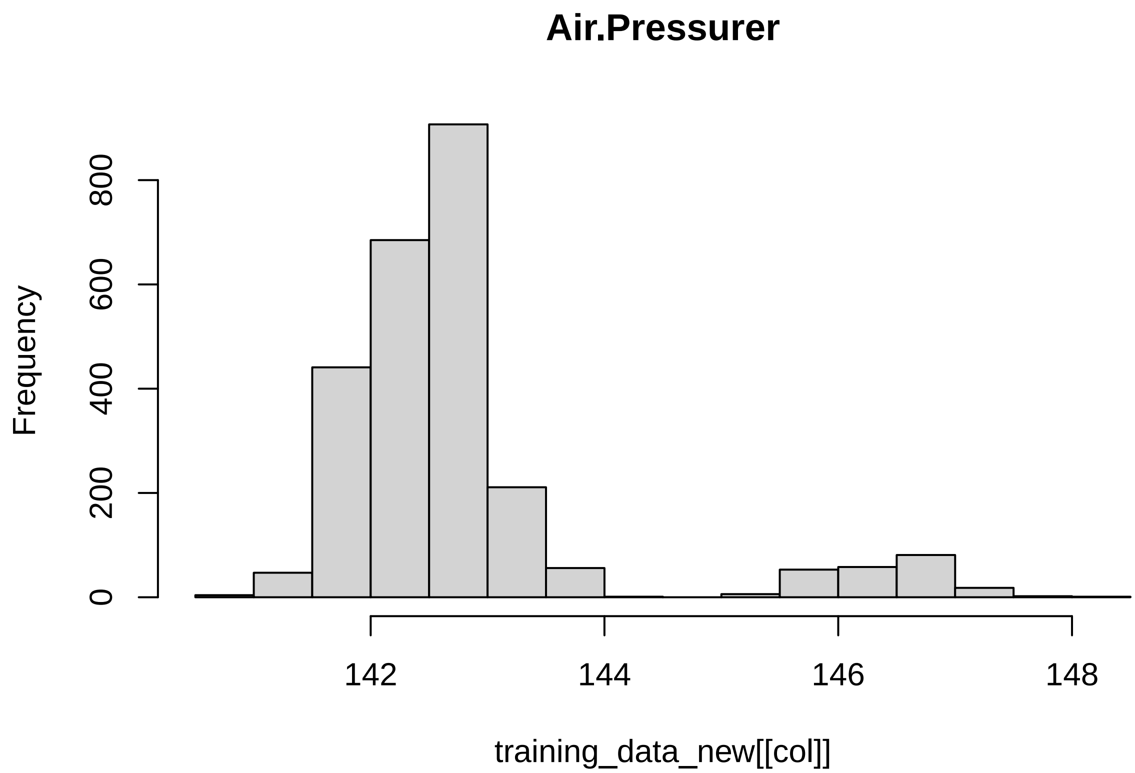

Usage.cont
Frequency
0
200
400
600
800
12
14
16
18
20
22
24
26
training_data_new[[col]]



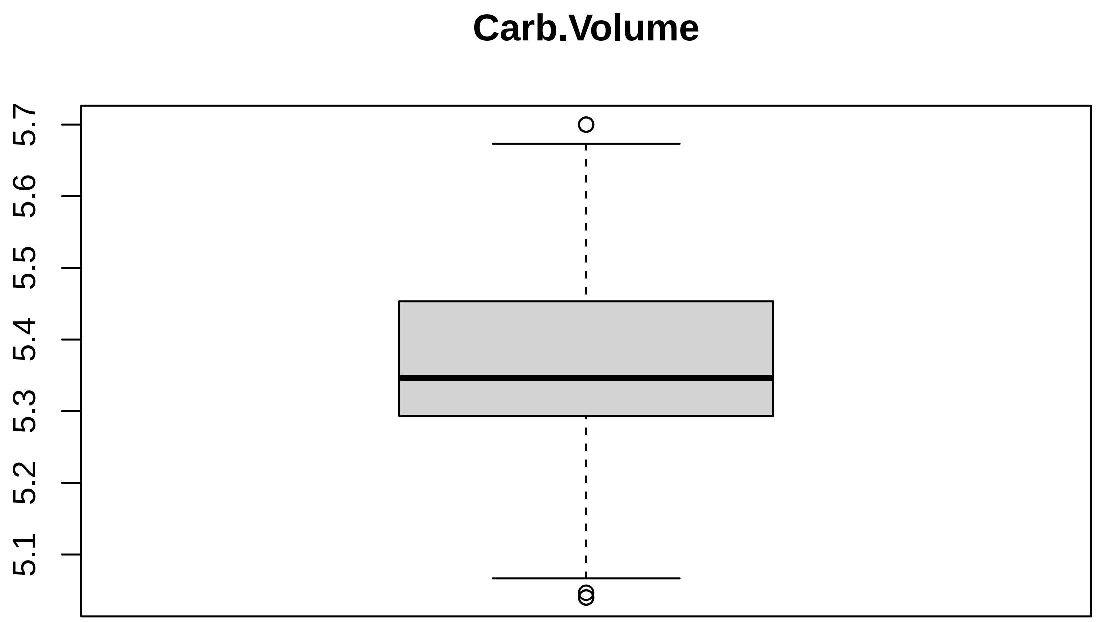
Carb.Flow
Frequency
0
200
600
1000
0
1000
2000
3000
4000
5000
training_data_new[[col]]

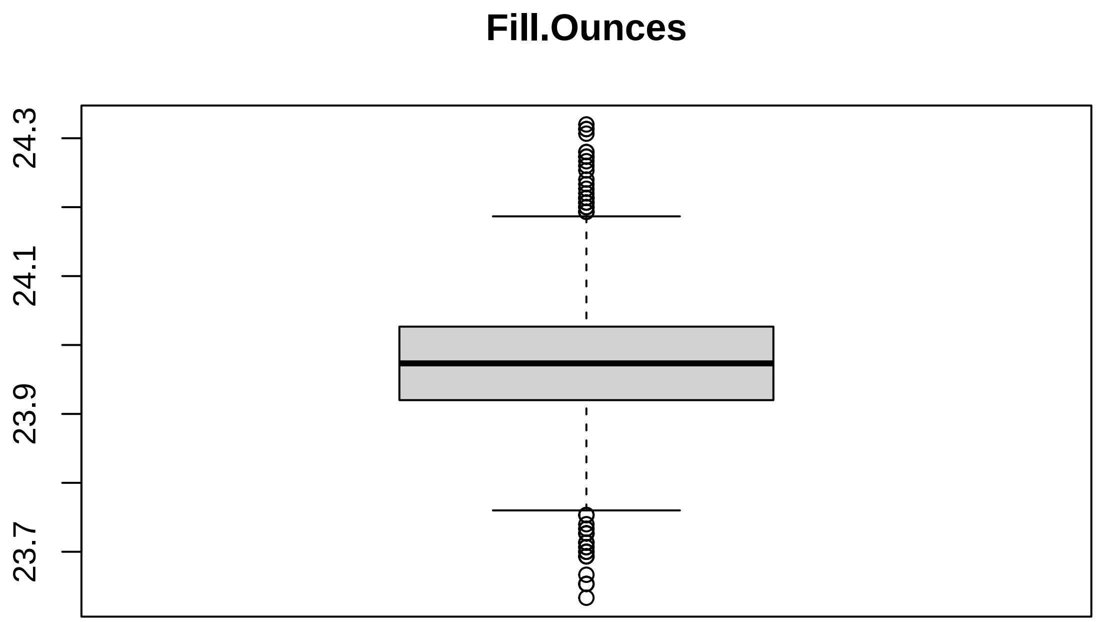

PH
Frequency
0
100
200
300
400
500
8.0
8.5
9.0
training_data_new[[col]]

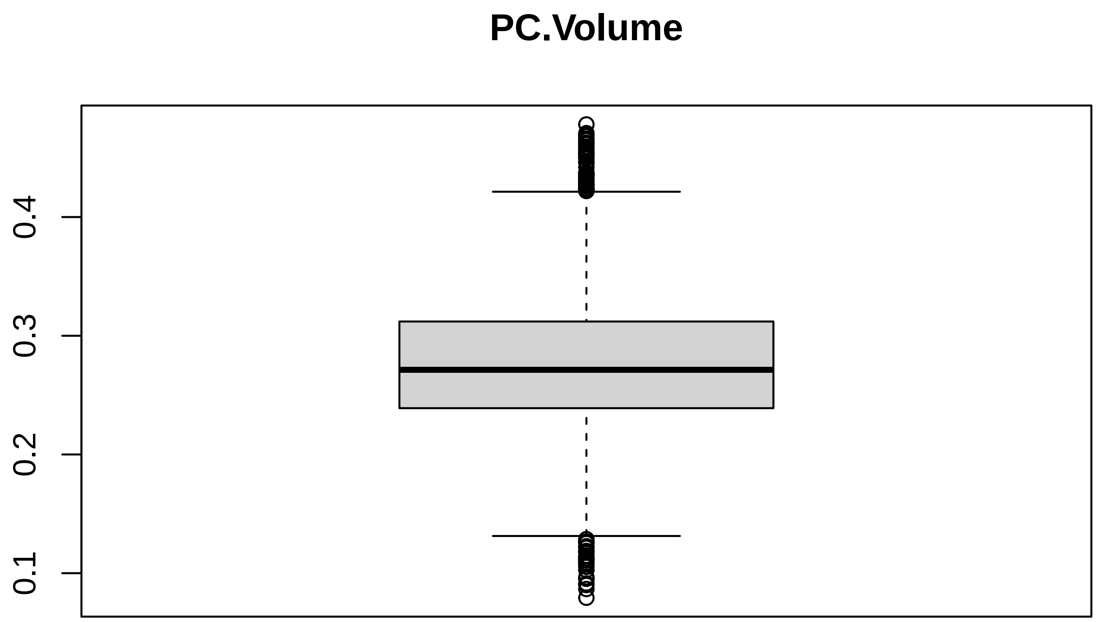

Oxygen.Filler
Frequency
0
500
1000
1500
0.0
0.1
0.2
0.3
0.4
training_data_new[[col]]

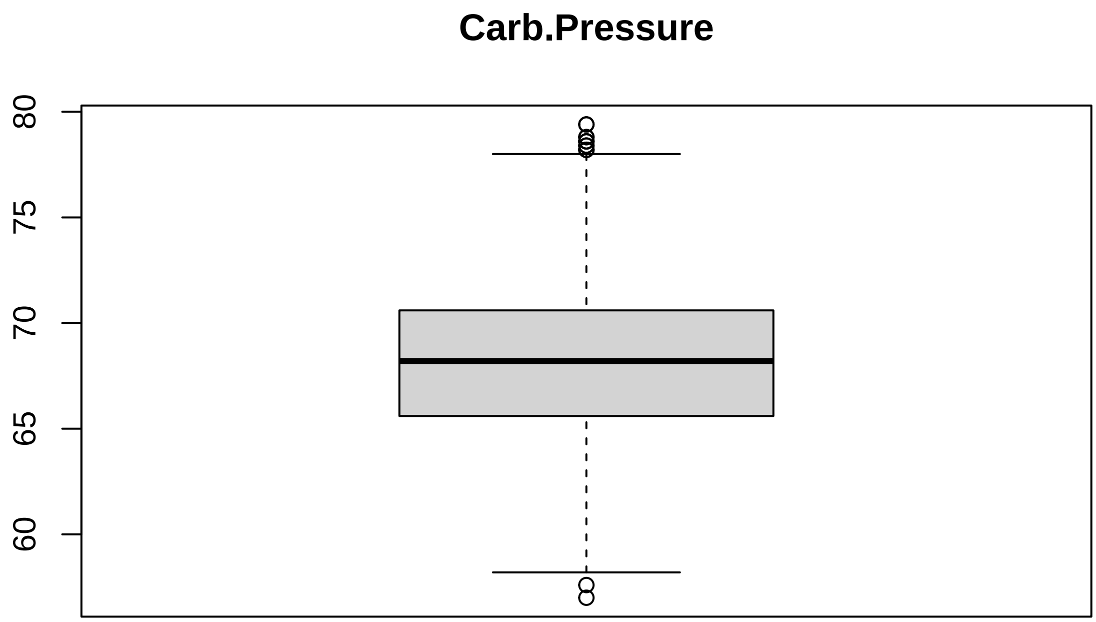


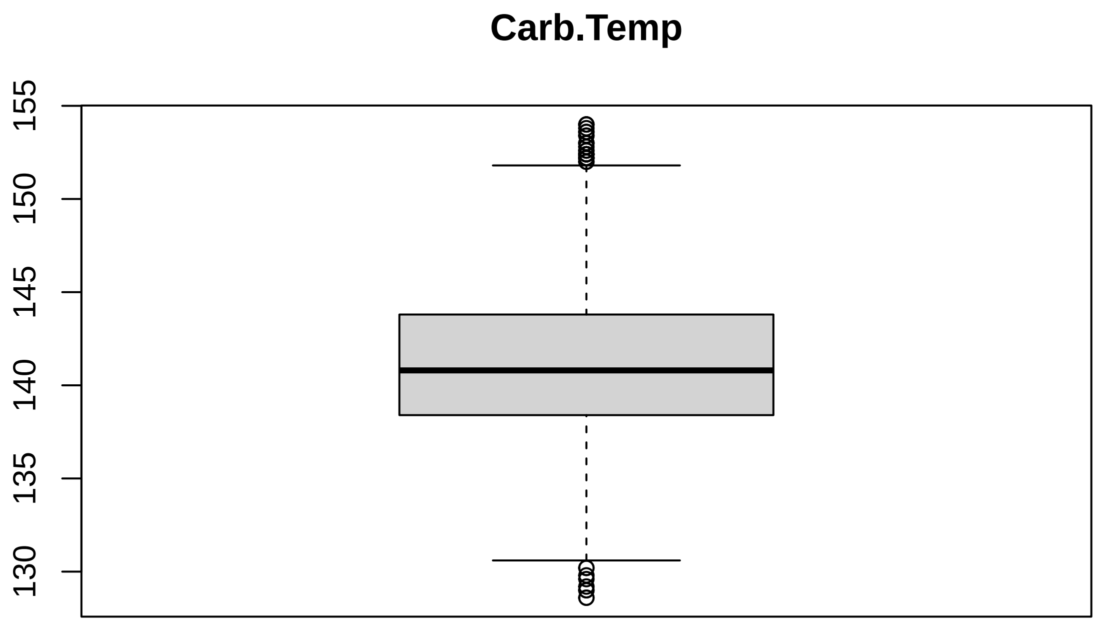

```
#box plots
for (col in names(training_data_new %>% select(-Brand.Code))) {
  boxplot(training_data_new[[col]], main=col)
}
```

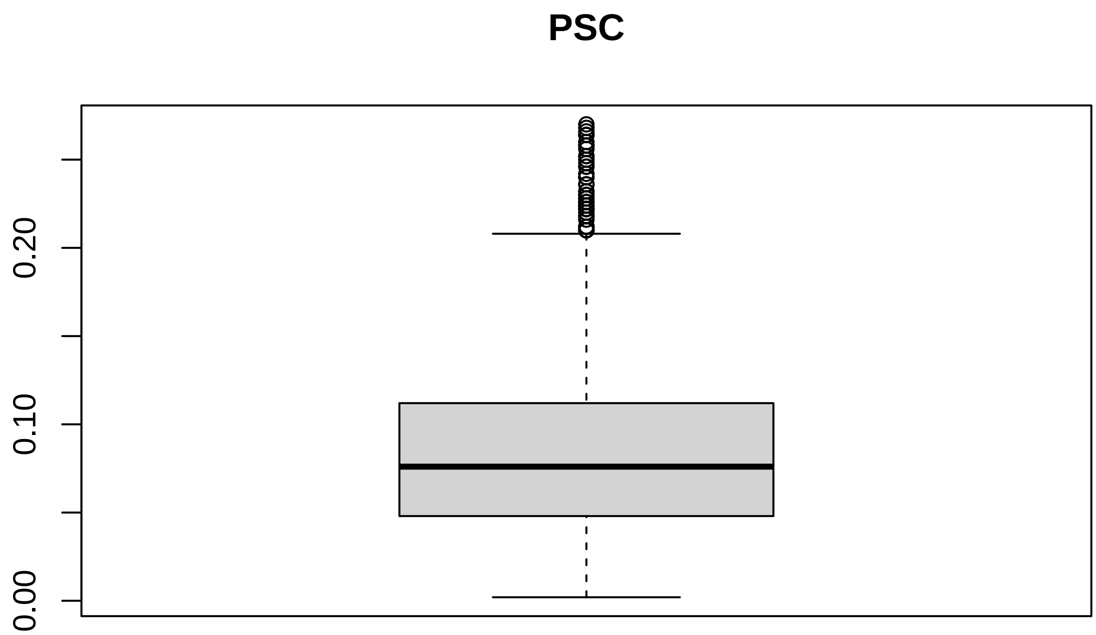

Carb.Volume
5.1
5.2
5.3
5.4
5.5
5.6
5.7

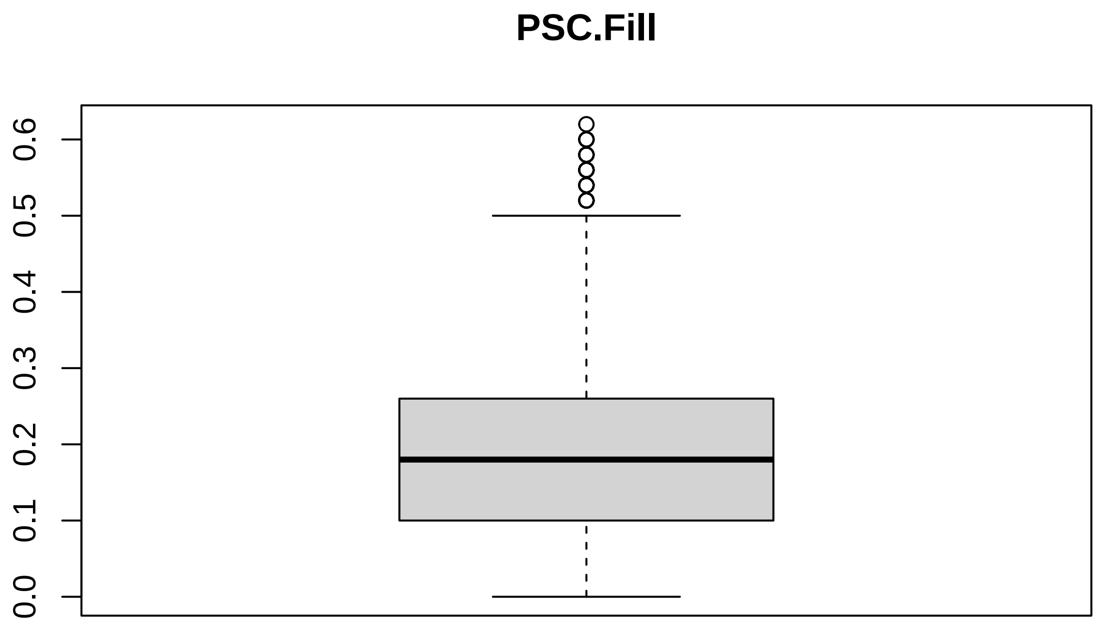

Fill.Ounces
24.3
24.1
23.9
23.7

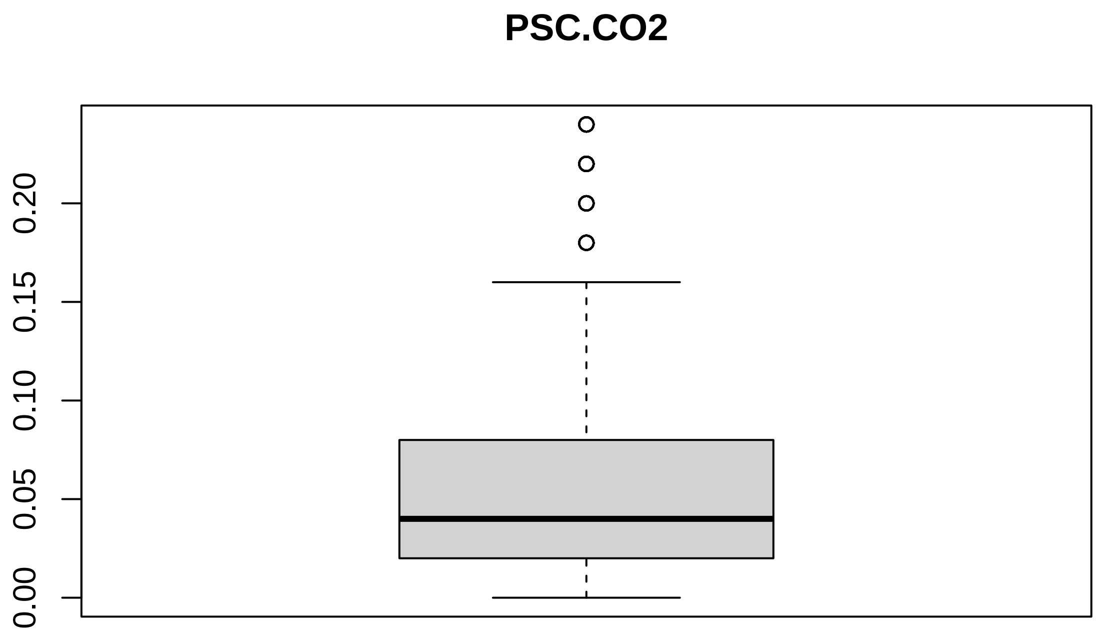

PC.Volume
0.1
0.2
0.3
0.4

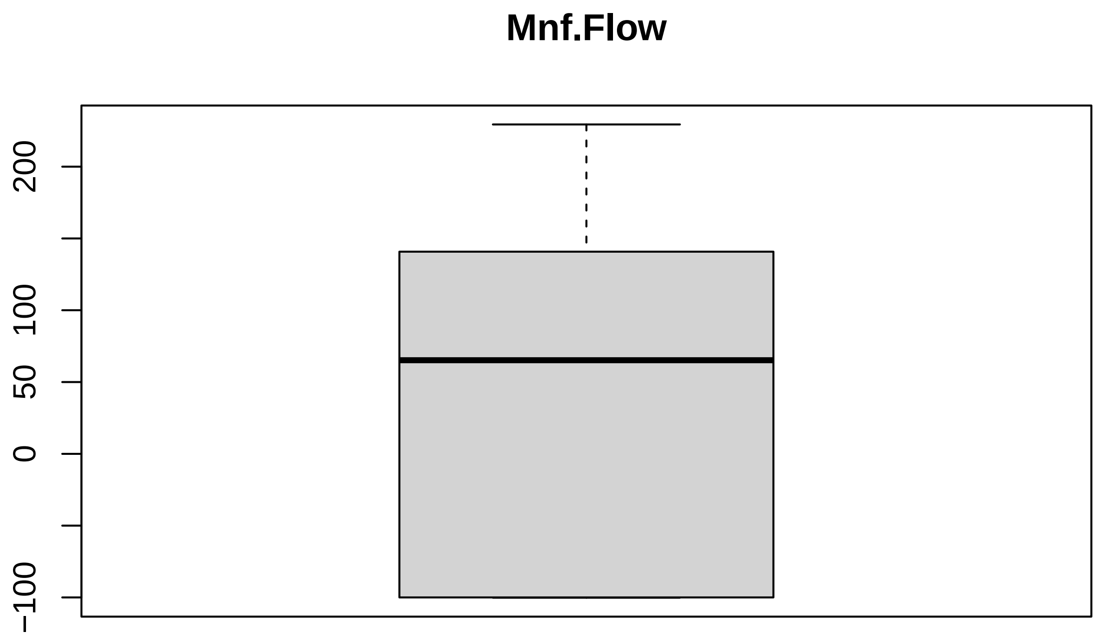

Carb.Pressure
80
75
70
65
60

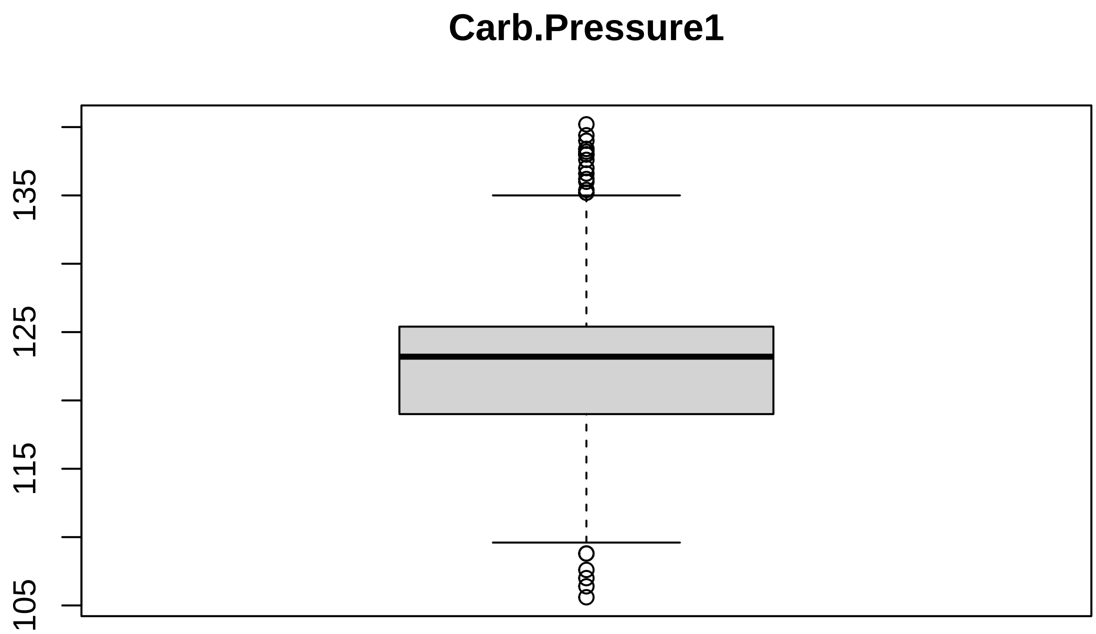

Carb.Temp
155
150
145
140
135
130

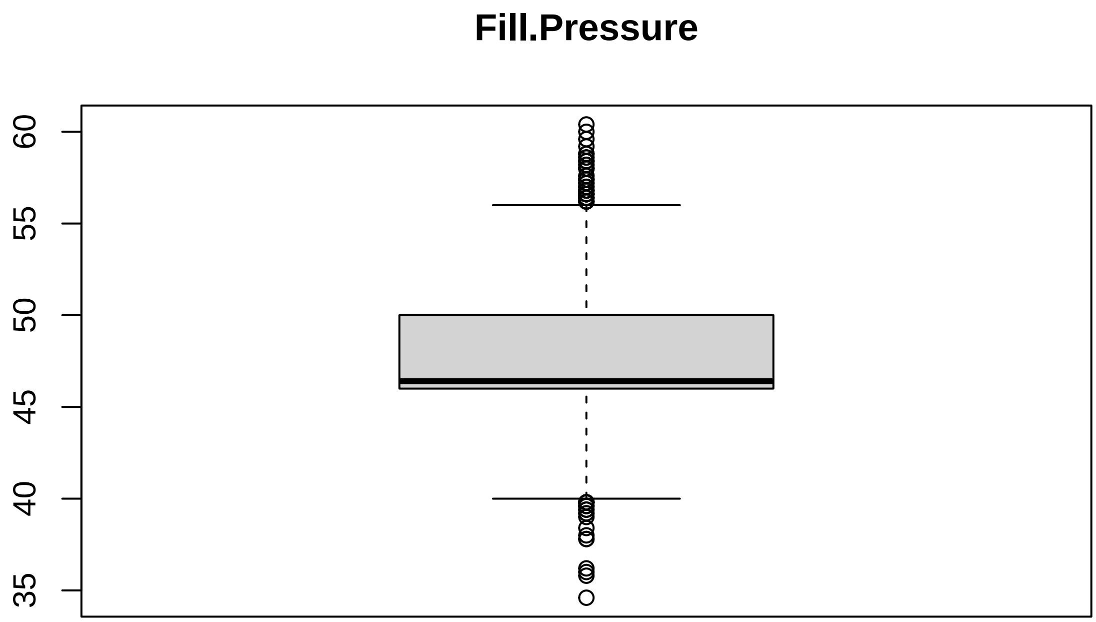

PSC
0.00
0.10
0.20

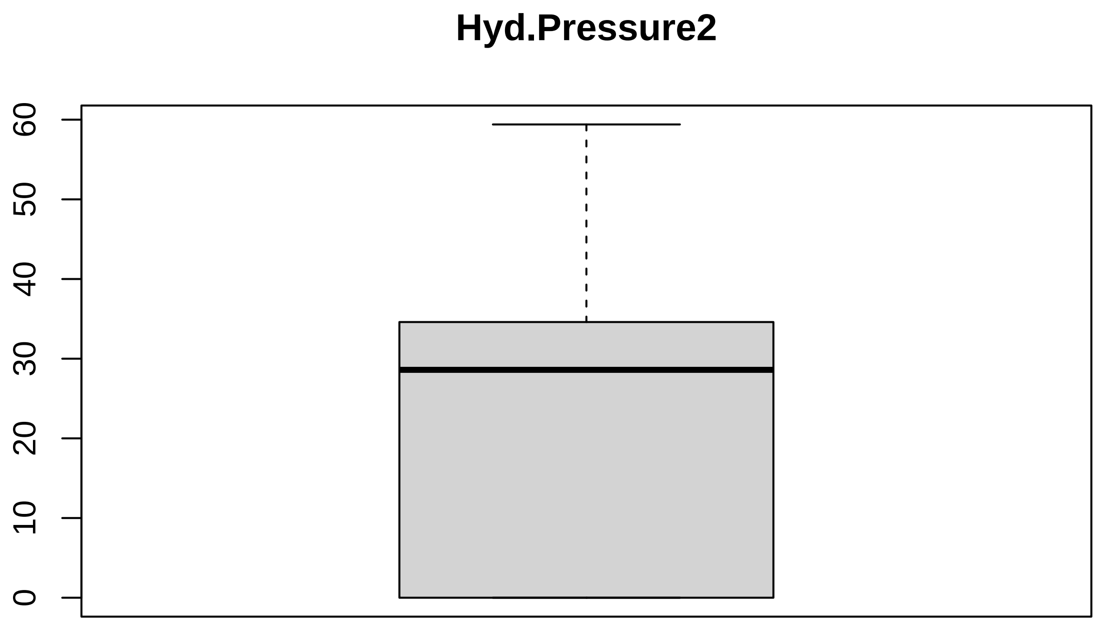

PSC.Fill
0.0
0.1
0.2
0.3
0.4
0.5
0.6

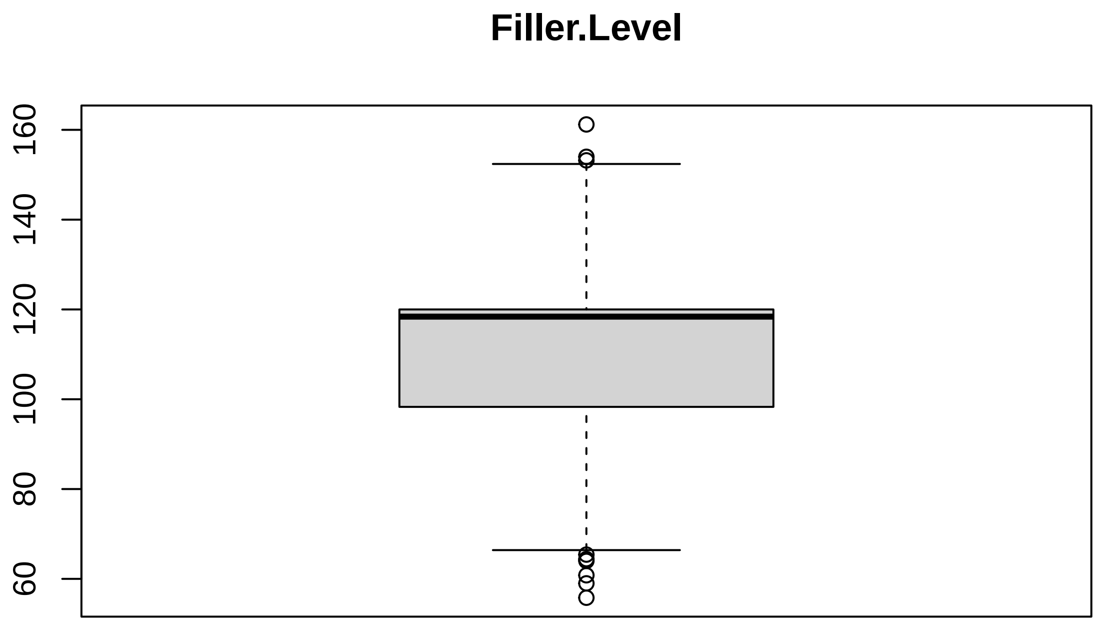

PSC.CO2
0.00
0.05
0.10
0.15
0.20

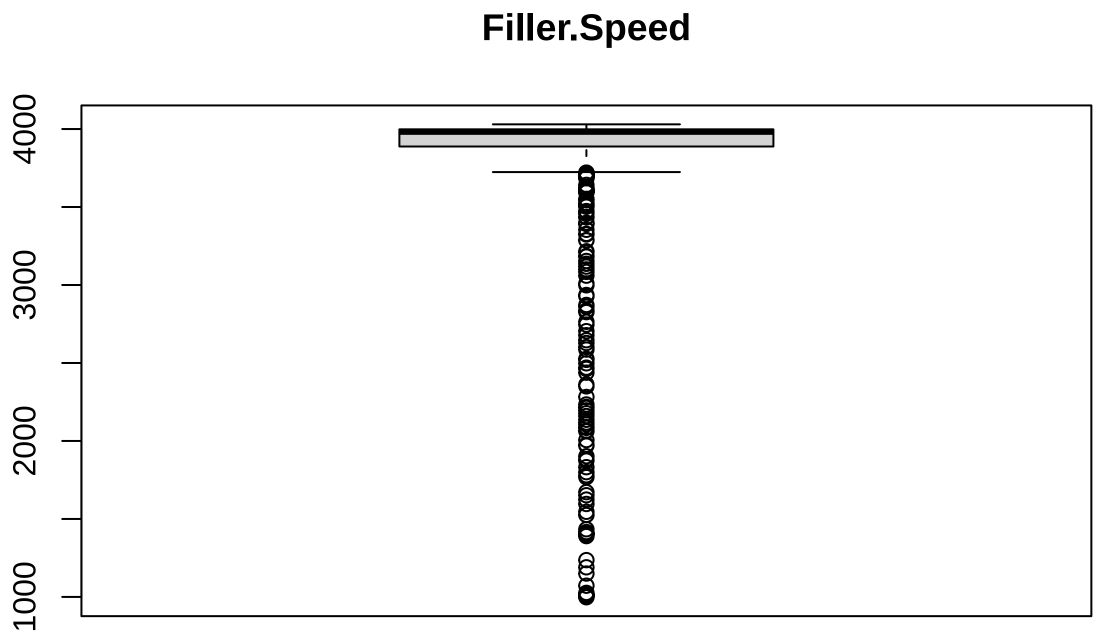

Mnf.Flow
200
100
50
0
−100

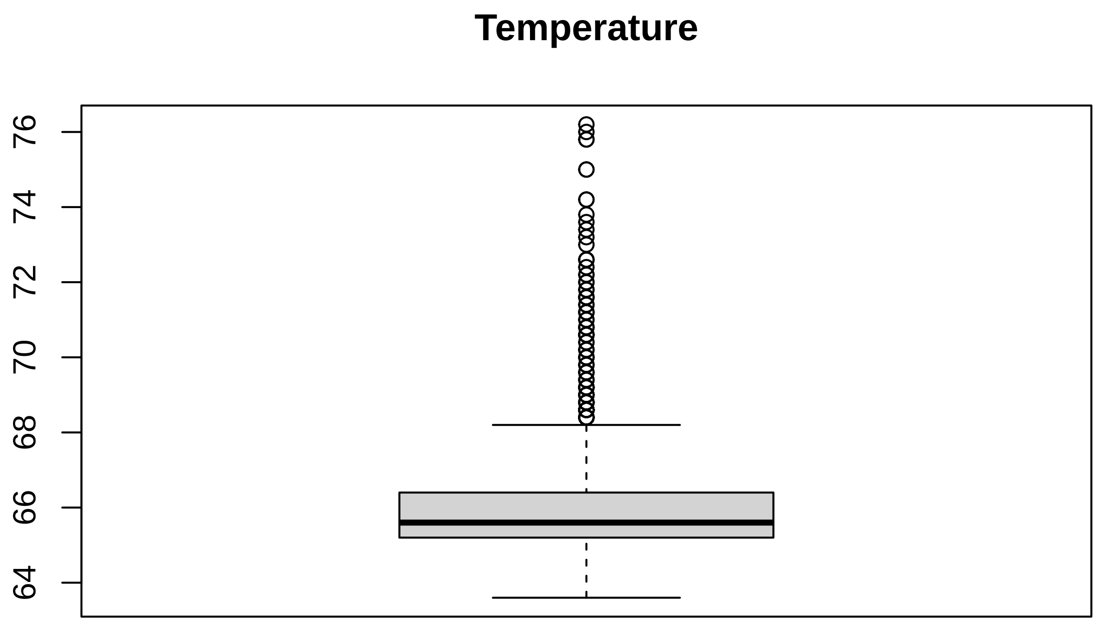

Carb.Pressure1
135
125
115
105

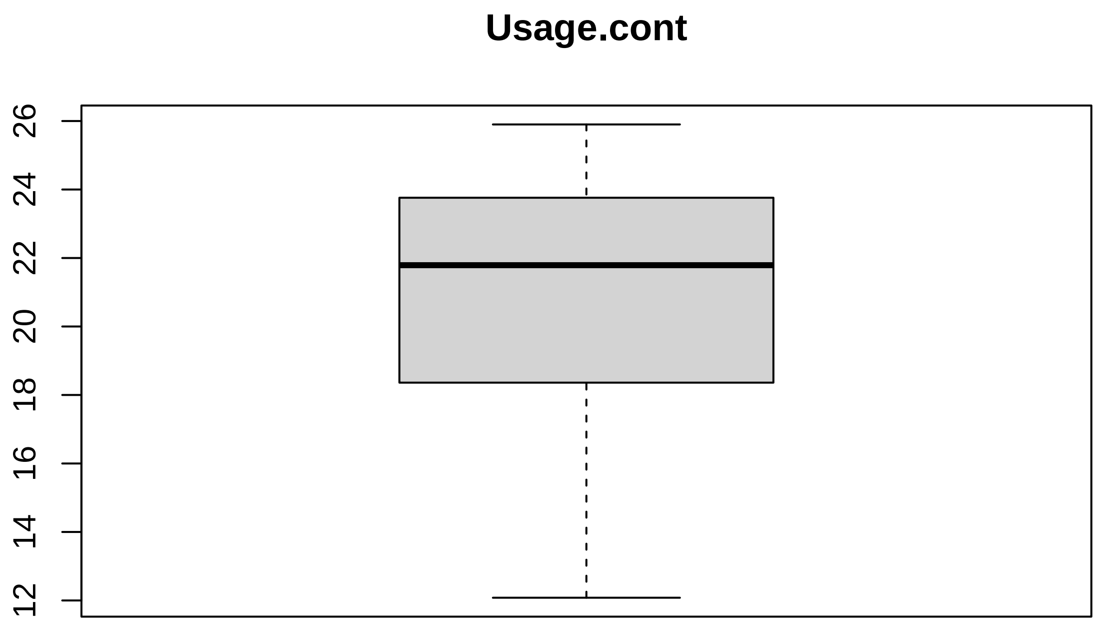

Fill.Pressure
60
55
50
45
40
35

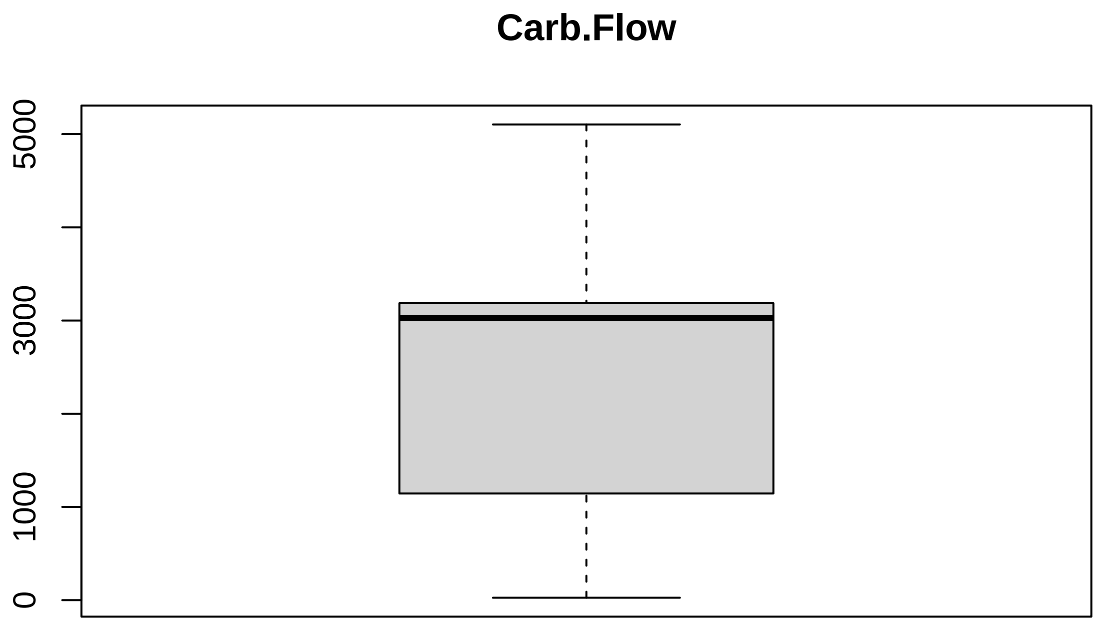

Hyd.Pressure2
0
10
20
30
40
50
60

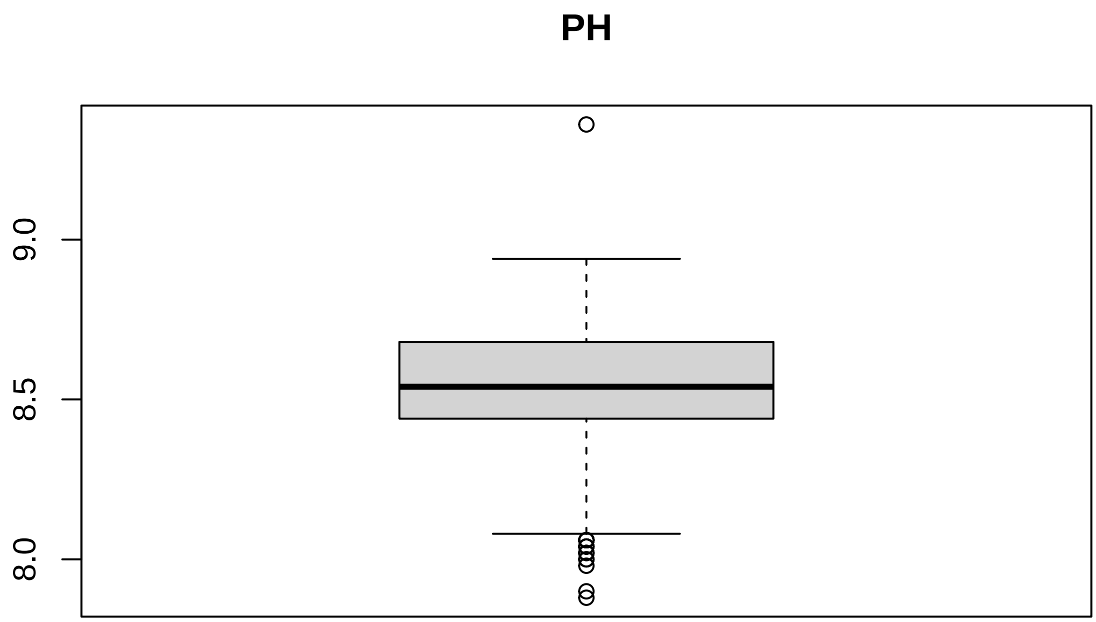

Filler.Level
60
80
100
120
140
160

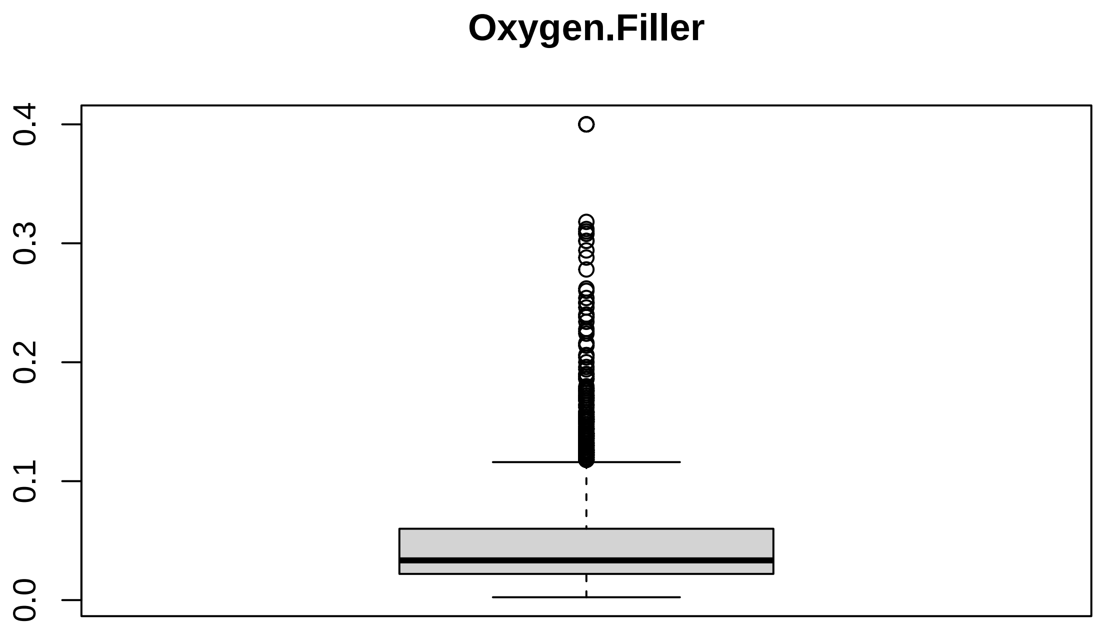

Filler.Speed
4000
3000
2000
1000

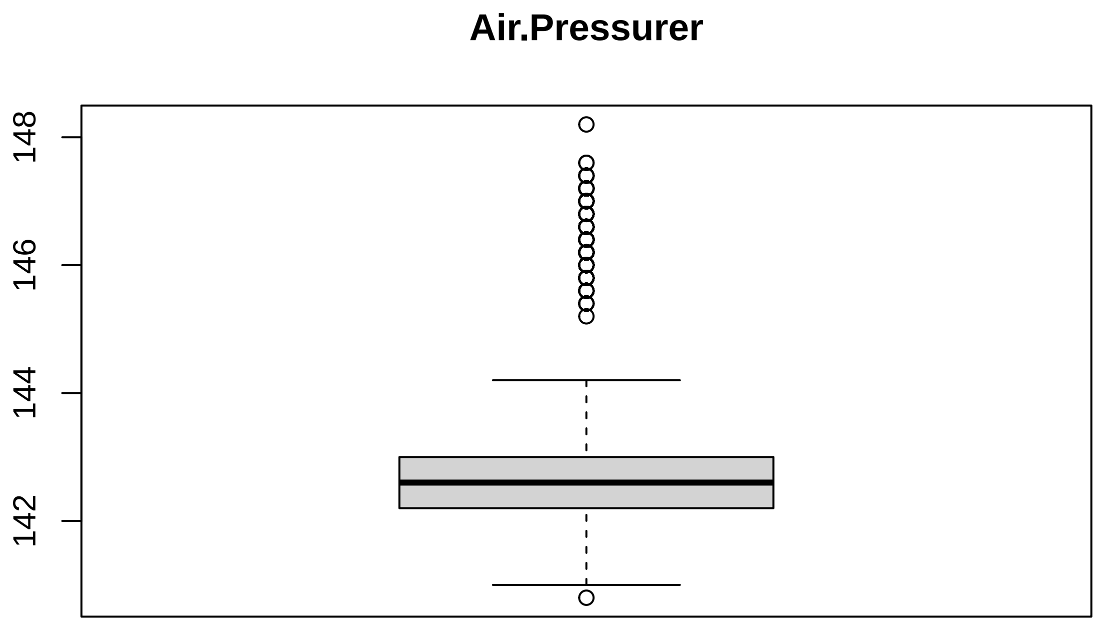

Temperature
76
74
72
70
68
66
64


Usage.cont
26
24
22
20
18
16
14
12


Carb.Flow
5000
3000
1000
0


PH
9.0
8.5
8.0


Oxygen.Filler
0.4
0.3
0.2
0.1
0.0



Calculating the number of outliers in each column based on the IQR method

```r
outliers <- sapply(training_data_new |> select(-Brand.Code), function(x) {
  Q1 <- quantile(x, 0.25, na.rm=TRUE)
  Q3 <- quantile(x, 0.75, na.rm=TRUE)
  IQR <- Q3 - Q1
  sum(x < Q1 - 1.5*IQR | x > Q3 + 1.5*IQR, na.rm=TRUE)
})
outliers
```

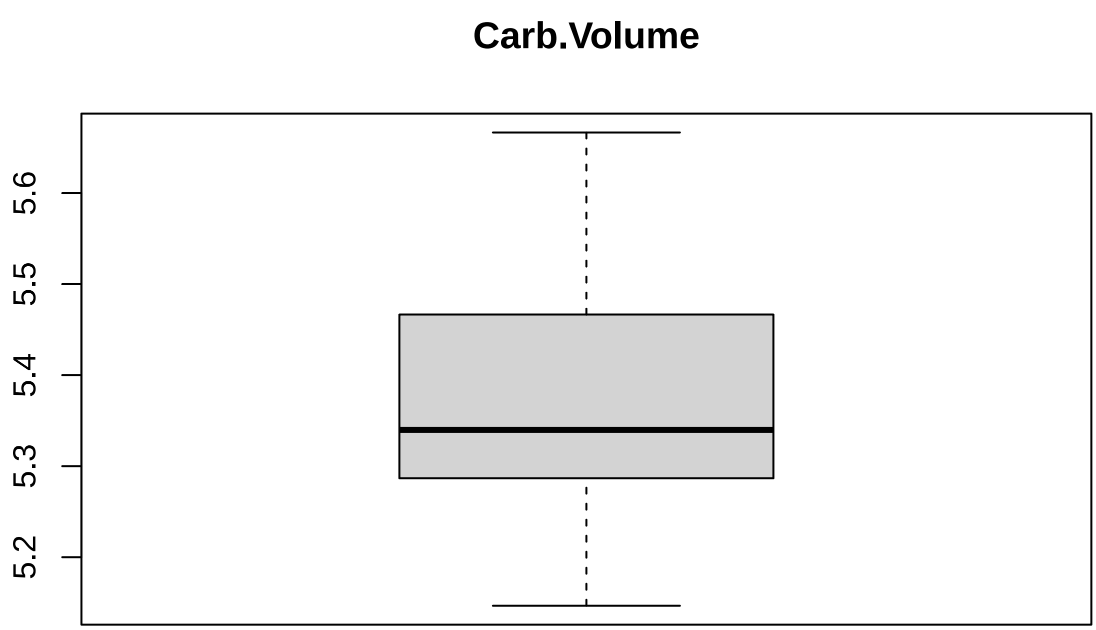
```
##    Carb.Volume    Fill.Ounces      PC.Volume  Carb.Pressure      Carb.Temp
##              5             55             86             14             36
##            PSC       PSC.Fill        PSC.CO2       Mnf.Flow Carb.Pressure1
##             52             54             77              0             24
##  Fill.Pressure  Hyd.Pressure2   Filler.Level   Filler.Speed    Temperature
##             79              0              9            439            123
##     Usage.cont      Carb.Flow             PH Oxygen.Filler  Air.Pressurer
##              0              0             18            189            220
```

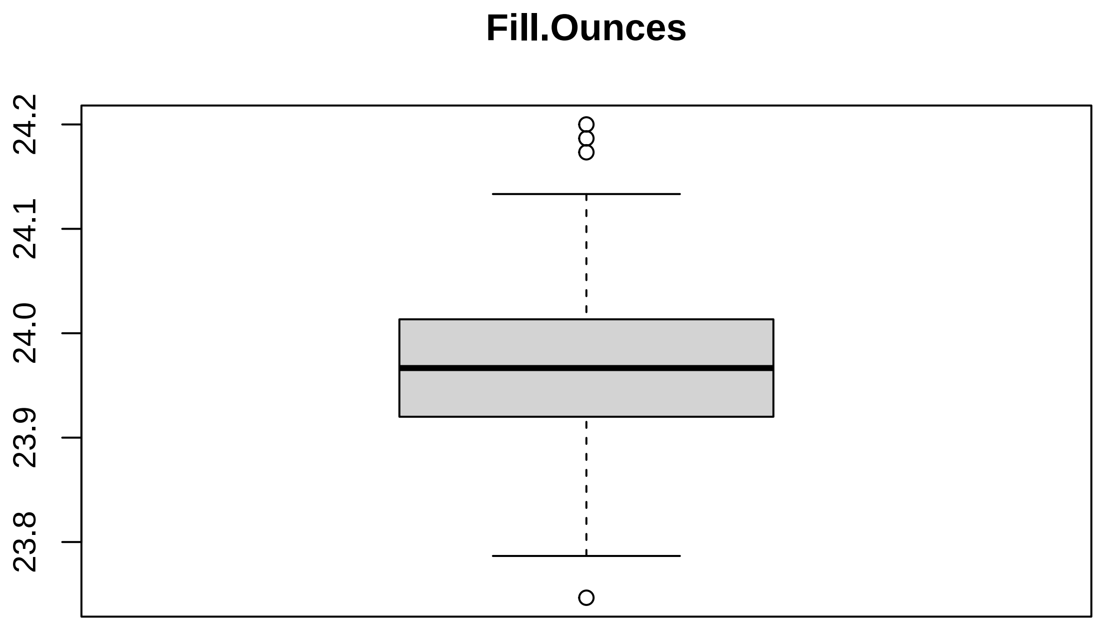
Cross-checking the graphs with the data set, we could remove the one Filler.Level value that appears on the top of the box plot, 161.2 when the next-highest is 154. pH has one comparatively high value (9.36) that's nowhere close to the next-highest value. We can remove that row - it's missing 4 other variables. Other outlier values are one of several similar values, and could be categorized that way due to the shape of the distribution and wide range of values. For example, it looks like there's one particularly high value for Oxygen.Filler, but when we check there are 5 cases with the same value.

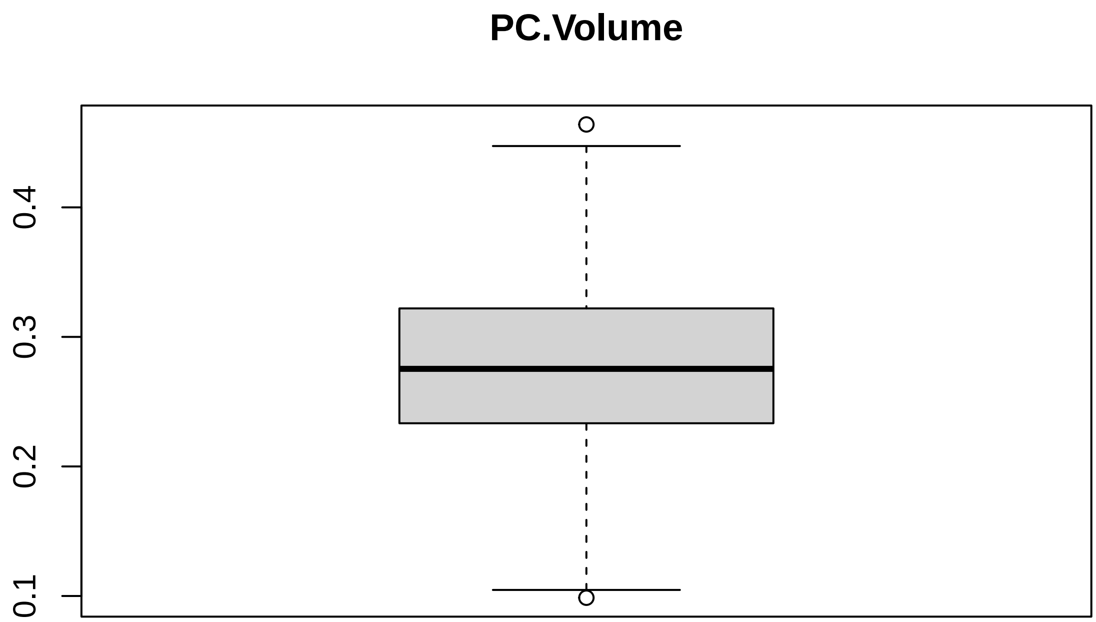
```
#removing the one outliers
training_data_new <- training_data_new |>
  filter(PH < 9 | is.na(PH)) |>
  filter(Filler.Level < 160 | is.na(Filler.Level))

#removing pH NAs because we shouldn't impute the response variable
training_data_no_na <- training_data_new |>
  filter(!is.na(PH))
```

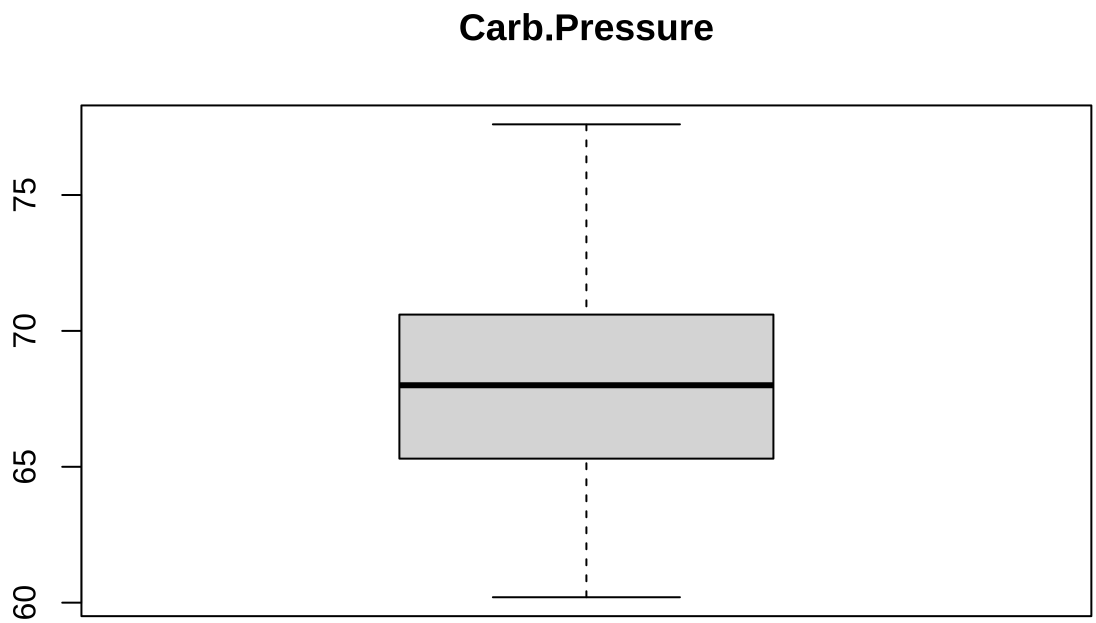
1.3.5 Looking at outliers in the test data set

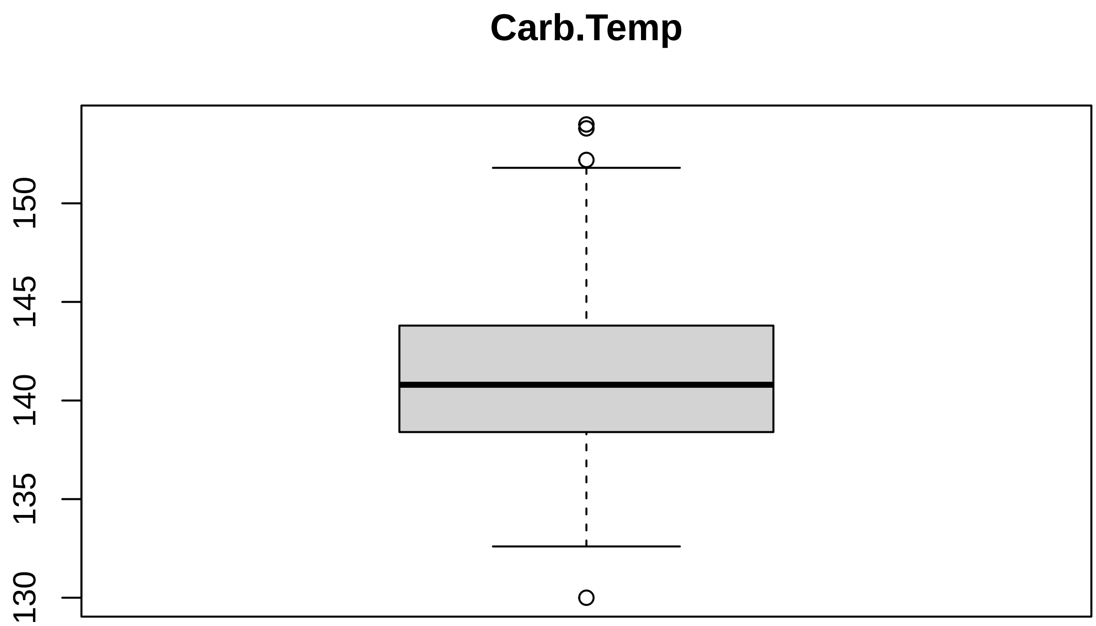
```
#leaving out the categorical and blank response variable
for (col in names(test_data_new %>% select(-Brand.Code, -PH))) {
  boxplot(test_data_new[[col]], main=col)
}
```

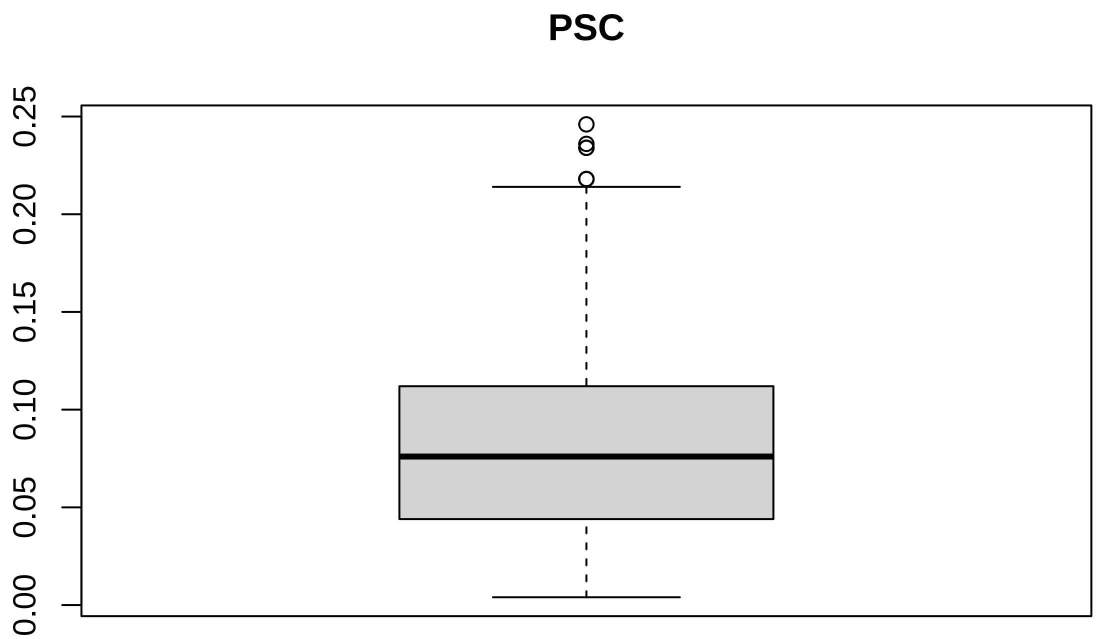


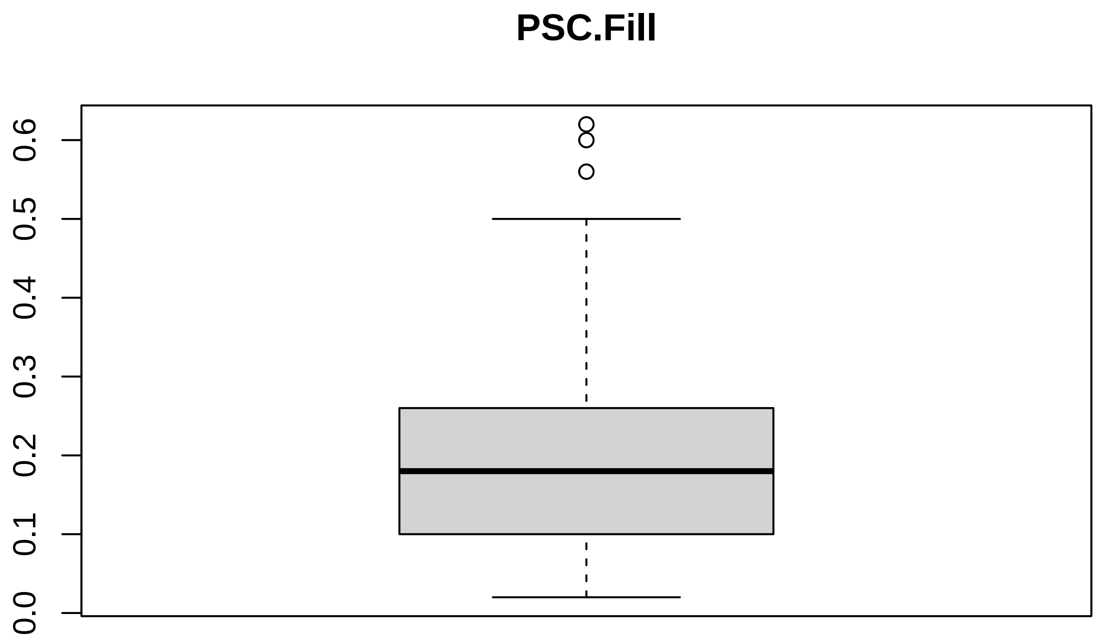

Fill.Ounces
24.2
24.1
24.0
23.9
23.8

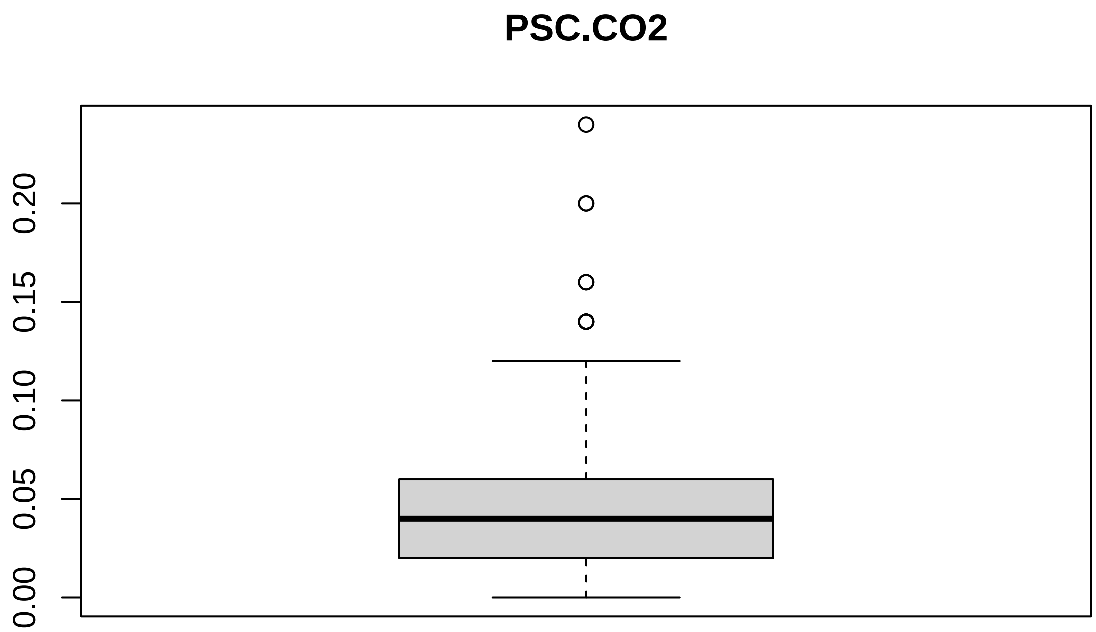

PC.Volume
0.1
0.2
0.3
0.4

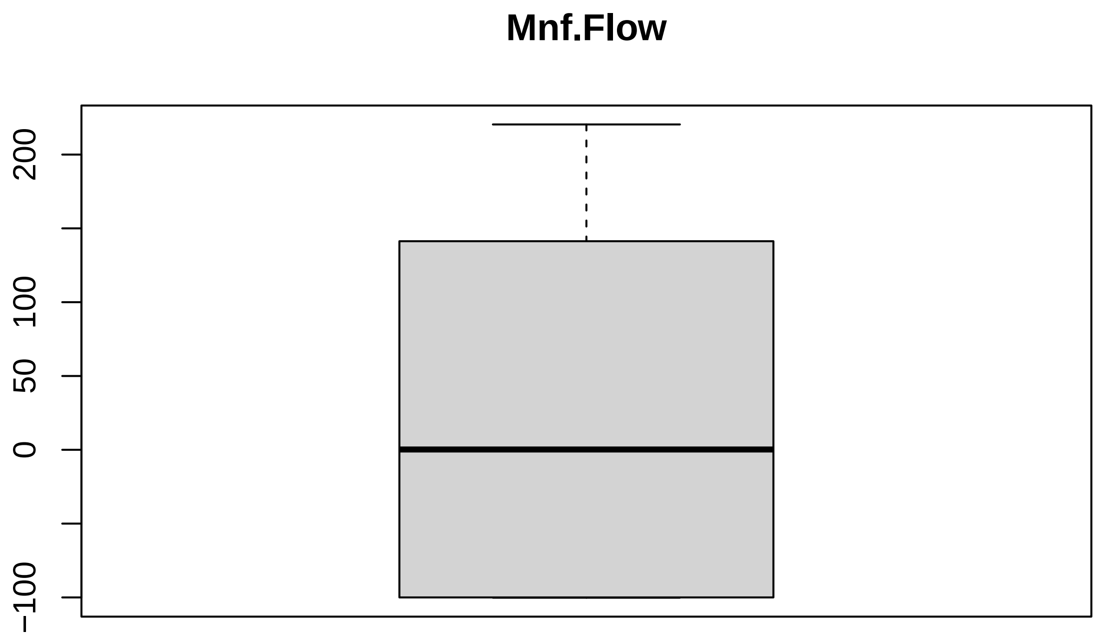

Carb.Pressure
75
70
65
60


Carb.Temp
130
135
140
145
150

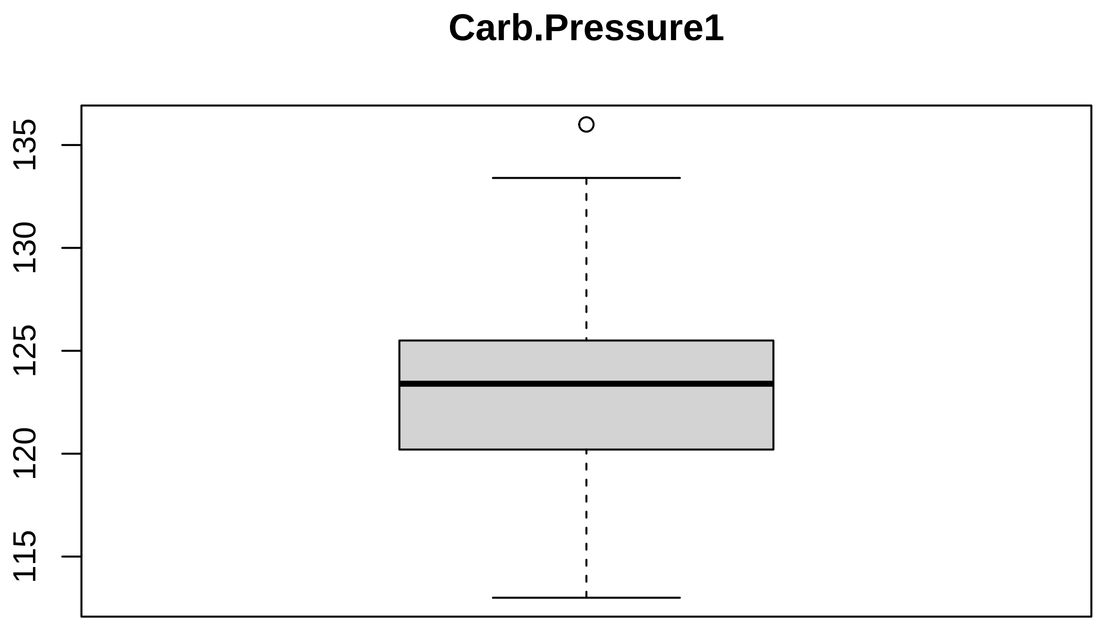

PSC
0.00
0.05
0.10
0.15
0.20
0.25

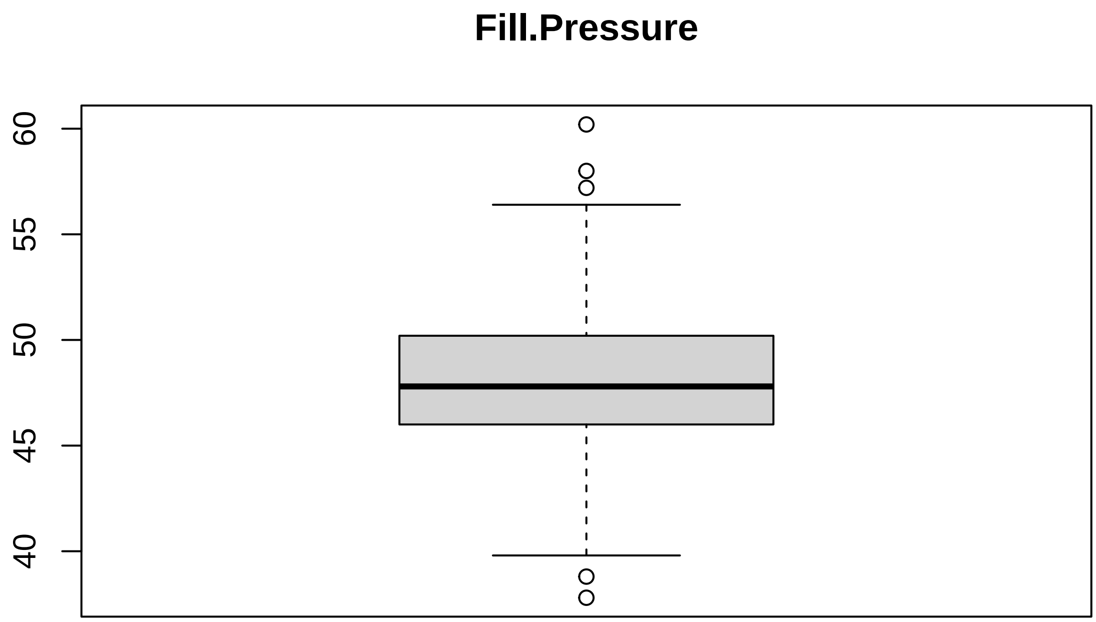

PSC.Fill
0.0
0.1
0.2
0.3
0.4
0.5
0.6

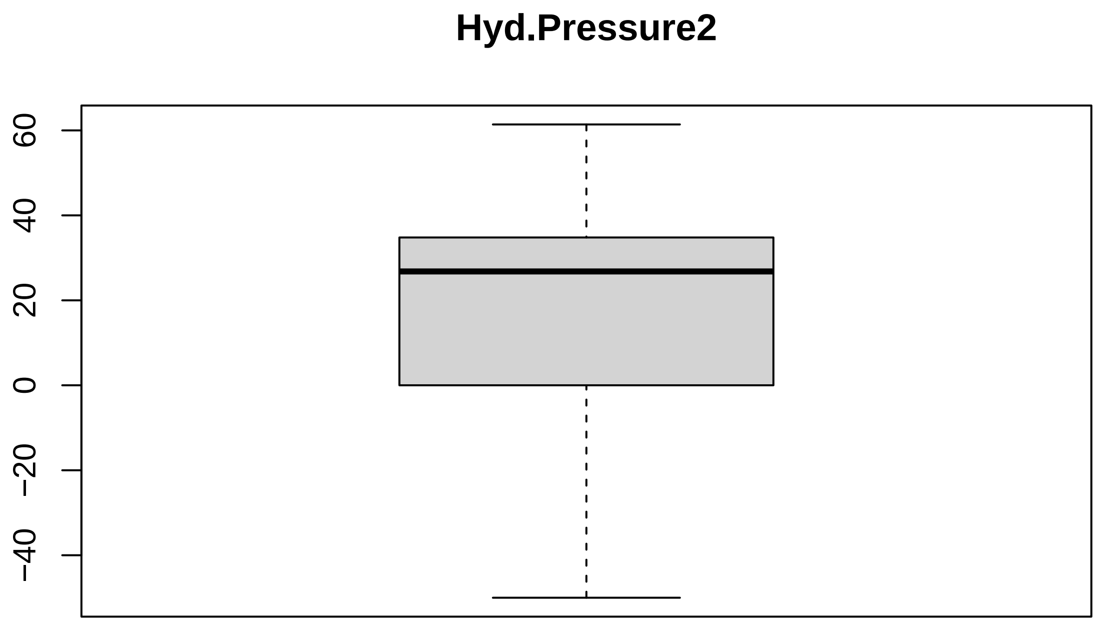

PSC.CO2
0.00
0.05
0.10
0.15
0.20

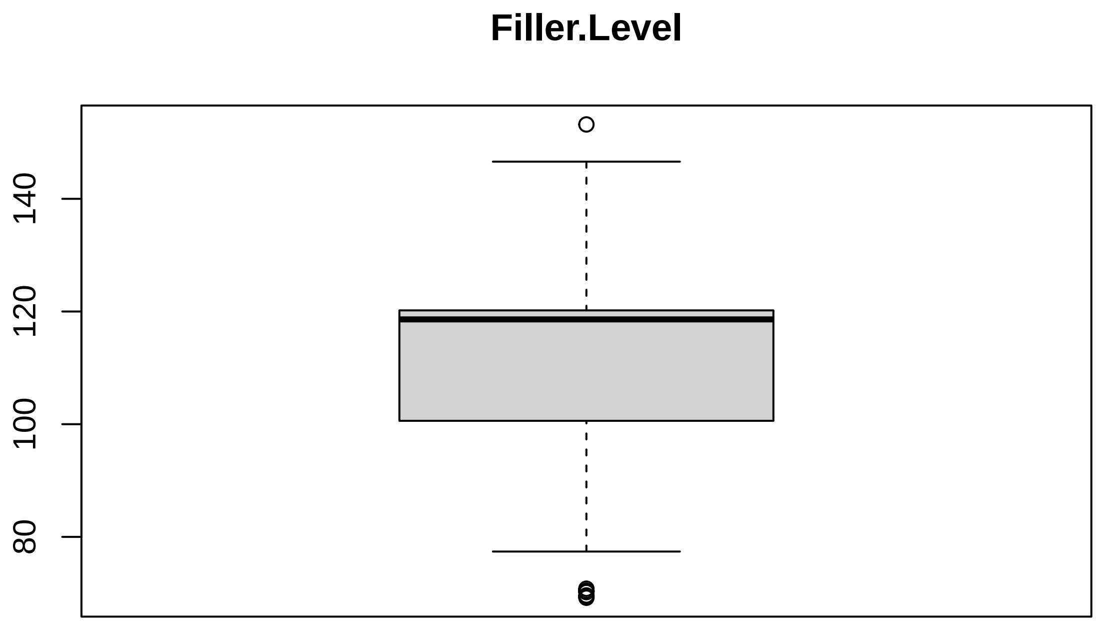


Mnf.Flow
200
100
50
0
−100

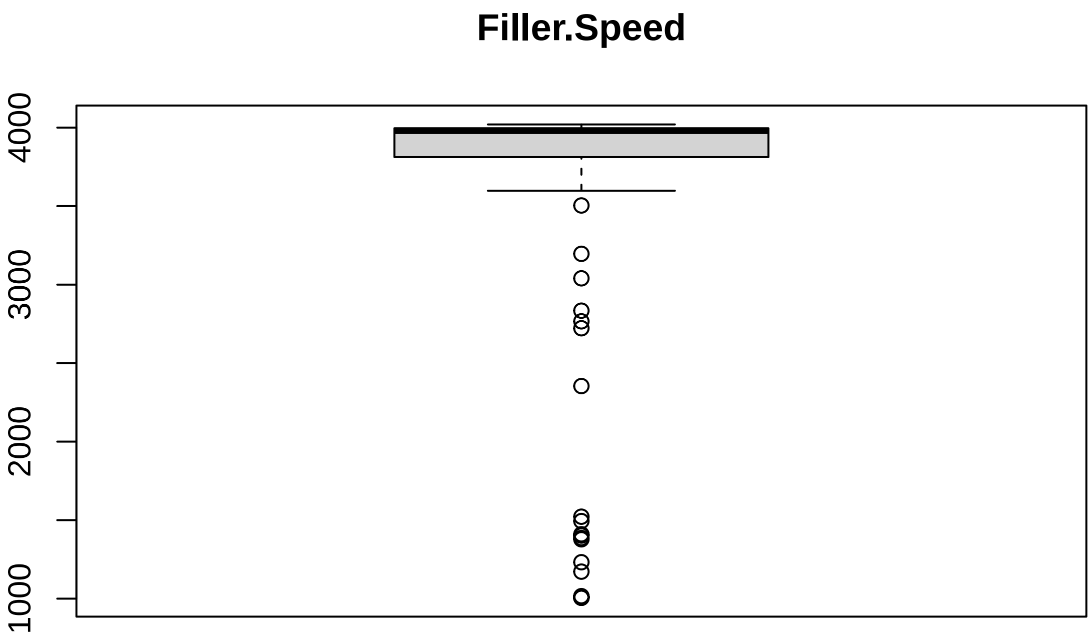

Carb.Pressure1
135
130
125
120
115


Fill.Pressure
60
55
50
45
40

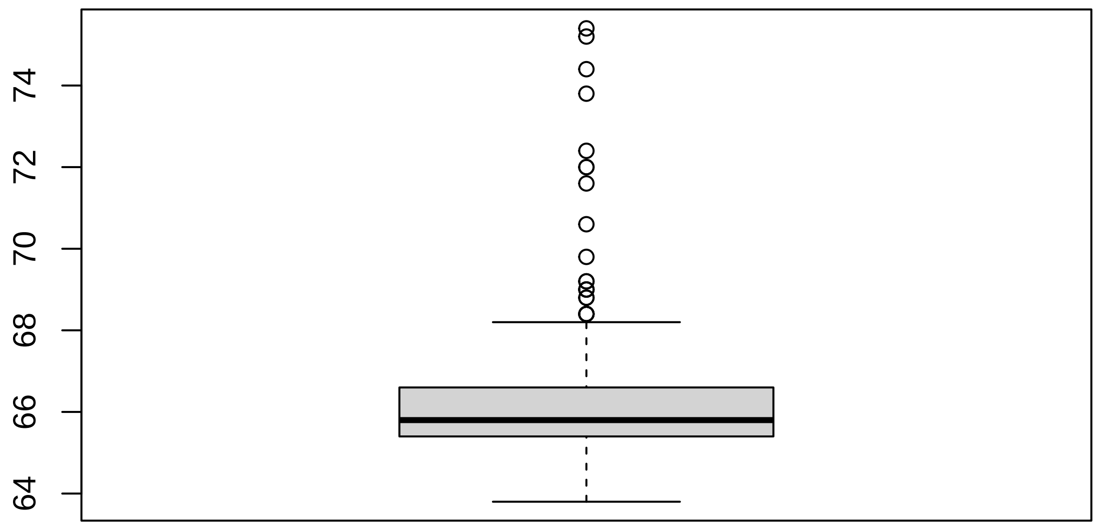

Hyd.Pressure2
60
40
20
0
−20
−40

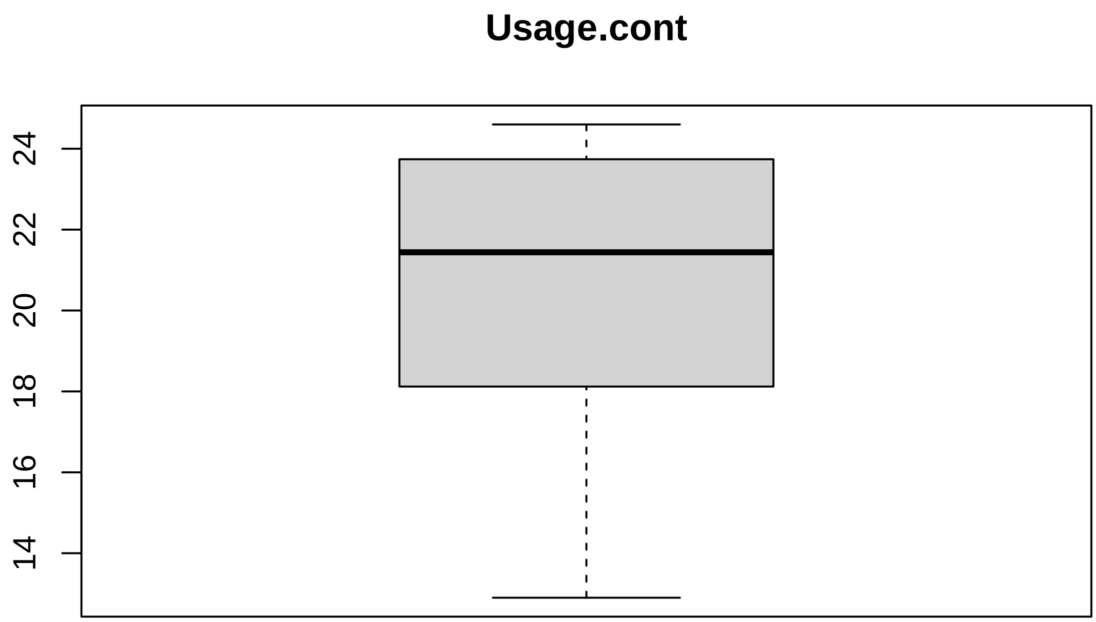


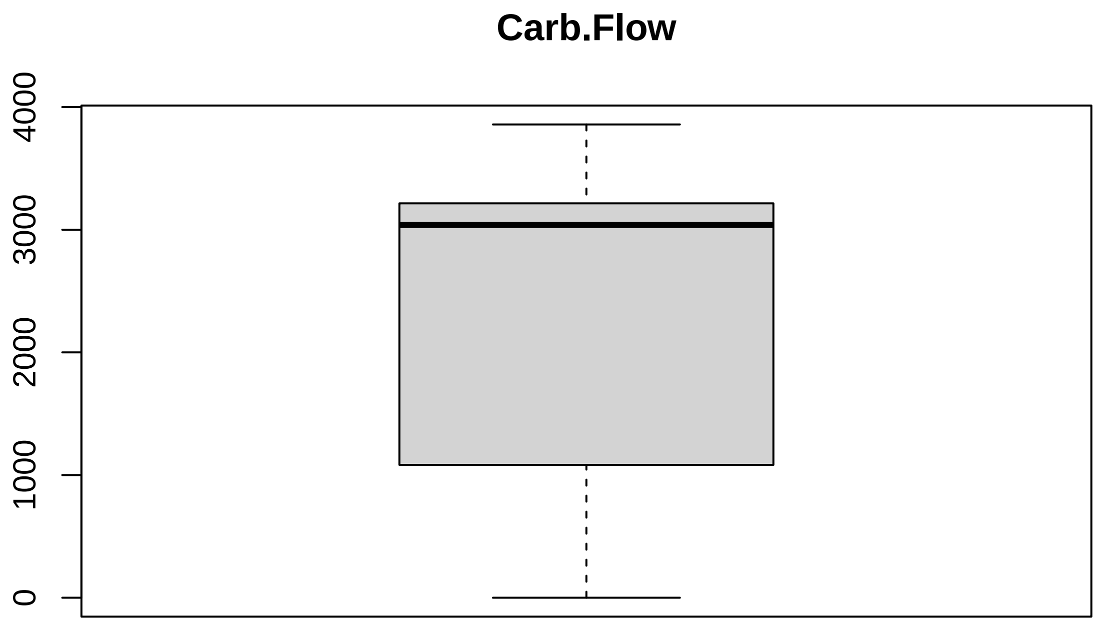
Filler.Level
140
120
100
80

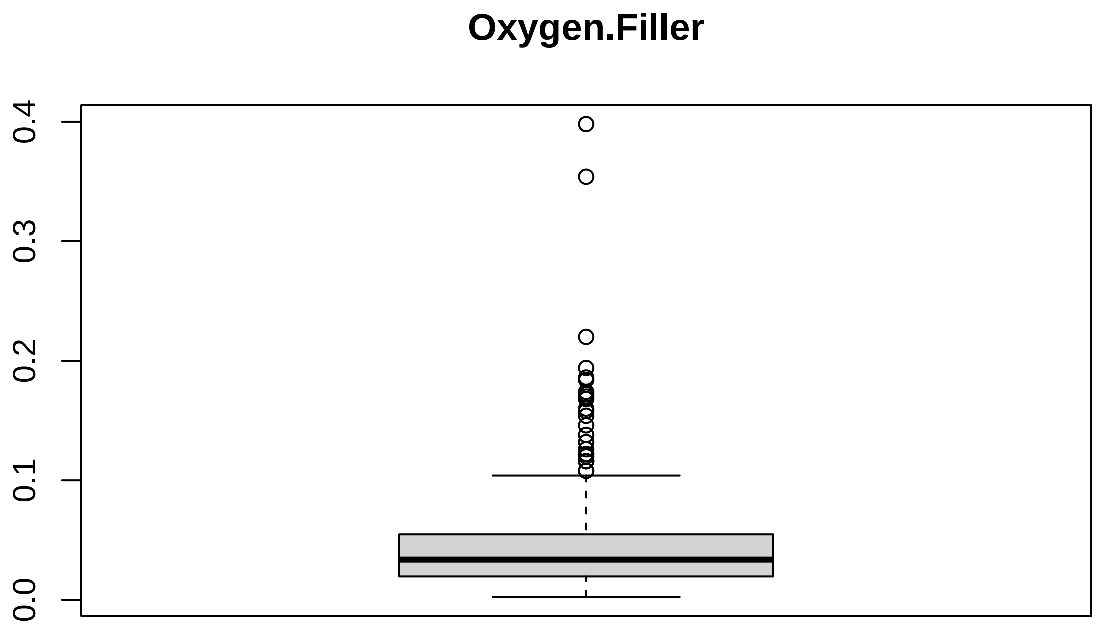

Filler.Speed
4000
3000
2000
1000

Temperature

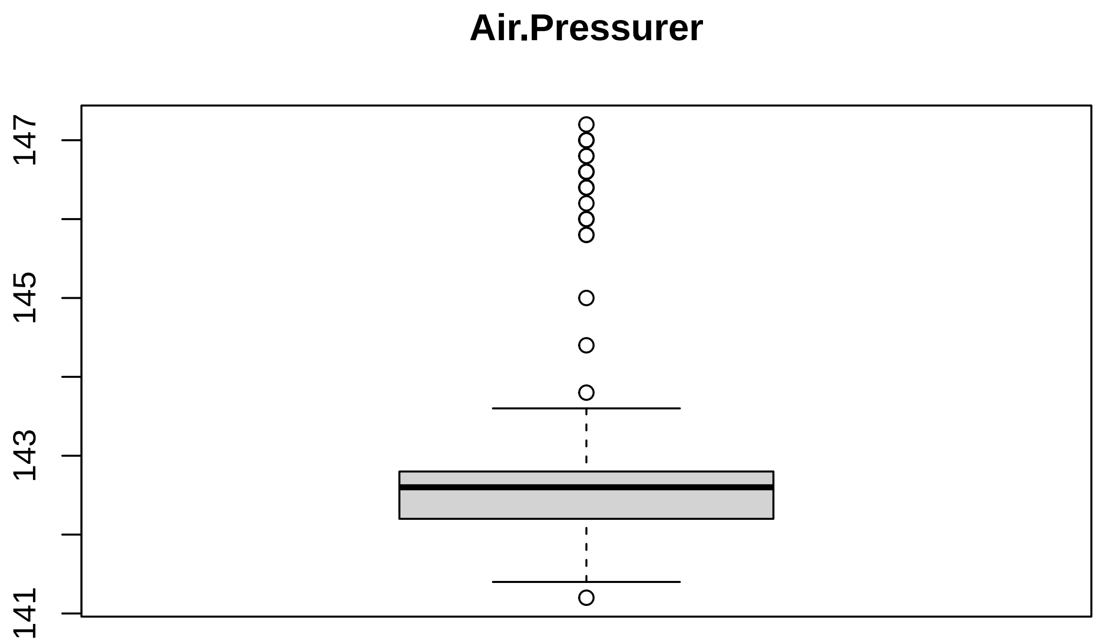



Usage.cont
24
22
20
18
16
14


Carb.Flow
4000
3000
2000
1000
0



Oxygen.Filler
0.0
0.1
0.2
0.3
0.4




```
test_data_new |> count(Brand.Code)
```

```
##   Brand.Code   n
## 1          A  35
## 2          B 129
## 3          C  31
## 4          D  64
## 5       <NA>   8
```

Oxygen filler shows up again with a few outliers, which just seems to be the case for this variable; the outliers in filler speed are all clustered; filler level has outliers on both ends, as does fill pressure. Carb.pressure1 has one high value, PSC.CO2 has a few higher values designated outliers due to the clustering on the lower end, same is true for PSC.Fill, and the rest are similar.

We won't remove any of these because they don't look like errors, and in some cases represent a different, smaller cluster of values.

1.4 Dealing with missing values

```
sum(is.na(training_data_no_na))
```

```
## [1] 515
```

```r
sort(colSums(is.na(training_data_no_na)), decreasing = TRUE)
```

```
##     Brand.Code   Filler.Speed      PC.Volume        PSC.CO2    Fill.Ounces
##            120             52             39             39             38
##            PSC Carb.Pressure1  Carb.Pressure      Carb.Temp       PSC.Fill
##             33             32             27             26             23
##  Fill.Pressure  Hyd.Pressure2   Filler.Level    Temperature  Oxygen.Filler
##             17             15             15             11             11
##    Carb.Volume     Usage.cont      Carb.Flow       Mnf.Flow             PH
##             10              5              2              0              0
##  Air.Pressurer
##              0
```

```r
#checking the number of complete cases
table(rowSums(is.na(training_data_no_na)))
```

```
##
##    0    1    2    3    4    5    6
## 2169  309   64   18    2    2    1
```

516 missing values across 2566 cases and 21 variables - 516/53886 is about .9% overall missingness, which is very low. Notably, brand code is missing in 120 cases.

2169 cases our of 2566 are complete, or about 85%. If we dropped all NAs, we would be dropping 15% of the data. Instead, we will use the VIM package for simple KNN imputation. This package just fills in the nearest neighbor-–it doesn't change the other variables. Vim can handle categorical variables, like brand code. We'll use K=5.

```r
training_data_imputed <- kNN(training_data_no_na, k = 5)

#making sure it worked
sum(is.na(training_data_imputed))
```

```
## [1] 0
```

1.4.0.1 Imputing values for the test data First, let's check how many missing values there are after we've removed columns in the test data.

```r
table(rowSums(is.na(test_data_new)))
```

```
##
##   1   2   3   4   5   9
## 225  33   6   1   1   1
```

```r
colSums(is.na(test_data_new))
```

```
##     Brand.Code    Carb.Volume    Fill.Ounces      PC.Volume  Carb.Pressure
##              8              1              6              4              0
##      Carb.Temp            PSC       PSC.Fill        PSC.CO2       Mnf.Flow
##              1              5              3              5              0
```

```
## Carb.Pressure1  Fill.Pressure  Hyd.Pressure2   Filler.Level   Filler.Speed
##              4              2              1              2             10
##    Temperature     Usage.cont      Carb.Flow             PH  Oxygen.Filler
##              2              2              0            267              3
##   Air.Pressurer
##              1
```

```
#total NAs (subtracting 267 for the missing pH values)
sum(is.na(test_data_new)) - 267
```

```
## [1] 60
```

Now there are only 60 missing values total. 42 rows have missing values. One row has 8 missing values (the one that had 14 before). Let's remove that:

```
test_data_new <- test_data_new[rowSums(is.na(test_data_new)) != 9, ]

#confirming that only one value was removed
count(test_data_new)
```

```
##     n
## 1 266
```

Imputing with the same method we used for the training data:

```
#list all columns except PH
cols_to_impute <- setdiff(colnames(test_data_new), "PH")

test_data_imputed <- kNN(test_data_new, variable = cols_to_impute, k = 5)

#we should get 266 cases
sum(is.na(test_data_imputed))
```

```
## [1] 266
```

1.5 Break into training and test data

Breaking the training data set into an 80%/20% training and test set:

```
set.seed(1122)

train_index <- sample(nrow(training_data_imputed),
                      0.8 * nrow(training_data_imputed))

#create a training set
train_y <- training_data_imputed$PH[train_index]
train_x <- training_data_imputed[train_index, ] |> dplyr::select(-PH)

#test set
test_y <- training_data_imputed$PH[-train_index]
test_x  <- training_data_imputed[-train_index, ] |> dplyr::select(-PH)
```

1.5.1 Centering and scaling

I did not scale anything individually, every package should have an option to center/scale automatically, and use a Box-Cox for the data with less normal distributions.